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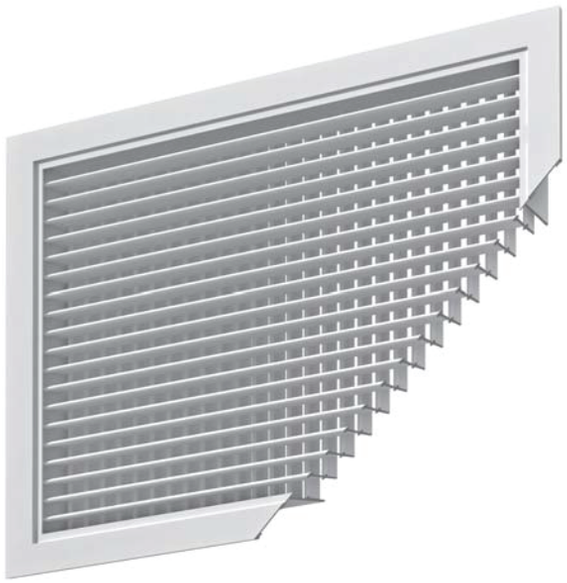
Double Deflection Grille DD/DD1



DD

Description
Aluminium ventilation grille equipped with individually adjustable blades that allow air flow direction. This can be provided with two rows of adjustable blades - called double deflection grille **DD** or a single row of adjustable blades - called **DD1**.

Application
Supply/ exhaust of the air flow

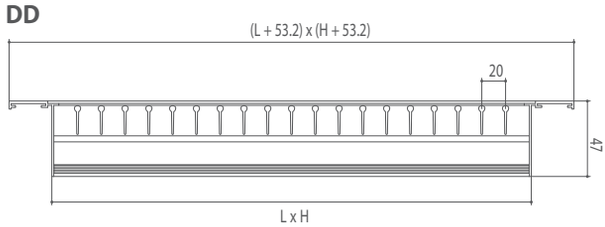
Characteristics
The front row blades are mounted perpendicular to the length **"L"**.
The grille is provided with mounting holes (standard).
A masked settlement frame **CC** can be provided (optional)
The mounting of the grille into the settlement frame is carried out using a clip mechanism to ease the mounting/ removal process.
Specially designed for paneled ceiling insertion (600 x 600 mm), the 595 x 595 mm (exterior) grille is provided without mounting holes.
The air flow can be adjusted by mounting an air control damper **OBD** on the grille inlet.
The grilles with length more than 500 mm are provided with intermediate shaft for reinforcement.

Accessories
Settlement frame made of galvanized sheet metal **CC**
Air control damper with opposable blades **OBD**
Insulated/Non-insulated adapter **ADP IZ/N**

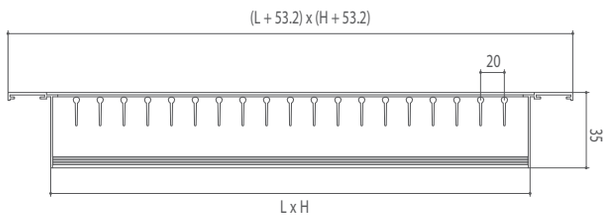
Materials and finishing
Extruded aluminum material, electrostatic field painted white RAL 9016.
Other RAL colors available on request.
Natural anodised aluminum.

Technical design

DD



DD1



Assembly Methods
Visible mounting holes on grille frame (standard).
Settlement frame (optional)

Selection table DD/DD1

Air flow (m^3/h)	LxH	200 x 100	300 x100	300 x 150	400 x 100	500 x 100	300 x 200	400 x 150	600 x 100	400 x 200	500 x 150	800x100	500x200	600 x 150	600 x200	800 x 150	500 x 300	800 x 200	600 x 300	800 x 300	1000 x 300
	Ak(m^2)	0.011	0.016	0.024	0.024	0.03	0.04	0.04	0.04	0.049	0.049	0.049	0.06	0.06	0.078	0.078	0.105	0.105	0.132	0.16	0.21
150	Veff[m/s]	3.8	2.6																		
	ΔPt [pa]	10.0	5.0																		
	X [m]	4.1	3.4																		
	NR [dB(A)]	-	-																		
200	Veffm/s]	5.0	3.5	2.3	2.3																
	ΔPt [pa]	18.0	8.0	4.0	4.0																
	X [m]	5.5	4.5	3.7	3.7																
	NR [dB(A)]	19.0	-	-	-																
300	Veff[m/s]	7.6	5.2	3.5	3.5	2.8	2.1	2.1	2.1												
	ΔPt [pa]	40.0	19.0	8.0	8.0	5.0	3.0	3.0	3.0												
	X [m]	8.2	6.8	5.6	5.6	5.0	4.3	4.3	4.3												
	NR [dB(A)]	28.0	21.0	-	-	-	-	-	-												
400	Veff[m/s]		6.9	4.6	4.6	3.7	2.8	2.8	2.8	2.3	2.3	2.3									
	ΔPt [pa]		3.4	15.0	15.0	10.0	5.0	5.0	5.0	4.0	4.0	4.0									
	X [m]		9.1	7.4	7.4	6.0	5.7	5.7	5.7	5.2	5.2	5.2									
	NR [dB(A)]		28.0	20.0	20.0	16.0	-	-	-	-	-	-									
500	Veff[m/s]			5.8	5.8	4.6	3.5	3.5	3.5	2.8	2.8	2.8	2.3	2.3							
	ΔPt [pa]			23.0	23.0	15.0	8.0	8.0	8.0	6.0	6.0	6.0	4.0	4.0							
	X [m]			9.3	9.3	8.3	7.2	7.2	7.2	6.5	6.5	6.5	5.9	5.9							
	NR [dB(A)]			25.0	25.0	21.0	-	-	-	-	-	-	-	-							
700	Veff[m/s]					6.5	4.9	4.9	4.9	4.0	4.0	4.0	3.2	3.2	2.5	2.5					
	ΔPt [pa]					29.0	17.0	17.0	17.0	11.0	11.0	11.0	7.0	7.0	4.0	4.0					
	X [m]					12.0	10.0	10.0	10.0	9.1	9.1	9.1	8.2	8.2	7.2	7.2					
	NR [dB(A)]					29.0	23.0	23.0	23.0	19.0	19.0	19.0	15.0	15.0	-	-					
900	Veff[m/s]						6.2	6.2	6.2	5.1	5.1	5.1	4.2	4.2	3.2	3.2	2.4	2.4			
	ΔPt [pa]						27.0	27.0	27.0	18.0	18.0	18.0	12.0	12.0	7.0	7.0	4.0	4.0			
	X [m]						13.0	13.0	13.0	12.0	12.0	12.0	11.0	11.0	9.2	9.2	8.0	8.0			
	NR [dB(A)]						29.0	29.0	29.0	25.0	25.0	25.0	21.0	21.0	16.0	16.0	-	-			
1200	Veff[m/s]									6.8	6.8	6.8	5.5	5.5	4.3	4.3	3.2	3.2	2.5	2.1	
	ΔPt [pa]									32.0	32.0	32.0	22.0	22.0	13.0	13.0	7.0	7.0	4.0	3.0	
	X [m]									16.0	16.0	16.0	14.0	14.0	12.0	12.0	11.0	11.0	9.5	8.6	
	NR [dB(A)]									32.0	32.0	32.0	22.0	22.0	23.0	23.0	17.0	17.0	-	-	
1500	Veff[m/s]												6.9	6.9	5.3	5.3	4.0	4.0	3.2	2.6	2.0
	ΔPt [pa]												34.0	34.0	20.0	20.0	11.0	11.0	7.0	5.0	3.0
	X [m]												18.0	18.0	15.0	15.0	13.0	13.0	12.0	11.0	9.4
	NR [dB(A)]												34.0	34.0	29.0	29.0	23.0	23.0	18.0	-	-

Annotation

- Ak [m²] - Free surface of the grille
- Veff[m/s] - Air velocity in grille
- Δ Pt [pa] - Pressure loss
- X [m] - Air jet length
- NR [dB(A)] - Noise level without the room attenuation
- Vk [m/s] - Speed at which airflow length was calculated "x" [m] - Vk [m/s] = 0.375 m/s

NR < 25 | 25≤NR≤35

Correction coefficients for other air velocities

Vk[m/s]		0.25	0.375	0.5
x[m]	A	1.5	1	0.75
	B	1.05	0.7	0.53

- A = with Coanda effect
- B = without Coanda effect

Order codes

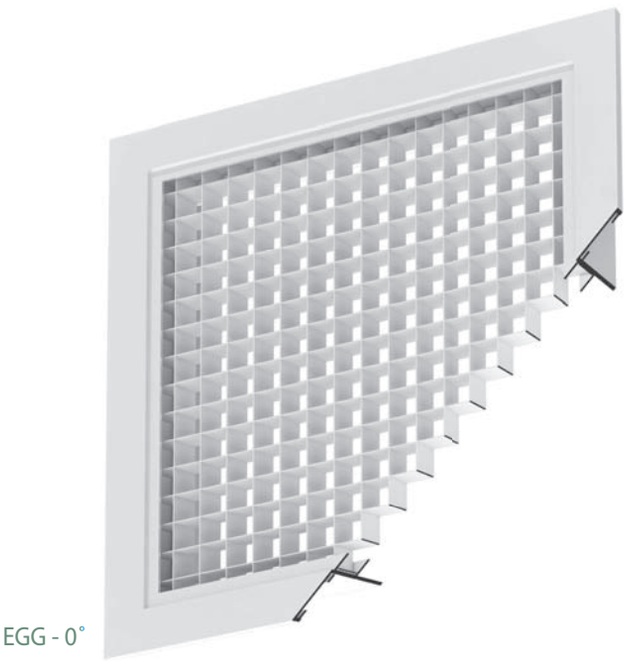
Type + **OBD** + **CC** + **ADP IZ/N** / **Dimensions** / **L X H** / **O**

- **Type**: DD / DD-1
- **OBD**: Air control damper
- **CC**: Settlement frame
- **ADP IZ/N**: Insulated/ Non-insulated adapter
- **Dimensions**: **L X H**

L X H	L X H
200 x 100	500 x 300
300 x 100	600 x 100
300 x 150	600 x 150
300 x 200	600 x 200
400 x 100	600 x 300
400 x 150	800 x 100
400 x 200	800 x 150
500 x 100	800 x 200
500 x 150	800 x 300
500 x 200	1000 x 300

- **L X H**: Other dimensions on request
- **O**: **0** Painted white RAL 9016; **R** Other RAL paints; Natural anodised

Perforated Grille EGG/EGGD



EGG - 0˚

Description

Rectangular ventilation grille with grid that allows the evacuation of high air flow rates with minimum pressure loss **EGG/EGGD**.

Application

Exhaust and recirculation of the air flow.

Characteristics

The grid measures 12.5 x 12.5mm, at 0 or 45 degrees.
Grille is provided with countersunk holes for mounting (standard) **EGG/EGGD.**
Specifically designed for integration into the ceiling box dimensions 600 x 600 mm, the 595 x 595 mm grille (external mesurement) is provided without mounting holes.
Optionally for **EGG,** a settlement frame can be used for concealed mounting and mounting holes on inlet connection for **EGGD**.
The grille is mounted into the settlement frame using a clip mechanism, allowing fast detachment/installation.
EGGD is opened via a click type system (open/close by pressing).
Optionally, an air filter **FIL** can be mounted on the inlet connection of the grille.

Accessories

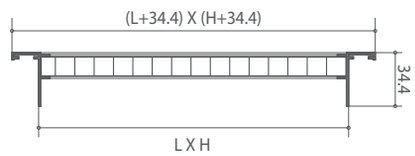
Air filter **FIL**
Air control damper with opposable blades **OBD**
Settlement frame made of galvanized metal sheet **CC**
Insulated/Non-insulated adapter **ADP IZ/N**

Materials and finishing

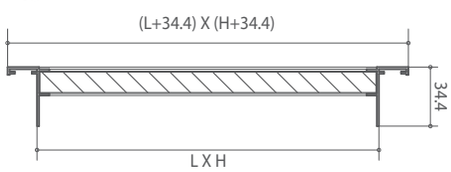
Aluminum material, electrostatic field painted white RAL 9016.
Other RAL colors available on request.
Natural anodised aluminum.

Technical design

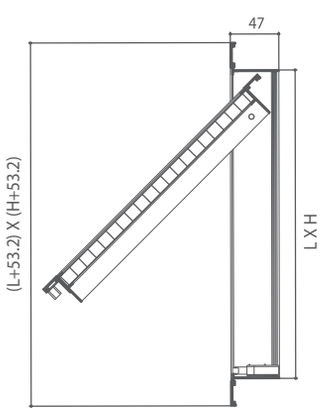
EGG - 0˚



EGG - 45˚



EGGD-0˚



Assembly Methods

Mounting screws (standard)
Settlement frame (optional) **EGG.**
For **EGGD,** with screws on the inlet connection of the grille (optional).

Selection table EGG/EGGD

Air flow (m^3/h)	LxH Ak [m^2]	200 x 100 0.0135	300 x 150 0.0324	400 x 200 0.0675	300 x 300 0.0729	500 x 300 0.1296	600 x 300 0.1539	800 x 300 0.206	600 x 600 0.3249	1000 x 600 0.55575
100	Veff NR [dB(A)] ΔPt [pa]	2.1 < 20 7.0								
200	Veff NR [dB(A)] ΔPt [pa]	4.1 29.0 27.0	1.7 < 20 5.0							
300	Veff NR [dB(A)] ΔPt [pa]	6.2 40.0 61.0	2.6 21.0 11.0	1.2 < 20 2.0	1.1 < 20 2.0					
400	Veff NR [dB(A)] ΔPt [pa]	8.2 48.0 108.0	3.4 29.0 19.0	1.6 < 20 4.0	1.5 < 20 4.0					
500	Veff NR [dB(A)] ΔPt [pa]		4.3 35.0 29.0	2.1 20.0 7.0	1.9 < 20 6.0	1.1 < 20 2.0				
600	Veff NR [dB(A)] ΔPt [pa]		5.1 40.0 42.0	2.5 25.0 10.0	2.3 23.0 8.0	1.3 < 20 3.0	1.1 < 20 2.0			
700	Veff NR [dB(A)] ΔPt [pa]		6.0 44.0 58.0	2.9 29.0 13.0	2.7 27.0 11.0	1.5 < 20 4.0	1.3 < 20 3.0			
800	Veff NR [dB(A)] ΔPt [pa]		6.9 48.0 75.0	3.3 33.0 17.0	3.0 31.0 15.0	1.7 < 20 5.0	1.4 < 20 3.0	1.1 < 20 2.0		
900	Veff NR [dB(A)] ΔPt [pa]		7.7 51.0 95.0	3.7 36.0 22.0	3.4 34.0 19.0	1.9 22.0 6.0	1.6 < 20 4.0	1.2 < 20 2.0		
1000	Veff NR [dB(A)] ΔPt [pa]			4.1 39.0 27.0	3.8 37.0 23.0	2.1 25.0 7.0	1.8 21.0 5.0	1.3 < 20 3.0		
1500	Veff NR [dB(A)] ΔPt [pa]			6.2 50.0 61.0	5.7 48.0 52.0	3.2 36.0 17.0	2.7 32.0 12.0	2.0 26.0 7.0	1.3 < 20 3.0	
2000	Veff NR [dB(A)] ΔPt [pa]				7.6 56.0 93.0	4.3 44.0 29.0	3.6 40.0 21.0	2.7 34.0 12.0	1.7 24.0 5.0	1.0 < 20 2.0
3000	Veff NR [dB(A)] ΔPt [pa]					6.4 55.0 66.0	5.4 51.0 47.0	4.0 45.0 26.0	2.6 35.0 11.0	1.5 24.0 4.0
4000	Veff NR [dB(A)] ΔPt [pa]						7.2 59.0 83.0	5.4 53.0 47.0	3.4 43.0 19.0	2.0 32.0 6.0
5000	Veff NR [dB(A)] ΔPt [pa]							6.7 59.0 73.0	4.3 49.0 29.0	2.5 38.0 10.0
6000	Veff NR [dB(A)] ΔPt [pa]								5.1 54.0 42.0	3.0 43.0 14.0
7000	Veff NR [dB(A)] ΔPt [pa]								6.0 59.0 57.0	3.5 47.0 20.0
8000	Veff NR [dB(A)] ΔPt [pa]									4.0 51.0 26.0
9000	Veff NR [dB(A)] ΔPt [pa]									4.5 54.0 32.0
10000	Veff NR [dB(A)] ΔPt [pa]									5.0 57.0 40.0

Annotation

- Ak [m²] - Free surface of the grille
- Veff [m/s] - Air velocity inside the grille
- NR [dB(A)] - Noise level without the room attenuation
- ΔPt [pa] - Pressure loss

NR < 35 | 35≤NR≤45 | NR>45

Order codes

Type - 0/45° + FIL + OBD + CC + ADP IZ/N / Dimensions / L X H / O

- **EGG/EGGD**
- Air filter
- Air control damper
- Settlement frame
- Insulated/ non-insulated adapter
- Dimensions
  - L X H: 200 x 100, 300 x 150, 400 x 200, 300 x 300, 500 x 300
  - L X H: 600 x 300, 800 x 300, 600 x 600, 1000 x 600
- Other dimensions on request
- **0** Painted white RAL 9016
  R Other RAL paints
  Natural anodised

EILAT Grille

Description
Rectangular ventilation grille with two rows of adjustable blades and operating lever.

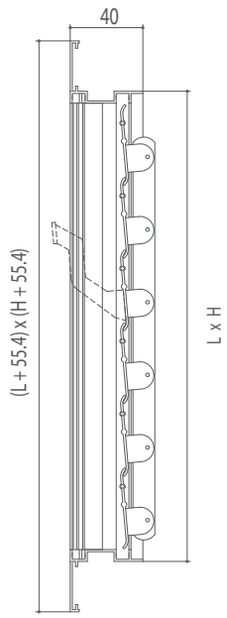
Application
Supply/ exhaust of the air flow

Characteristics
The front row blades are mounted perpendicular to the length "L" and are individually adjustable.
Additional blades are mounted perpendicular to the "H" (high) and allow adjustment of air flow through the operating lever.
The action lever operates on a maximum length of 500 mm. For larger dimensions, additional operating lever will be mounted.
The grille is provided with mounting holes (standard).

Accessories
Insulated/Non-insulated adapter **ADP IZ/N**.

Materials and finishing
Extruded aluminum material, electrostatic field painted white RAL 9016.
Other RAL colors available on request.

Technical design



Assembly Methods
Standard mounting, with screws.

Selection table EILAT

Air flow (m^3/h)	LxH	200 x 100	300 x100	300 x 150	400 x 100	500 x 100	300 x 200	400 x 150	600 x 100	400 x 200	500 x 150	800x100	500x200	600 x 150	600 x200	800 x 150	500 x 300	800 x 200	600 x 300	800 x 300	1000 x 300
	Ak(m^2)	0.011	0.016	0.024	0.024	0.03	0.04	0.04	0.04	0.049	0.049	0.049	0.06	0.06	0.078	0.078	0.105	0.105	0.132	0.16	0.21
150	Veff[m/s]	3.8	2.6																		
	ΔPt [pa]	10.0	5.0																		
	X [m]	4.1	3.4																		
	NR [dB(A)]	-	-																		
200	Veffm/s]	5.0	3.5	2.3	2.3																
	ΔPt [pa]	18.0	8.0	4.0	4.0																
	X [m]	5.5	4.5	3.7	3.7																
	NR [dB(A)]	19.0	-	-	-																
300	Veff[m/s]	7.6	5.2	3.5	3.5	2.8	2.1	2.1	2.1												
	ΔPt [pa]	40.0	19.0	8.0	8.0	5.0	3.0	3.0	3.0												
	X [m]	8.2	6.8	5.6	5.6	5.0	4.3	4.3	4.3												
	NR [dB(A)]	28.0	21.0	-	-	-	-	-	-												
400	Veff[m/s]		6.9	4.6	4.6	3.7	2.8	2.8	2.8	2.3	2.3	2.3									
	ΔPt [pa]		3.4	15.0	15.0	10.0	5.0	5.0	5.0	4.0	4.0	4.0									
	X [m]		9.1	7.4	7.4	6.0	5.7	5.7	5.7	5.2	5.2	5.2									
	NR [dB(A)]		28.0	20.0	20.0	16.0	-	-	-	-	-	-									
500	Veff[m/s]			5.8	5.8	4.6	3.5	3.5	3.5	2.8	2.8	2.8	2.3	2.3							
	ΔPt [pa]			23.0	23.0	15.0	8.0	8.0	8.0	6.0	6.0	6.0	4.0	4.0							
	X [m]			9.3	9.3	8.3	7.2	7.2	7.2	6.5	6.5	6.5	5.9	5.9							
	NR [dB(A)]			25.0	25.0	21.0	-	-	-	-	-	-	-	-							
700	Veff[m/s]					6.5	4.9	4.9	4.9	4.0	4.0	4.0	3.2	3.2	2.5	2.5					
	ΔPt [pa]					29.0	17.0	17.0	17.0	11.0	11.0	11.0	7.0	7.0	4.0	4.0					
	X [m]					12.0	10.0	10.0	10.0	9.1	9.1	9.1	8.2	8.2	7.2	7.2					
	NR [dB(A)]					29.0	23.0	23.0	23.0	19.0	19.0	19.0	15.0	15.0	-	-					
900	Veff[m/s]						6.2	6.2	6.2	5.1	5.1	5.1	4.2	4.2	3.2	3.2	2.4	2.4			
	ΔPt [pa]						27.0	27.0	27.0	18.0	18.0	18.0	12.0	12.0	7.0	7.0	4.0	4.0			
	X [m]						13.0	13.0	13.0	12.0	12.0	12.0	11.0	11.0	9.2	9.2	8.0	8.0			
	NR [dB(A)]						29.0	29.0	29.0	25.0	25.0	25.0	21.0	21.0	16.0	16.0	-	-			
1200	Veff[m/s]									6.8	6.8	6.8	5.5	5.5	4.3	4.3	3.2	3.2	2.5	2.1	
	ΔPt [pa]									32.0	32.0	32.0	22.0	22.0	13.0	13.0	7.0	7.0	4.0	3.0	
	X [m]									16.0	16.0	16.0	14.0	14.0	12.0	12.0	11.0	11.0	9.5	8.6	
	NR [dB(A)]									32.0	32.0	32.0	22.0	22.0	23.0	23.0	17.0	17.0	-	-	
1500	Veff[m/s]												6.9	6.9	5.3	5.3	4.0	4.0	3.2	2.6	2.0
	ΔPt [pa]												34.0	34.0	20.0	20.0	11.0	11.0	7.0	5.0	3.0
	X [m]												18.0	18.0	15.0	15.0	13.0	13.0	12.0	11.0	9.4
	NR [dB(A)]												34.0	34.0	29.0	29.0	23.0	23.0	18.0	-	-

NR < 25 | 25≤NR≤35

Annotation

* Ak [m²] - Free surface of the grille
* Veff[m/s] - Air velocity in grille
* ΔPt [pa] - Pressure loss
* X [m] - Air jet length
* NR [dB(A)] - Noise level without room attenuation
* Vk [m/s] - Speed at which airflow length was calculated
  "x" [m] - Vk [m/s] = 0.375 m/s

Order codes

Type + **ADP IZ/N** / **Dimensions** / **L X H** / **O**

- **Type**: **EILAT**
- **ADP IZ/N**: Insulated/Non-insulated adapter
- **Dimensions**: **L X H** / **L X H**

L X H	L X H
200 x 100	500 x 300
300 x 100	600 x 100
300 x 150	600 x 150
300 x 200	600 x 200
400 x 100	600 x 300
400 x 150	800 x 100
400 x 200	800 x 150
500 x 100	800 x 200
500 x 150	800 x 300
500 x 200	1000 x300

- **L X H**: Other dimensions on request
- **O**: **0** Painted white RAL 9016; **R** Other RAL paints

Gravitational Grille NRD/NRDD

NRD

Description
Rectangular aluminum grille for air overpressure with mobile blades (normally closed) .

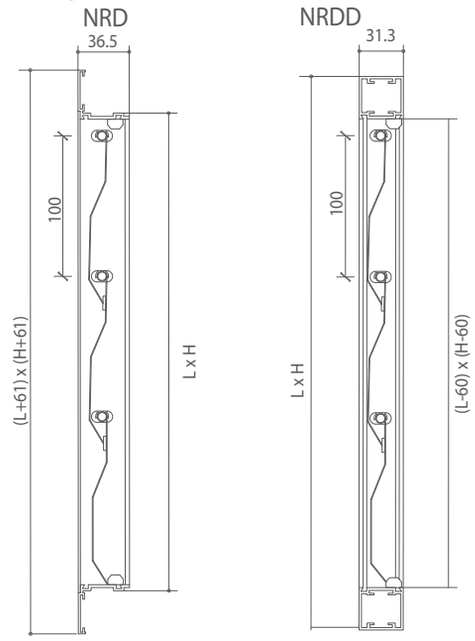
Application
Exhaust/ supply of the airflow

Characteristics
The standard grille is produced for air flow exhaust **NRD/NRDD**
Upon request, if the grid is used for introduction of air, the blades will be placed in reverse.
Plastic bearing sections to avoid friction.
Sealing gasket.
Central reinforcement for lengths over 500 mm.
Provided with mounting holes.

Accessories
Insulated/Non-insulated adapter **ADP IZ/N.**

Materials and finishing
Anodised frame with natural aluminum blades
Plastic bearing sections

Technical design



Assembly Methods
Countersinking holes on the visible frame of the grille (standard) **NRD**

Selection table NRD

Air velocity inside the grille Veff(m/s)	1	2	3	4	5	6	7
Pressure loss ΔPt (Pa)	10	18	25	30	36	44	52

Discharge area AK [m²]									
Height (mm)	Length (mm)								
	200	300	400	500	600	700	800	900	1000
200	0.03	0.05	0.07	0.08	0.10	0.12	0.14	0.15	0.17
300	0.05	0.08	0.10	0.13	0.16	0.19	0.21	0.24	0.27
400	0.07	0.10	0.14	0.18	0.22	0.25	0.29	0.33	0.37
500	0.08	0.13	0.18	0.23	0.27	0.32	0.37	0.42	0.46
600	0.10	0.16	0.22	0.27	0.33	0.39	0.45	0.50	0.56
700	0.12	0.19	0.25	0.32	0.39	0.46	0.52	0.59	0.66
800	0.14	0.21	0.29	0.37	0.45	0.52	0.60	0.68	0.76
900	0.15	0.24	0.33	0.42	0.50	0.59	0.68	0.77	0.85
1000	0.17	0.27	0.37	0.46	0.56	0.66	0.76	0.85	0.95

Selection table NRD - depending on air flow for velocity Veff = 3 m/s

Air flow [mc/h]									
Height (mm)	Length (mm)								
	200	300	400	500	600	700	800	900	1000
200	324	540	756	864	1080	1296	1512	1620	1836
300	540	864	1080	1404	1728	2052	2268	2592	2916
400	756	1080	1512	1944	2376	2700	3132	3564	3996
500	864	1404	1944	2484	2916	3456	3996	4536	4968
600	1080	1728	2376	2916	3564	4212	4860	5400	6048
700	1296	2052	2700	3456	4212	4968	5616	6372	7128
800	1512	2268	3132	3996	4860	5616	6480	7344	8208
900	1620	2592	3564	4536	5400	6372	7344	8316	9180
1000	1836	2916	3996	4968	6048	7128	8208	9180	10260

Annotation

* Veff[m/s] - Air velocity inside the grille
* ΔPt [pa] - Pressure loss
* Ak [m²] - Free surface

Note: Optimal velocity for net dimensions selection is between 2 and 4 m/s.
By applying the formula Ak[m²] = Air flow [mc/h] / 3600[s] / Veff [m/s] and using the selection table, the dimensions of the grille can be determined.

Order codes

Type + **ADP IZ/N** / **Dimensions** / **L X H**

NRD
NRDD

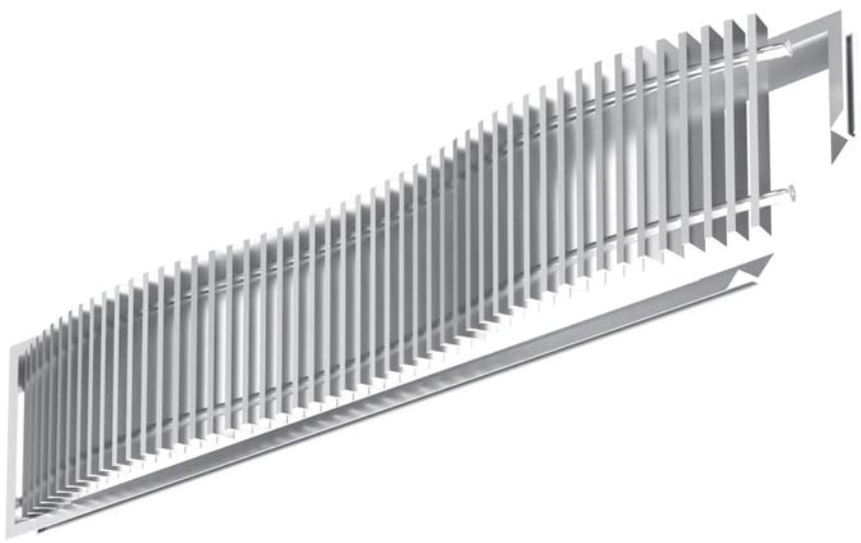
Insulated/Non-insulated adapter

L X H

200 x 200
300 x 300
400 x 400
500 x 500
600 x 600
700 x 700
800 x 800
900 x 900
1000 x 1000

Other dimensions on request

Floor Linear Diffuser LBDF



Description
Rectangular aluminum grille for floor mounting

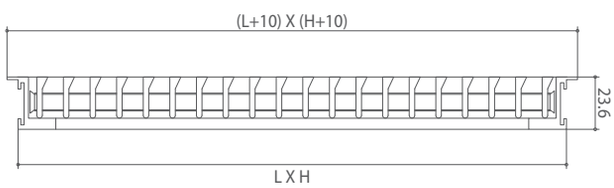
Application
Various air supply/ exhaust systems (eg. floor convector).

Characteristics
15 degree shaped blades perpendicular to the length "L" of the grille
The grille consists of an outdoor frame and a removable inner rolling core.
Removable core allows easy installation and easy cleaning.

Accessories
Insulated/Non-insulated adapter **ADP IZ/N**.

Materials and finishing
Extruded aluminum with natural anodised finishing

Technical design



Selection table LBDF

Supply		Veff [m/s]	2
		ΔPt [pa]	4
		NR [dB(A)]	21
Length (mm)	Height (mm)	Air flow (m^3/h)	Ak (mp)
450	100	100	0.014
550		125	0.017
650		150	0.021
800		180	0.025
1050		250	0.035
1300		300	0.042
450	150	180	0.025
550		230	0.032
650		270	0.038
800		350	0.049
1050		450	0.063
1300		570	0.079
450	200	270	0.038
550		330	0.046
650		400	0.056
800		500	0.069
1050		670	0.093
1300		840	0.117

Exhaust		Veff [m/s]	3
		ΔPt [pa]	7.5
		NR [dB(A)]	24
Length (mm)	Height (mm)	Air flow (m^3/h)	Ak (mp)
450	100	150	0.014
550		180	0.017
650		220	0.021
800		270	0.025
1050		370	0.035
1300		460	0.042
450	150	270	0.025
550		350	0.032
650		410	0.038
800		510	0.049
1050		690	0.063
1300		860	0.079
450	200	400	0.038
550		500	0.046
650		600	0.056
800		750	0.069
1050		1010	0.093
1300		1250	0.117

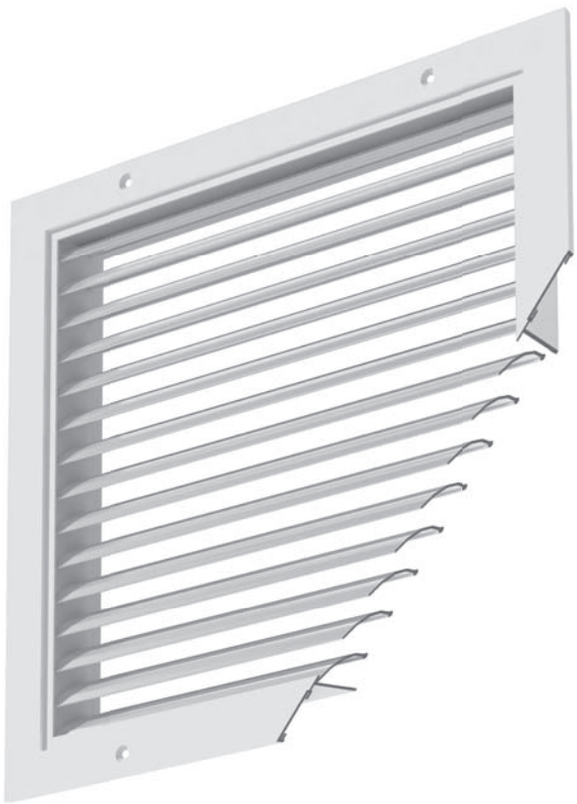
Annotation

* Veff[m/s] - Air velocity in grille
* Δ Pt [pa] - Pressure loss
* NR [dB(A)] - Noise level without the room attenuation
* Ak [m^2] - Free surface of the grille

Order codes

Type	+	ADP IZ/N	/	Dimensions
LBDF		Insulated/Non-insulated adapter		On request

Aspiration Grille SD



Description
Aluminum grille for air exhaust/ recirculation with fixed blades at 45 degrees **SD**.

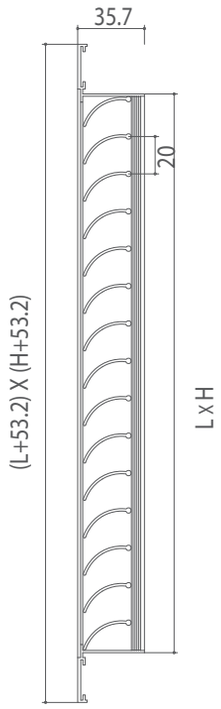
Application
Exhaust/recirculation of the air.

Characteristics
The grille is provided with fixed blades and countersinking holes for mounting, except size 595 x 595 mm (paneled ceiling).
Optionally, can be used a settlement frame for concealed mounting.
The air flow can be adjusted by mounting an air control damper **OBD** on the diffuser inlet.

Accessories
Air filter **FIL**
Air control damper with opposable blades **OBD**.
Settlement frame **CC** made of galvanized steel sheet.
Insulated/Non-insulated adapter **ADP IZ/N**.

Materials and finishing
Extruded aluminum electrostatic field painted white RAL 9016.
Other RAL colors available on request.
Natural anodised aluminum.

Technical design



Assembly Method
With screws (standard).
Settlement frame for mounting (optional).

Aspiration Grille SDD

Description
Aluminum grille for air exhaust/ recirculation with fixed blades at 45 degrees **SDD**.

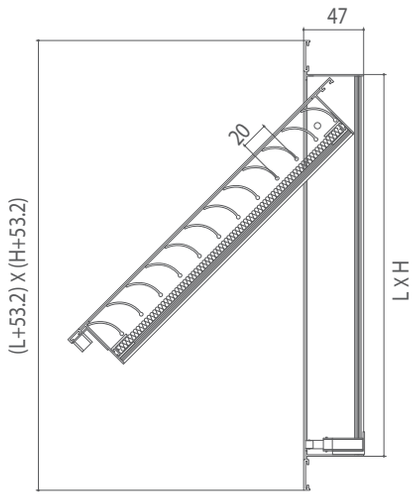
Application
Exhaust/recirculation of the air

Characteristics
The grille is provided with fixed blades and countersinking holes for mounting, except size 595 x 595 mm (paneled ceiling).
Optionally can be used a settlement frame for concealed mounting or mounting holes on inlet connection.
SDD is provided with access door and air filter.
The opening method of **SDD** consists of a click type system (open/close by pressing).
The air flow can be adjusted by mounting an air control damper **OBD** on the diffuser inlet.

Accessories
Air control damper with opposable blades **OBD.**
Settlement frame **CC** made of galvanized steel sheet.
Insulated/Non-insulated adapter **ADP IZ/N.**

Materials and finishing
Extruded aluminum electrostatic field painted white RAL 9016.
Other RAL colors available on request.
Natural anodised aluminum.

Technical design



Assembly Method
Mounting with screws on the visible frame (standard)
Mounting with screws on the inlet connection of the grille (optional).

Selection table SD/SDD

Air flow	L x H	200 x 100	300 x 150	400 x 200	300 x 300	500 x 300	600 x 300	800 x 300	600 x 600	1000 x 600
(m^3/h)	Ak [m^2]	0.0076	0.0189	0.0345	0.0416	0.0721	0.0915	0.1173	0.1759	0.2934
60	Veff[m/s]	2.2								
	ΔPt [pa]	5.0								
	NR [dB(A)]	22.0								
80	Veff[m/s]	2.9								
	ΔPt [pa]	9.0								
	NR [dB(A)]	29.0								
100	Veff[m/s]	3.7								
	ΔPt [pa]	14.0								
	NR [dB(A)]	35.0								
140	Veff[m/s]	5.1	2.1							
	ΔPt [pa]	27.0	2.0							
	NR [dB(A)]	44.0	16.0							
180	Veff[m/s]	6.6	2.6							
	ΔPt [pa]	45.0	3.0							
	NR [dB(A)]	51.0	22.0							
200	Veff[m/s]	7.3	2.9	1.6						
	ΔPt [pa]	56.0	4.0	3.0						
	NR [dB(A)]	53.0	25.0	22.0						
300	Veff[m/s]		4.4	2.4	2.0					
	ΔPt [pa]		9.0	6.0	2.0					
	NR [dB(A)]		36.0	32.0	22.0					
400	Veff[m/s]		5.9	3.2	2.7					
	ΔPt [pa]		16.0	12.0	4.0					
	NR [dB(A)]		43.0	40.0	30.0					
500	Veff[m/s]		7.3	4.0	3.3	1.9				
	ΔPt [pa]		25.0	18.0	7.0	2.0				
	NR [dB(A)]		49.0	46.0	36.0	24.0				
600	Veff[m/s]				4.0	2.3	1.8	1.4		
	ΔPt [pa]				11.0	3.0	2.0	1.0		
	NR [dB(A)]				41.0	29.0	24.0	18.0		
800	Veff[m/s]				5.3	3.1	2.4	1.9		
	ΔPt [pa]				19.0	6.0	4.0	2.0		
	NR [dB(A)]				48.0	36.0	32.0	26.0		
1000	Veff[m/s]				6.7	3.9	3.0	2.4	1.6	
	ΔPt [pa]				30.0	10.0	6.0	4.0	1.0	
	NR [dB(A)]				54.0	42.0	38.0	32.0	21.0	
1400	Veff[m/s]					5.4	4.3	3.3	2.2	
	ΔPt [pa]					20.0	13.0	7.0	2.0	
	NR [dB(A)]					51.0	46.0	40.0	29.0	
1800	Veff[m/s]						5.5	4.3	2.8	
	ΔPt [pa]						22.0	12.0	4.0	
	NR [dB(A)]						53.0	47.0	36.0	
2000	Veff[m/s]							4.7	3.2	
	ΔPt [pa]							15.0	5.0	
	NR [dB(A)]							50.0	39.0	
3000	Veff[m/s]								4.7	2.8
	ΔPt [pa]								13.0	1.0
	NR [dB(A)]								49.0	22.0
4000	Veff[m/s]									3.8
	ΔPt [pa]									1.0
	NR [dB(A)]									30.0

NR < 25 | 25≤NR≤35 | NR>35

Annotation

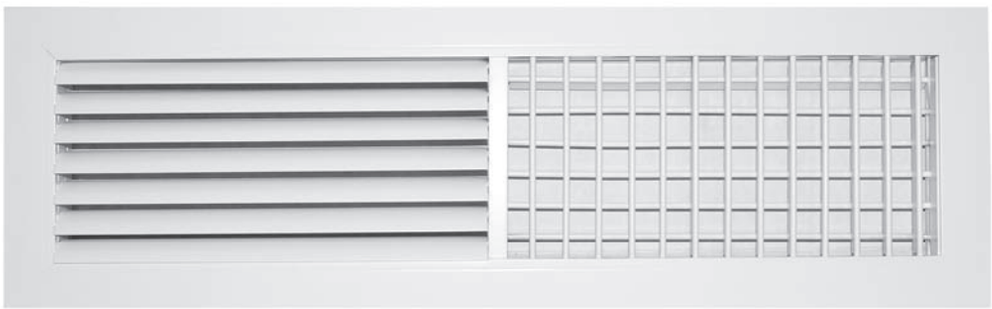
* Ak [m²] - Free surface
* Veff[m/s] - Air velocity inside the grille
* Δ Pt [pa] - Pressure loss
* NR [dB(A)] - Noise level without the room attenuation

Order codes

Type + **FIL** + **OBD** + **CC** + **ADP IZ/N** / **Dimensions** / **L x H** / **O**

- **Type**: **SD** **SDD**
- **FIL**: Air filter (for SD)
- **OBD**: Air control damper
- **CC**: Settlement frame
- **ADP IZ/N**: Insulated/Non-insulated adapter
- **Dimensions**: **L x H** 200 x 100, 300 x 150, 400 x 200, 300 x 300, 500 x 300, 600 x 300, 800 x 300, 600 x 600, 1000 x 600
- **L x H**: Other dimensions on request
- **O**: **0** Painted white RAL 9016; **R** Other RAL paints; Natural anodised

Combined Grille SD-DD



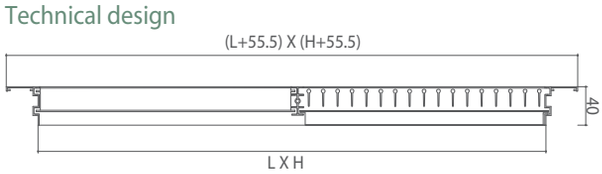
Description
Aluminum rectangular grill that consists in simple deflection grille **SD** combined with a double deflection grille **DD**

Application
Supply/ exhaust of the airflow

Characteristics
The DD grille blades are individually adjustable and allow airflow direction and the SD grille blades are fixed.
Grille is provided with countersunk holes for mounting.
A settlement frame CC is recommended for concealed installation
The settlement frame can only be used if the SD blades are perpendicular to the length scale L
The air flow can be adjusted by mounting an air control damper OBD on the grille inlet connection.

Accessories
Air control damper with opposable blades **OBD**
Settlement frame made of galvanized metal sheet **CC**
Insulated/Non-insulated adapter **ADP IZ/N**

Materials and finishing
Extruded aluminum material, electrostatic field painted white RAL 9016.
Other RAL colors available on request.

Technical design


Assembly Methods
Screws for mounting (standard)
Settlement frame for mounting (optional)

Order codes

Type	+	OBD	+	CC	+	ADP IZ/N	/	L X H	/	O
SD-DD		Air control damper		Settlement frame		Indulated/Non-insulated adapter		On request		**0** Painted white RAL 9016 **R** Other RAL paints

Transfer Grille TG / TGS

TG

Description
Aluminum transfer grille for door mounting **TG/TGS**.

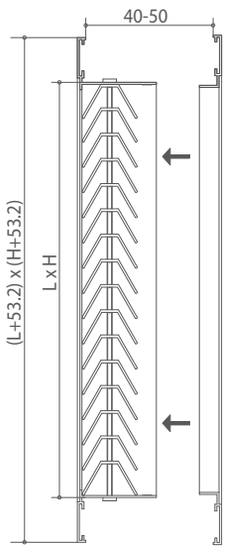
Application
Designed to allow air transfer and pressure equalization between rooms

Characteristics
Grid transfer is provided with "V"-shaped blades with no opposite visibility.
TG is used for doors with thickness of 40-50 mm, and **TGS** for doors with thickness of 20 mm.
The standard version of this product is provided with mounting holes.

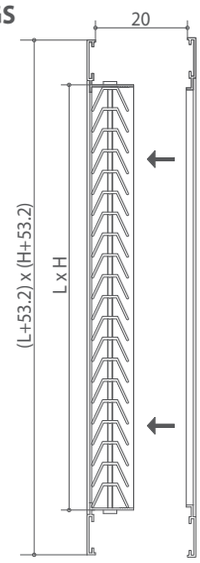
Materials and finishing
Extruded aluminum, electrostatic field painted white RAL 9016.
Other RAL colors available on request.
Natural anodised aluminum.

Technical design

TG



TGS



Assembly Method
Standard mounting, with screws.

Selection table TG

Air flow (m^3/h)	L x H	200 X 100	300 X 100	400 X 200	300 X 300	400 X 300	500 X 300	600 X 300	600 X 400
	Ak [m^2]	0.0099	0.0245	0.0447	0.0507	0.0681	0.0856	0.1031	0.138
50	Veff[m/s] NR [dB(A)] ΔPt [Pa]	1.4 30.0 10.0							
60	Veff[m/s] NR [dB(A)] ΔPt [Pa]	1.7 35.0 14.0	0.7 13.0 2.0						
70	Veff[m/s] NR [dB(A)] ΔPt [Pa]	2.0 39.0 19.0	0.8 16.0 3.0						
80	Veff[m/s] NR [dB(A)] ΔPt [Pa]	2.2 42.0 24.0	0.9 20.0 4.0						
90	Veff[m/s] NR [dB(A)] ΔPt [Pa]	2.5 45.0 31.0	1.0 23.0 5.0	0.6 8.0 2.0					
100	Veff[m/s] NR [dB(A)] ΔPt [Pa]	2.8 48.0 38.0	1.1 25.0 6.0	0.6 10.0 2.0					
120	Veff[m/s] NR [dB(A)] ΔPt [Pa]	3.4 52.0 55.0	1.4 30.0 9.0	0.7 15.0 3.0	0.7 12.0 2.0				
140	Veff[m/s] NR [dB(A)] ΔPt [Pa]		1.6 34.0 12.0	0.9 19.0 4.0	0.8 16.0 3.0	0.6 8.0 2.0			
160	Veff[m/s] NR [dB(A)] ΔPt [Pa]		1.8 37.0 16.0	1.0 22.0 5.0	0.9 19.0 4.0	0.7 12.0 2.0			
180	Veff[m/s] NR [dB(A)] ΔPt [Pa]		2.0 40.0 20.0	1.1 25.0 6.0	1.0 22.0 5.0	0.7 15.0 3.0	0.6 9.0 2.0		
200	Veff[m/s] NR [dB(A)] ΔPt [Pa]		2.3 43.0 25.0	1.2 28.0 8.0	1.1 25.0 6.0	0.8 17.0 3.0	0.6 12.0 2.0		
250	Veff[m/s] NR [dB(A)] ΔPt [Pa]		2.8 48.0 39.0	1.6 33.0 12.0	1.4 30.0 9.0	1.0 23.0 5.0	0.8 17.0 3.0	0.7 13.0 2.0	
300	Veff[m/s] NR [dB(A)] ΔPt [Pa]		3.4 53.0 57.0	1.9 38.0 17.0	1.6 35.0 13.0	1.2 27.0 7.0	1.0 22.0 5.0	0.8 17.0 3.0	
350	Veff[m/s] NR [dB(A)] ΔPt [Pa]			2.2 42.0 23.0	1.9 38.0 18.0	1.4 31.0 10.0	1.1 26.0 6.0	0.9 21.0 4.0	0.7 14.0 2.0
400	Veff[m/s] NR [dB(A)] ΔPt [Pa]			2.5 45.0 30.0	2.2 42.0 24.0	1.6 34.0 13.0	1.3 29.0 8.0	1.1 24.0 6.0	0.8 17.0 3.0
500	Veff[m/s] NR [dB(A)] ΔPt [Pa]			3.1 50.0 47.0	2.7 47.0 37.0	2.0 40.0 20.0	1.6 34.0 13.0	1.3 30.0 9.0	1.0 23.0 5.0
600	Veff[m/s] NR [dB(A)] ΔPt [Pa]				3.3 52.0 53.0	2.4 45.0 29.0	1.9 39.0 19.0	1.6 34.0 13.0	1.2 27.0 7.0
700	Veff[m/s] NR [dB(A)] ΔPt [Pa]					2.9 48.0 40.0	2.3 43.0 25.0	1.9 38.0 17.0	1.4 31.0 10.0
800	Veff[m/s] NR [dB(A)] ΔPt [Pa]					3.3 52.0 52.0	2.6 46.0 33.0	2.2 41.0 23.0	1.6 34.0 13.0
900	Veff[m/s] NR [dB(A)] ΔPt [Pa]						2.9 49.0 42.0	2.4 44.0 29.0	1.8 37.0 16.0
1000	Veff[m/s] NR [dB(A)] ΔPt [Pa]						3.2 52.0 52.0	2.7 47.0 36.0	2.0 40.0 20.0
1200	Veff[m/s] NR [dB(A)] ΔPt [Pa]							3.2 52.0 51.0	2.4 44.0 29.0
1400	Veff[m/s] NR [dB(A)] ΔPt [Pa]								2.8 48.0 39.0
1600	Veff[m/s] NR [dB(A)] ΔPt [Pa]								3.2 51.0 51.0

Annotation

* Ak [m²] - Free surface of the grille
* Veff[m/s] - Air velocity in grille
* NR [dB(A)] - Noise level without the room attenuation
* ΔPt [pa] - Pressure loss

NR < 35 | 35≤NR≤45 | NR>45

Order codes

Type / **Dimensions** / **L x H** / **O**

Type
TG
TGS

Dimensions
LxH
200 x 100
300 x 150
400 x 200
300 x 300
400 x 300
600 x 200
500 x 300
600 x 300
600 x 400

L x H
Other dimensions on request

O
0 Painted white RAL 9016
R Other RAL paints
Natural anodised

Exterior Grille WPL/WPLS

Description
Weatherproof rectangular aluminum ventilation grille, providing protection against ingress of rain, snow, leaves, birds and insects.

Application
Supply/ exhaust of the airflow

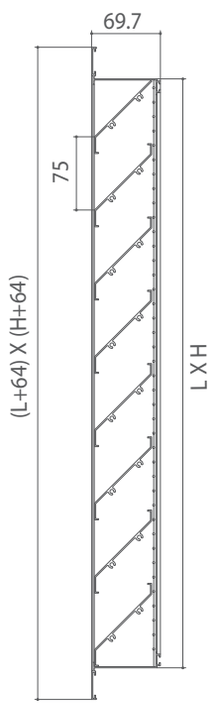
Characteristics
The grille is provided with weatherproof blades, galvanized steel wire mesh and mounting holes for fixing.
Wire mesh used for **WPL** grille is 10 x 10 mm and 6.3 mm x 6.3 mm for **WPLS**
Available sizes:
WPL any size from 800 x 800 mm to 2000 x 2000 mm.
WPLS any size from 100 x 100 mm to 800 x 800 mm.
Modular execution for large dimensions

Accessories
Air filter **FIL**
Insulated/Non-insulated adapter **ADP IZ/N**

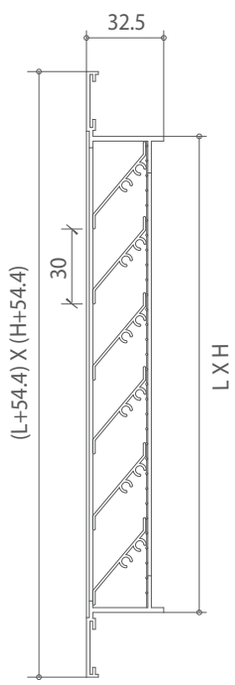
Materials and finishing
Extruded aluminum, electrostatic field painted white RAL 9016
Other RAL colors available on request
Natural anodised aluminum.

Technical design

WPL



WPLS



Assembly Method
Standard mounting, with screws.

Selection table WPL

Air flow (m^3/h)	Ak [m^2]	0.30	0.40	0.50	0.60	0.80	1.00	1.5	2.00	2.5	3.00
3000	Veff [m/s] ΔPt [pa]	2.9 10	2.2 12.0								
4000	Veff [m/s] ΔPt [pa]	4 15	2.9 25.0	2.4 14.0							
5000	Veff [m/s] ΔPt [pa]	4.4 25	3.6 40.0	2.9 24.0	2.4 15.0						
6000	Veff [m/s] ΔPt [pa]	5.6 30	4.2 54.0	3.3 35.0	2.9 24.0	2.1 8					
7000	Veff [m/s] ΔPt [pa]	6.5 28	4.9 77.0	4.0 48.0	3.5 37.0	2.4 15.0					
8000	Veff [m/s] ΔPt [pa]			4.6 63.0	3.9 53.0	2.8 24.0	2.3 10				
9000	Veff [m/s] ΔPt [pa]			5.4 90.0	4.5 63.0	3.2 34.0	2.7 20.0				
10000	Veff [m/s] ΔPt [pa]				4.9 77.0	3.5 37.0	2.8 24.0				
20000	Veff [m/s] ΔPt [pa]						5.4 90.0	3.8 45.0	2.8 23.0	2.3 14.0	
30000	Veff [m/s] ΔPt [pa]							5.7 92.0	4.0 53.0	3.3 35.0	2.8 23.0
40000	Veff [m/s] ΔPt [pa]								5.4 90.0	4.5 63.0	3.8 45.0

Discharge area Ak [m^2]									
Height (mm)	Length (mm)								
	800	900	1000	1100	1200	1400	1600	1800	2000
800	0.36	0.40	0.45	0.50	0.55	0.64	0.74	0.83	0.93
900	0.40	0.45	0.51	0.56	0.61	0.72	0.83	0.93	1.04
1000	0.45	0.51	0.57	0.63	0.69	0.81	0.94	1.06	1.18
1100	0.51	0.57	0.64	0.71	0.78	0.91	1.05	1.08	1.32
1200	0.55	0.62	0.70	0.77	0.84	0.99	1.14	1.28	1.43
1400	0.66	0.74	0.83	0.92	1.01	1.18	1.36	1.53	1.71
1600	0.75	0.85	0.95	1.05	1.15	1.36	1.56	1.76	1.96
1800	0.85	0.96	1.08	1.19	1.30	1.53	1.76	1.98	2.21
2000	0.95	1.08	1.21	1.34	1.47	1.72	1.98	2.23	2.49

Selection table WPLS

Air flow (m^3/h)	Ak [m^2]	0.01	0.02	0.03	0.04	0.05	0.06	0.08	0.10	0.15	0.20	0.25	0.30	0.40
200	Veff [m/s] ΔPt [pa]	5.6 50	2.8 8.0											
300	Veff [m/s] ΔPt [pa]		4.2 28.0	2.8 8.0										
400	Veff[m/s] ΔPt [pa]		5.6 50.0	3.9 26.0	2.8 8.0	2.3 3.0								
500	Veff [m/s] ΔPt [pa]			4.8 38.0	3.6 20.0	2.8 8	2.3 3.0							
600	Veff [m/s] ΔPt [pa]			5.8 53.0	4.3 29.0	3.4 18.0	3.0 14.0	2.2 2						
700	Veff [m/s] ΔPt [pa]				5 40	4.0 25.0	3.4 18.0	2.5 7.0	2.2 2					
800	Veff [m/s] ΔPt [pa]				5.8 53.0	4.7 35.0	3.8 25	2.8 8	2.4 4.0					
1000	Veff [m/s] ΔPt [pa]					5.5 50	4.8 38.0	3.5 18.0	2.8 8.0					
2000	Veff [m/s] ΔPt [pa]								5.8 53.0	3.8 25.0	2.8 8.0	2.4 4.0		
3000	Veff [m/s] ΔPt [pa]									5.8 53.0	4.4 30	3.5 19.0	2.8 8.0	2.2 2.0
4000	Veff [m/s] ΔPt [pa]										5.6 51.0	4.6 32.0	3.8 25	2.8 8.0
5000	Veff [m/s] ΔPt [pa]											5.5 50.0	4.8 37.0	3.5 18.0

Discharge area Ak [m^2]								
Height (mm)	Length (mm)							
	150	200	300	400	500	600	700	800
100	0.001	0.008	0.012	0.017	0.024	0.028	0.040	0.050
150	0.013	0.017	0.025	0.030	0.040	0.050	0.060	0.070
200	0.018	0.020	0.030	0.040	0.060	0.070	0.080	0.090
300	0.027	0.035	0.050	0.070	0.090	0.100	0.120	0.140
400	0.036	0.045	0.070	0.090	0.120	0.140	0.170	0.190
500	0.045	0.060	0.090	0.120	0.150	0.180	0.210	0.240
600	0.050	0.070	0.100	0.130	0.180	0.210	0.250	0.290
700	0.060	0.080	0.110	0.160	0.210	0.250	0.300	0.330
800	0.070	0.090	0.130	0.180	0.240	0.300	0.340	0.400

Annotation

* Ak [m^2] - Free surface of the grille
* ΔPt [pa] - Pressure loss
* Veff[m/s] - Air velocity inside the grille

Note: For different discharge areas than those recorded in selection tables, air velocity and pressure drop will be obtained by interpolation.

Order codes

Type + **FIL** + **ADP IZ/N** / **Dimensions** / **L x H** / **O**

Type
- **WPL**
- **WPLS**

FIL
- Air filter

ADP IZ/N
- Insulated/Non-insulated adapter

Dimensions

WPL	WPLS
800 x 800	150 x 150
900 X 900	200 x 200
1000 X 1000	300 x 300
1100 X 1100	400 x 400
1200 X 1200	500 x 500
1400 X 1400	600 x 600
1600 X 1600	700 x 700
1800 X 1800	800 x 800
2000 X 2000	

L x H
- Other dimensions on request

O
- **0** Painted white RAL 9016
- **R** Other RAL paints
- Natural anodised

Exterior Grille With Mobile Blades WPLH

Description
Weatherproof rectangular aluminum grille with drive system/ control for the blades. Ensures protection against intrusion of rain, snow, leaves and birds.

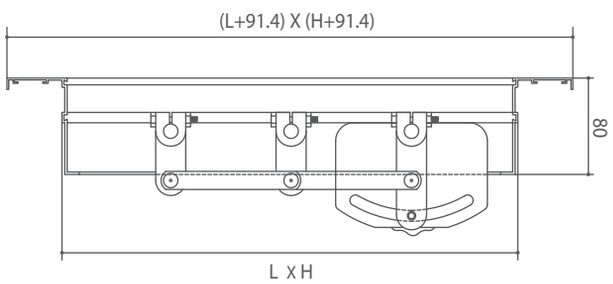
Application
Supply/ exhaust of the airflow

Characteristics
The grille is equipped with weatherproof blades, drive system for adjustment/ locking of the blades, wire mesh and countersinking holes for fixing.
The drive allows adjustment and locking of the blades in the desired position by operating the control lever.
Galvanized steel wire mesh with grids of 10 x 10 mm.

Materials and finishing
Natural anodised aluminum

Assembly Method
Standard mounting, with screws.

Technical design



Order codes

Type	/	Dimensions	/	0
WPLH		On request		Natural anodised

Circular Exterior Grille WPLSR, WPLR

WPLSR

Description
Weatherproof circular aluminum grille. Ensures protection against intrusion of rain, snow, leaves, birds and insects.

Application
Supply/exhaust of the airflow

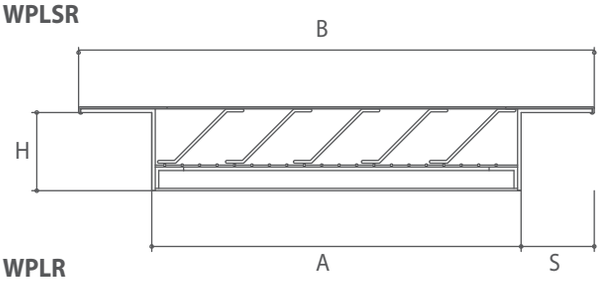
Characteristics
The grille is equipped with weatherproof blades, wire mesh and countersinking holes for fixing.
The wire mesh is available in multiple sizes (grid: 1.5 x1.5, 6.3 x 6.3 and 10 x 10 mm).

Materials and finishing
Extruded aluminum electrostatic field painted white RAL 9016.
Other RAL colors available on request.

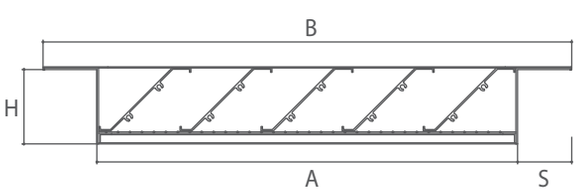
Assembly Method
Standard mounting, with screws.

Technical design

WPLSR



WPLR



Type	Ø	A	B	S	H	Mesh dimension (mm)
WPLSR	125	121	181	30	33	1.5 x 1.5
	160	156	216	30	33	
	200	196	256	30	33	
	250	246	306	30	33	
	315	295	375	40	33	6.3 x 6.3
	400	380	460	40	33	
	450	430	510	40	33	
	500	480	560	40	33	
	630	610	690	40	33	
	710	690	770	40	33	
WPLR	800	782	882	50	70	10 x 10
	1000	982	1082	50	70	

Selection table WPLSR, WPLR

Air flow	Diameter (mm)	125	160	200	250	315	400	450	500	630	710	800	1000
(m^3/h)	Ak(m^2)	0.006	0.01	0.015	0.024	0.038	0.063	0.08	0.1	0.16	0.2	0.27	0.43
60	Veff [m/s] NR [dB(A)] △Pt [pa]	2.6 34.0 25.0											
100	Veff [m/s] NR [dB(A)] △Pt [pa]	4.3 37.0 29.0	2.8 34.0 27.0	1.9 <20 4.0									
150	Veff [m/s] NR [dB(A)] △Pt [pa]		4.2 40.0 36.0	2.8 34.0 25.0									
200	Veff [m/s] NR [dB(A)] △Pt [pa]			3.7 40.0 40.0	2.3 34.0 18.0	2.0 20.0 9.0							
400	Veff [m/s] NR [dB(A)] △Pt [pa]				4.6 38.0 30.0	2.9 34.0 30.0	1.8 30.0 12.0	1.4 23.0 7.0	1.1 <16 4.0				
700	Veff [m/s] NR [dB(A)] △Pt [pa]					5.1 54.0 97.0	3.1 34.0 34.0	2.4 30.0 20.0	1.9 30.0 12.5	1.2 19.0 5.0			
800	Veff [m/s] NR [dB(A)] △Pt [pa]						3.5 44.0 45.0	2.8 34.0 26.0	2.2 32.0 18.0	1.4 24.0 7.0	1.1 <15 4.0		
1000	Veff [m/s] NR [dB(A)] △Pt [pa]						4.4 50.0 68.0	3.5 45.0 43.0	2.8 34.0 27.0	1.7 29.0 10.0	1.4 22.0 6.0	1.0 <15 1.5	
1600	Veff [m/s] NR [dB(A)] △Pt [pa]								4.4 51.0 70.0	2.8 34.0 25.0	2.2 30.0 14.0	1.6 25.0 4.0	1.0 <15 1.5
2000	Veff [m/s] NR [dB(A)] △Pt [pa]									3.5 45.0 40.0	2.8 34.0 25.0	2.1 30.0 6.0	1.3 22.0 2.5
2500	Veff [m/s] NR [dB(A)] △Pt [pa]									4.3 52.0 63.0	3.5 45.0 36.0	2.6 34.0 9.0	1.6 28.0 3.5
4000	Veff [m/s] NR [dB(A)] △Pt [pa]											4.1 48.0 23.0	2.6 34.0 9.0
6000	Veff [m/s] NR [dB(A)] △Pt [pa]												3.9 48.0 20.0

Annotation

* Ak [m²] - Free surface
* Veff[m/s] - Air velocity inside the grille
* NR [dB(A)] - Noise level
* ΔPt [pa] - Pressure loss

Order codes

Type / **Ø** / **0**

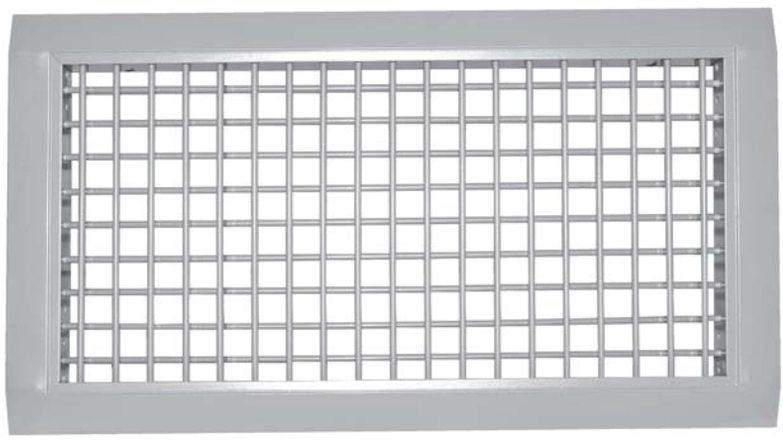
WPLSR
WPLR

WPLSR: 125, 160, 200, 250, 315, 400, 450, 500, 630, 710

WPLR: 800, 1000

0 Painted white RAL 9016
R Other RAL paints

Ventilation Grille for Circular Air Ducts RDD/RDD1



RDD

Description
Rectangular aluminum grille for mounting on circular air ducts.

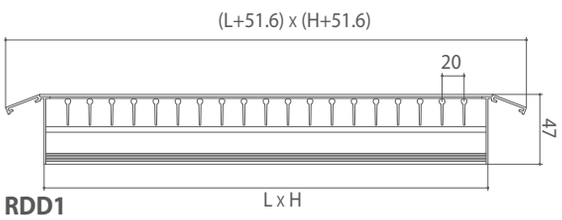
Application
Supply/ exhaust of the air flow.

Characteristics
The grille is equipped with individually adjustable blades and countersinking holes for mounting.
Movable blades allow airflow direction in desired areas.
RDD has two rows of adjustable blades and **RDD1** has a single row of blades.
Grids over 500 mm in length are provided with intermediate reinforcement rods.
The air flow can be adjusted by mounting a air control damper **OBD** on the grille's inlet.

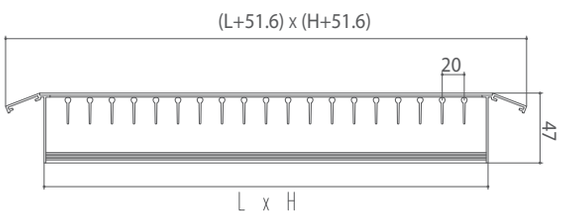
Accessories
Air control damper with opposable blades **OBD.**

Materials and finishing
Extruded aluminum electrostatic field painted white RAL 9016
Other RAL colors available on request

Technical design
RDD



RDD1



Grill Dimensions	Ø tube	
	min.	max.
425 x 75	160	400
525 x 75	160	400
625 x 75	160	400
425 x 125	315	900
525 x 125	315	900
625 x 125	315	900
425 x 225	630	1400
525 x 225	630	1400
625 x 225	630	1400
825 x 225	630	1400
1025 x 225	630	1400

Assembly Method
Standard mounting, with screws

Selection table RDD/RDD1

Air flow m³/h	LxH	425 x 75	525 x 75	625 x 75	425 x 125	525 x 125	625 x 125	425 x 225	525 x 225	625 x 225	825 x 225	1025 x 225
	Ak[m²]	0.013	0.016	0.019	0.025	0.031	0.037	0.049	0.061	0.073	0.097	0.122
200	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]	5.1 24.0 4.3 12.1	4.6 20.0 3.5 8.0									
250	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]	6.4 29.0 5.3 18.8	5.7 25.0 4.3 12.4	5.3 22.0 3.7 8.8								
300	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]	7.6 33.0 6.4 27.1	6.9 29.0 5.2 17.9	6.3 26.0 4.4 12.7	5.5 21.0 3.3 7.3							
350	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]	8.9 37.0 7.5 36.9	8.0 33.0 6.1 24.4	7.4 30.0 5.1 17.3	6.4 24.0 3.9 10.0	5.8 20.0 3.1 6.5						
400	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]	10.2 40.0 8.5 48.2	9.2 36.0 6.9 31.8	8.4 33.0 5.8 22.6	7.3 28.0 4.4 13.0	6.6 23.0 3.6 8.5	6.0 20.0 3.0 6.0					
450	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]	11.5 43.0 9.6 61.0	10.3 39.0 7.8 40.3	9.5 36.0 6.6 18.6	8.3 31.0 5.0 16.5	7.4 26.0 4.0 10.7	6.8 23.0 3.4 7.5					
500	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]		11.5 42.0 8.7 49.7	10.5 38.0 7.3 35.3	9.2 33.0 5.6 20.4	8.2 29.0 4.5 13.2	7.6 25.0 3.8 9.3	6.6 20.0 2.8 5.3				
600	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]			12.6 43.0 8.8 50.8	11.0 38.0 6.7 29.3	9.9 33.0 5.4 19.1	9.1 30.0 4.5 13.4	7.9 24.0 3.4 7.6	7.1 20.0 2.7 4.9			
700	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]				12.9 41.0 7.8 39.9	11.5 37.0 6.3 26.0	10.6 34.0 5.3 18.2	9.2 28.0 4.0 0.4	8.2 24.0 3.2 6.7	7.5 20.0 2.7 4.7		
800	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]				14.7 45.0 8.9 52.1	13.2 40.0 7.2 33.9	12.1 37.0 6.0 23.8	10.5 31.0 4.5 3.6	9.4 27.0 3.6 8.8	8.6 24.0 3.0 6.1		
900	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]					14.8 43.0 8.1 42.9	13.6 40.0 6.8 30.1	11.8 34.0 5.1 17.2	10.6 30.0 4.1 11.1	9.7 26.0 3.4 7.7	8.4 21.0 2.6 4.4	
1000	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]						15.1 42.0 7.5 37.2	13.1 37.0 5.7 21.2	11.8 33.0 4.6 13.7	10.8 29.0 3.8 9.6	9.3 23.0 2.9 5.4	
1200	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]							15.7 41.0 6.8 30.5	14.1 37.0 5.5 19.7	12.9 33.0 4.6 13.8	11.2 28.0 3.4 7.8	10.0 23.0 2.7 4.9
1400	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]							18.4 45.0 7.9 41.6	16.5 41.0 6.4 26.8	15.1 37.0 5.3 18.7	13.1 32.0 4.0 10.6	11.6 27.0 3.2 6.7
1600	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]								18.8 44.0 7.3 35.0	17.2 40.0 6.1 24.5	14.9 35.0 4.6 13.9	13.3 30.0 3.6 8.8
1800	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]									19.4 43.0 6.8 31.0	16.8 38.0 5.2 17.5	15.0 33.0 4.1 11.1
2000	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]										18.7 40.0 5.7 21.6	16.6 36.0 4.6 13.7
2500	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]											20.8 41.0 5.7 21.4

Vk = 0.2 (m/s)

NR < 25 | 25≤NR≤35 | NR>35

Annotation

* Ak [m²] - Free surface
* ΔPt [pa] - Pressure loss
* X [m] - Air jet length
* NR [dB(A)] - Noise level without the room attenuation
* Vk [m/s] - Velocity at which the air jet length was calculated "x" [m]

Order codes

Type	+	OBD	/	Dimensions	/	L x H	/	0
RDD / RDD1		Air control damper		L x H: 425 x 75, 525 x 75, 625 x 75 / 425 x 125, 525 x 125, 625 x 125 / 425 x 225, 525 x 225, 625 x 225, 825 x 225, 1025 x 225		Other dimensions on request		**0** Painted white RAL 9016 **R** Other RAL paints

Ventilation Grille for Circular Air Ducts RDR/RDR1

RDR

Description
Rectangular aluminum grille for mounting on circular air ducts.

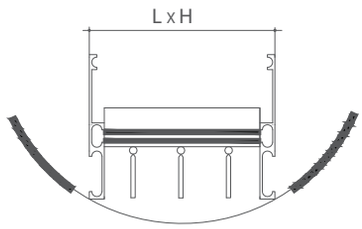
Application
Supply/ exhaust of the air flow.

Characteristics
The grille is equipped with individually adjustable blades and countersinking holes for mounting.
Movable blades allow airflow direction in desired areas.
RDR has two rows of adjustable blades and **RDR1** has a single row of blades.
Grids over 500 mm in length are provided with intermediate reinforcement rods.
The air flow can be adjusted by mounting a air control damper **OBD** on the grille's inlet.

Accessories
Air control damper with opposable blades **OBD**.

Materials and finishing
Extruded aluminum electrostatic field painted white RAL 9016
Other RAL colors available on request .

Technical design
RDR



Grill Dimension	Ø tube
400 x 100	200
500 x 100	200
600 x 100	250
400 x 150	250
500 x 150	315
600 x 150	400
400 x 200	315
500 x 200	400
600 x 200	400
800 x 200	500
1000 x 200	630

Assembly Method
Standard mounting, with screws.

Selection table RDR/RDR1

Air flow m³/h	LxH	400x100	500x100	600x100	400x150	500x150	600x150	400x200	500x200	600x200	800x200	1000x200
	Ak[m²]	0.013	0.016	0.019	0.025	0.031	0.037	0.033	0.042	0.05	0.067	0.083
350	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]	9.3 - 7.48 16.0	8.0 - 6.08 10.0	7.1 - 5.12 7.0	5.9 - 3.89 5.0							
400	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]	10.7 - 8.55 21.0	9.2 - 6.94 13.0	8.1 - 5.85 9.0	6.7 - 4.44 6.0	5.8 - 3.58 4.0		5.6 - 3.37 3.0				
450	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]	12.0 21.0 9.62 27.0	10.3 - 7.81 17.0	9.1 - 6.58 12.0	7.6 - 5.00 8.0	6.5 - 4.03 5.0		6.3 - 3.79 4.0				
500	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]	13.3 25.0 10.7 33.0	11.5 19.0 8.7 21.0	10.2 - 7.3 14.0	8.4 - 5.6 10.0	7.3 - 4.5 6.0	6.4 - 3.8 4.0	6.9 - 4.2 5.0	6.0 - 3.3 3.0			
600	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]	16.0 30.0 12.8 47.0	13.8 24.0 10.4 30.0	12.2 19.0 8.8 21.0	10.1 - 6.7 14.0	8.7 - 5.4 9.0	7.7 - 4.5 6.0	8.3 - 5.1 8.0	7.2 - 4.0 5.0	6.4 - 3.3 3.0	5.2 - 2.5 2.0	
700	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]	18.7 35.0 15.0 64.0	16.1 29.0 12.2 41.0	14.2 24.0 10.2 28.0	11.8 19.0 7.8 19.0	10.2 - 6.3 12.0	9.0 - 5.3 8.0	9.7 - 5.9 10.0	8.4 - 4.6 7.0	7.4 - 3.9 5.0	6.1 - 2.9 3.0	5.3 - 2.3 2.0
800	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]	21.3 39.0 17.1 84.0	18.4 33.0 13.9 53.0	16.2 28.0 11.7 37.0	13.5 23.0 8.9 25.0	11.6 - 7.2 16.0	10.3 - 6.0 11.0	11.1 - 6.7 14.0	9.6 - 5.3 9.0	8.5 - 4.4 6.0	7.0 - 3.3 3.0	6.0 - 2.7 2.0
900	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]	24.0 42.0 19.2 106.0	20.7 36.0 15.6 67.0	18.3 32.0 13.2 46.0	15.2 21.0 10.0 32.0	13.1 - 8.1 20.0	11.6 - 6.8 14.0	12.5 - 7.6 17.0	10.8 - 6.0 11.0	9.5 - 5.0 8.0	7.9 - 3.7 4.0	6.8 - 3.0 3.0
1000	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]		22.9 40.0 17.4 83.0	20.3 35.0 14.6 57.0	16.8 30.0 11.1 40.0	14.5 24.0 9.0 25.0	12.8 19.0 7.5 17.0	13.9 22.0 8.4 21.0	12.0 - 6.6 13.0	10.6 - 5.6 9.0	8.7 - 4.1 5.0	7.5 - 3.3 3.0
1200	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]			24.4 40.0 17.5 83.0	20.2 35.0 13.3 57.0	17.4 29.0 10.8 36.0	15.4 25.0 9.0 25.0	16.7 27.0 10.1 31.0	14.4 21.0 7.9 19.0	12.7 - 6.7 13.0	10.5 - 5.0 7.0	9.0 - 4.0 5.0
1500	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]			30.5 47.0 21.9 129.0	25.3 42.0 16.7 89.0	21.8 36.0 13.4 56.0	19.3 31.0 11.3 39.0	20.8 34.0 12.6 48.0	17.9 28.0 9.9 30.0	15.9 23.0 8.3 21.0	13.1 - 6.2 12.0	11.3 - 5.0 7.0
1750	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]					25.4 41.0 15.7 77.0	22.5 36.0 13.1 53.0	24.3 39.0 14.7 65.0	20.9 33.0 11.6 41.0	18.5 28.0 9.7 28.0	15.3 20.0 7.3 16.0	13.2 - 5.9 10.0
2000	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]						25.7 40.0 15.0 69.0	27.8 43.0 16.8 85.0	23.9 37.0 13.2 53.0	21.2 32.0 11.1 37.0	17.5 24.0 8.3 20.0	15.0 - 6.7 13.0
2500	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]								29.9 44.0 16.5 84.0	26.5 39.0 13.9 58.0	21.8 31.0 10.4 32.0	18.8 25.0 8.4 20.0
3000	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]									31.8 44.0 16.7 83.0	26.2 37.0 12.4 46.0	22.6 31.0 10.0 29.0
3500	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]									37.1 49.0 19.4 113.0	30.6 41.0 14.5 63.0	26.0 35.0 11.7 40.0
4000	X [m] NR [dB(A)] Veff(m/s) ΔPt [pa]											30.0 40.0 13.4 52.0

Vk = 0.25 (m/s) Room height = 2.7 m

NR < 25 | 25≤NR≤35 | NR>35

Annotation

* Ak [m²] - Free surface
* ΔPt [pa] - Pressure loss
* X [m] - Air jet length
* NR [dB(A)] - Noise level without the room attenuation
* Vk [m/s] - Velocity at which the air jet length was calculated "x" [m]

Order codes

Type	+	OBD	/	Dimensions	/	L x H	/	0
RDR / RDR1		Air control damper		L x H 400 x 100 500 x 100 600 x 100 400 x 150 500 x 150 600 x 150 400 x 200 500 x 200 600 x 200 800 x 200 1000 x 200		Other dimensions on request		**0** Painted white RAL 9016 **R** Other RAL paints

Decorative Grille D-SS1/D-SS2

Description
Supply/exhaust aluminum grille with fixed blades in 1 or 2 directions **D-SS1/ D-SS2**.

Application
Supply/ exhaust of the air flow

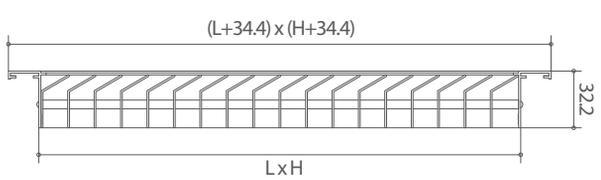
Characteristics
Fixed blades in 1 or 2 directions mounted parallel to length "L"
The grille is provided with mounting holes (standard).
The air flow can be adjusted by mounting an air control damper **OBD** on the grille inlet.

Accessories
Settlement frame made of galvanized sheet metal **CC**
Attachment spring **A**
Air control damper with opposable blades **OBD**
Air filter **FIL**
Insulated/Non-insulated adapter **ADP IZ/N**

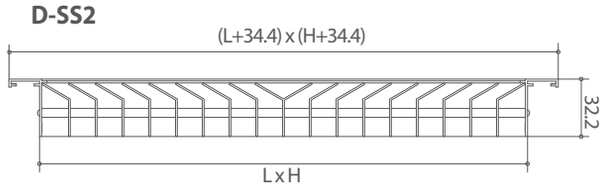
Materials and finishing
Extruded aluminum material, electrostatic field painted white RAL 9016
Other RAL colors available on request
Natural anodised aluminum

Technical design

D-SS1



D-SS2



Assembly Methods
Screws mounting (standard)
Settlement frame (optional)
Attachment spring (optional)

Selection table D-SS1 / D-SS2

L/H>2.1

Air flow (m^3/h)	Dimensions Ak [m^2]	500x200 0.05	600x200 0.06	800x200 0.08	1000x200 0.10	1200x200 800x300 0.12	1000x300 0.15	1200x300 1000x400 0.20	1200x400 0.25	1200x500 0.30
200	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]	<3 1.1 <20 <4	<3 0.9 <20 <4							
250	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]	3.0 1.4 <20 <4	<3 1.2 <20 <4	<3 0.9 <20 <4						
300	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]	3.5 1.7 <20 <4	3.3 1.4 <20 <4	3.0 1.0 <20 <4	<3 0.8 <20 <4					
350	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]	4.5 1.9 <20 <4	4.0 1.6 <20 <4	3.8 1.2 <20 <4	3.0 1.0 <20 <4	<3 0.8 <20 <4				
400	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]	5.0 2.2 <20 4.0	4.5 1.9 <20 <4	4.0 1.4 <20 <4	3.5 1.1 <20 <4	3.0 0.9 <20 <4				
450	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]		5.0 2.1 <20 <4	4.0 1.6 <20 <4	3.5 1.3 <20 <4	3.0 1.0 <20 <4				
500	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]		5.5 2.3 <20 4.5	5.0 1.7 <20 <4	4.5 1.4 <20 <4	4.0 1.2 <20 <4				
600	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]		6.5 2.8 20.0 6.0	6.0 2.1 <20 <4	5.3 1.7 <20 <4	5.0 1.4 <20 <4				
700	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]		9.0 3.2 25.0 8.5	8.0 2.4 <20 4.5	7.0 1.9 <20 <4	6.3 1.6 <20 <4				
800	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]			8.0 2.8 20.0 6.0	7.5 2.2 <20 4.0	7.0 1.9 <20 <4				
900	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]			9.5 3.1 23.0 8.0	8.5 2.5 <20 5.0	8.0 2.1 <20 <4				
1000	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]				9.0 2.8 20.0 6.0	8.0 2.3 <20 4.5				
1200	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]				12.5 3.3 28.0 9.0	11.0 2.8 20.0 6.0				
1400	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]				14.0 3.9 28.0 12.0	13.0 3.2 25.0 8.5	12.0 2.6 <20 5.5			
1600	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]					15.0 3.7 25.0 10.0	13.5 3.0 22.0 7.0	12.0 2.2 <20 4.0	11.0 1.8 <20 <4	
1800	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]					>15 4.2 30.0 14.0	15.0 3.3 28.0 9.0	13.0 2.5 <20 5.0	11.5 2.0 <20 <4	
2000	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]					>15 4.6 32.0 16.0	>15 3.7 25.0 10.0	14.0 2.8 20.0 6.0	12.5 2.2 <20 4.0	
2500	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]					>15 5.8 37.0 25.0	>15 4.6 32.0 16.0	>15 3.5 25.0 9.5	>15 2.8 20.0 6.0	15.0 2.3 <20 4.5
3000	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]					>15 6.9 43.0 38.0	>15 5.6 33.0 20.0	>15 4.2 30.0 14.0	>15 3.3 28.0 9.0	>15 2.8 20.0 6.0
3500	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]					>15 8.1 47.0 50.0	>15 6.5 40.0 30.0	>15 4.9 32.0 18.0	>15 3.9 28.0 12.0	>15 3.2 25.0 8.5
3600	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]							>15 5.0 37.0 24.0	>15 4.0 28.0 13.0	>15 3.3 28.0 9.0

Selection table D-SS1 , D-SS2

L/H<2.1

Air flow (m^3/h)	Dimensions	400x200	500x300	600x300	600x400	800x400
	Ak [m^2]	0.04	0.07	0.08	0.12	0.15
200	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]	<1 1.4 <20 <4	<1 0.8 <20 <4			
250	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]	<1 1.7 <20 <4	<1 1.0 <20 <4	<1 0.9 <20 <4		
300	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]	1.0 2.1 <20 <4	<1 1.2 <20 <4	<1 1.0 <20 <4	<1 0.7 <20 <4	
350	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]	1.2 2.4 <20 4.5	<1 1.4 <20 <4	<1 1.2 <20 <4	<1 0.8 <20 <4	<1 0.6 <20 <4
400	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]	1.4 2.8 20.0 6.0	1.0 1.6 <20 <4	<1 1.4 <20 <4	<1 0.9 <20 <4	<1 0.7 <20 <4
450	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]	1.6 3.1 23.0 8.0	1.2 1.8 <20 <4	1.0 1.6 <20 <4	<1 1.0 <20 <4	<1 0.8 <20 <4
500	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]	1.8 3.5 25.0 9.5	1.4 2.0 <20 <4	1.2 1.7 <20 <4	<1 1.2 <20 <4	<1 0.9 <20 <4
600	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]		1.8 2.4 <20 4.5	1.5 2.1 <20 <4	1.2 1.4 <20 <4	<1 1.1 <20 <4
700	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]		1.9 2.8 20.0 6.0	1.8 2.4 <20 4.5	1.5 1.6 <20 <4	1.2 1.3 <20 <4
800	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]		2.1 3.2 25.0 8.5	2.0 2.8 20.0 6.0	1.8 1.9 <20 <4	1.4 1.5 <20 <4
900	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]			2.2 3.1 23.0 8.0	1.9 2.1 <20 <4	1.5 1.7 <20 <4
1000	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]			2.5 3.5 25.0 9.5	1.9 2.3 <20 4.5	1.8 1.9 <20 <4
1200	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]			2.9 4.2 30.0 14.0	2.4 2.8 20.0 6.0	2.1 2.2 <20 4.0
1400	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]				2.8 3.2 25.0 8.5	2.4 2.6 <20 5.5
1600	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]				3.2 3.7 25.0 10.0	2.8 3.0 22.0 7.0
1800	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]				3.6 4.2 30.0 14.0	3.1 3.3 28.0 9.0
2000	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]				4.0 4.6 32.0 16.0	3.5 3.7 25.0 10.0
2500	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]					4.3 4.6 32.0 16.0
3000	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]					5.0 5.6 33.0 20.0
3500	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]					6.5 6.5 40.0 30.0
3600	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]					7.0 6.7 42.0 33.0

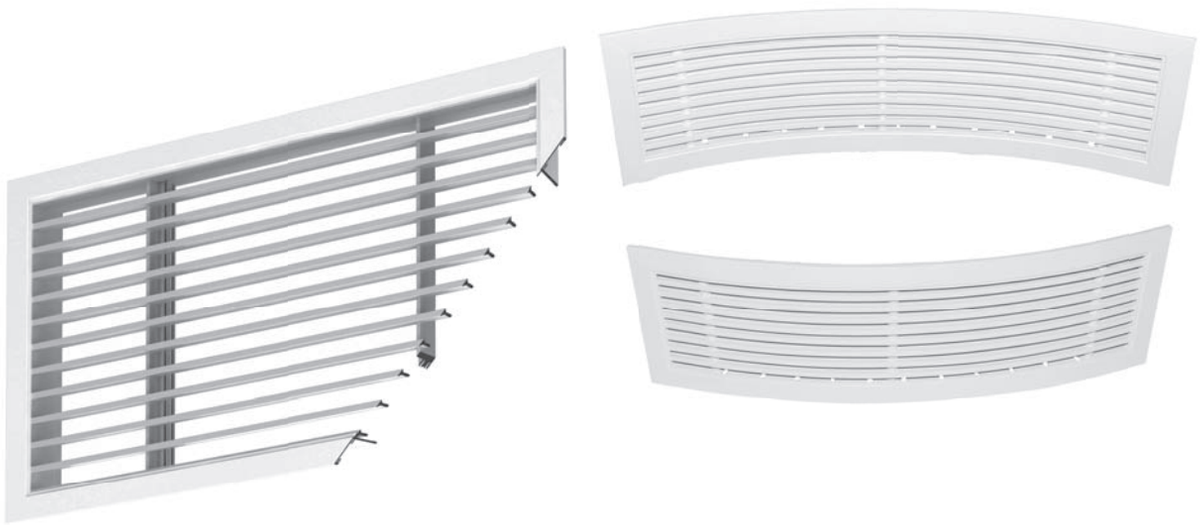
Annotation

* Ak [m²] - Free surface of the grille
* Veff[m/s] - Air velocity in grille
* ΔPt [pa] - Pressure loss
* X [m] - Air jet length
* NR [dB(A)] - Noise level without the room attenuation
* Vk = 0.5 m/s - Speed at which airflow length was calculated "x" [m]

Order codes

Type	+	CC	+	A (Attachment spring)	+	FIL	+	OBD	+	ADP IZ/N	/	Dimensions	/	O
D-SS1 / D-SS2		Settlement frame		No. of springs depending on grille dimensions		Air filter		Air control damper		Insulated/ Non-insulated adapter		On request		**O** Painted white RAL 9016 **R** Other RAL paints Natural anodised

Linear Decorative Grille LBD



Description
Aluminum grille for supply/exhaust of the air flow, with 0 or 15 degree fixed blades.

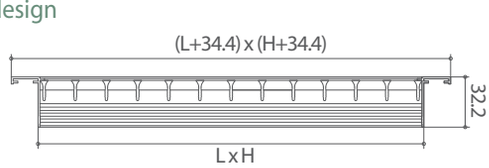
Application
Supply/exhaust of the air flow.

Characteristics
0 or 15 degree fixed blades, parallel with length "L".
The grille is provided with mounting holes (standard)
The air flow can be adjusted by mounting an air control damper OBD on the inlet connection.
On demand, the grille can be curved (concave/convex)

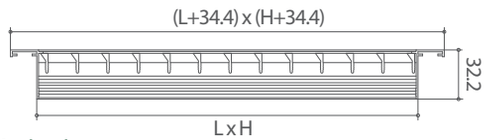
Accessories
Settlement frame made of galvanized sheet metal **CC**
Mounting spring **A.**
Air control damper with opposable blades **OBD**
Air filter **FIL**
Insulated/Non-insulated adapter **ADP IZ/N**

Materials and finishing
Extruded aluminum electrostatic field painted white RAL 9016.
Other RAL colors available on request.
Natural anodised aluminum.

Technical design
LBD - 0°

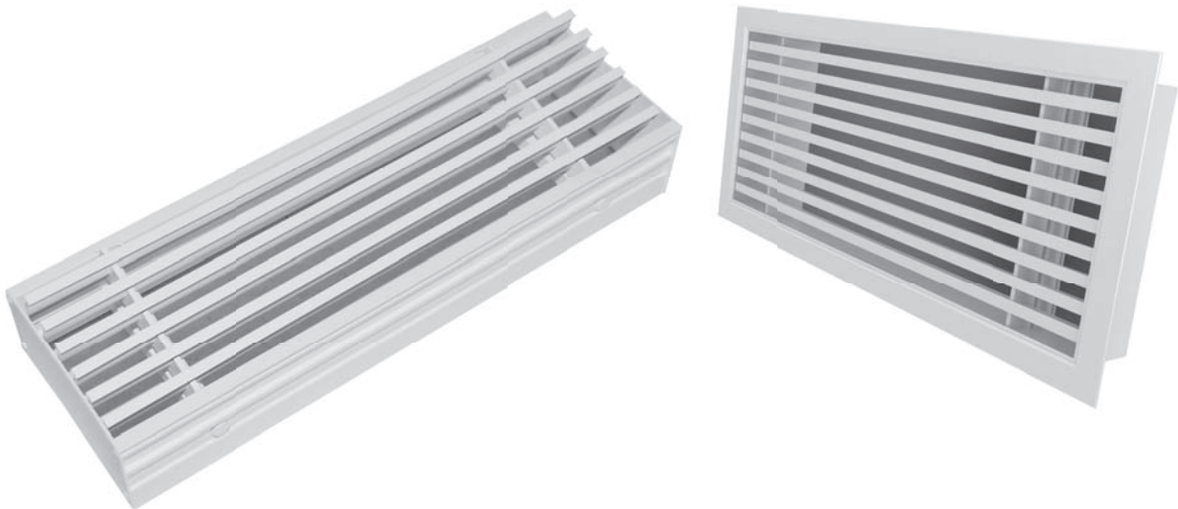


LBD - 15°



Assembly Methods
Screws (standard).
Settlement frame (optional)
Mounting spring (optional).

Linear Decorative Grille LBDB/LBDD



Description
Aluminum grille for supply/exhaust of the air flow, with 0 or 15 degree fixed blades.

Application
Supply/exhaust of the air flow.

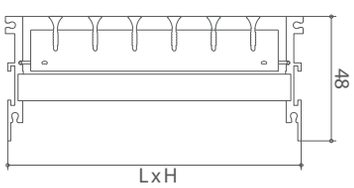
Characteristics
LBDB
- 0 or 15 degree fixed blades, parallel with length "L".
- Removable blades system
- Encased mounting into wall/ ceiling

LBDD
- Two rows of blades
- Frontal fixed blades at 0°/15°, parallel with lenght "L"
- Individually adjustable blades that allow air flow direction
- Provided with mounting holes (standard).

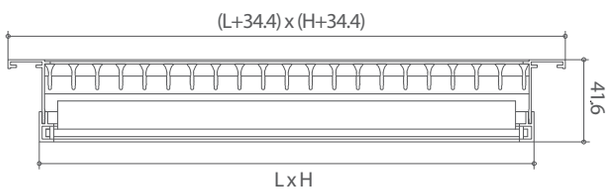
Accessories
Air control damper with opposable blades **OBD (LBDB)**
Mounting spring **A**
Insulated/Non-insulated adapter **ADP IZ/N**

Materials and finishing
Extruded aluminum electrostatic field painted white RAL 9016.
Other RAL colors available on request.

Technical design
LBDB



LBDD



Assembly Methods
LBDB - With screws on the inlet connection of the grille
LBDD - With screws on the frame (standard).
Mounting spring

Selection table LBD / LBDB / LBDD

Air flow m³/h	L x H	250x100	300x100	400x100	500x100	600x100	800x100	1000x100	1500x100	400x150	600x150	1000x150	600x200	800x200	1000x200
	Ak(m²)	0.011	0.013	0.019	0.024	0.029	0.041	0.05	0.075	0.033	0.045	0.081	0.064	0.085	0.12
140	X(m) NR (dB) ΔPt [pa] Veff (m/s)	3.0 <20 6.0 3.2													
200	X(m) NR (dB) ΔPt [pa] Veff (m/s)	4.2 25.0 10.0 4.6	3.8 21.0 10.0 4.0	3.2 <20 5.0 2.8											
300	X(m) NR (dB) ΔPt [pa] Veff (m/s)	6.3 35.0 30.0 6.9	5.8 30.0 18.0 6.0	5.0 24.0 10.0 4.2	4.3 18.0 7.0 3.3										
400	X(m) NR (dB) ΔPt [pa] Veff (m/s)	8.5 43.0 50.0 9.3	7.8 38.0 35.0 7.9	6.5 32.0 18.0 5.6	5.8 25.0 10.0 4.4	5.3 22.0 9.0 3.8	4.6 <20 5 2.6			5.1 25 8 3.4					
600	X(m) NR (dB) ΔPt [pa] Veff (m/s)	12.0 53.0 118.0 13.9	11.5 48.0 80.0 11.9	10.0 42.0 40.0 8.3	8.7 35.0 30.0 6.7	6.0 33.0 20.0 5.7	7 26 12 4.0	6 25 7 3.1		8 35 16 5.1	6.3 25 8 3.7				
800	X(m) NR (dB) ΔPt [pa] Veff (m/s)			15.0 51.0 75.0 11.1	12.0 45.0 48.0 8.9	10.5 45.0 35.0 7.7	9.3 33 20 5.3	8.5 32 15 4.2	6.5 20 6 3.2	11 42 32 6.7	8.5 30 15 4.9	6.8 21 5 2.7	7 25 8 3.5		
1000	X(m) NR (dB) ΔPt [pa] Veff (m/s)					13.0 45.0 55.0 9.6	11 42 30 6.6	10.2 37 20 5.2	8.3 28 9 4.0	13 46 48 8.4	11 38 22 6.2	8 27 8 3.3	9 30 12 4.3	7.8 25 8 3.3	7 <20 3 2.3
1400	X(m) NR (dB) ΔPt [pa] Veff (m/s)						16.5 50 55 9.3	14 45 37 7.3	12 37 18 5.6		15 48 40 8.6	12 35 15 4.7	12 40 22 6.1	11 33 12 4.6	11 33 12 3.2
2000	X(m) NR (dB) ΔPt [pa] Veff (m/s)											16 45 30 6.7	18 50 42 8.7	15 43 25 6.5	14 35 17 4.6
2500	X(m) NR (dB) ΔPt [pa] Veff (m/s)													20 48 37 8.2	18 45 25 5.8
3000	X(m) NR (dB) ΔPt [pa] Veff (m/s)														21 48 35 6.9

NR < 35 | 35≤NR≤45 | NR>45

Annotation

* Ak [m²] - Free surface
* X [m] - Air jet length
* Vk = 0.2 [m/s] - Velocity at which the air jet length was calculated "x" [m]
* NR [dB(A)] - Noise level without the room attenuation
* ΔPt [pa] - Pressure loss
* Veff[m/s] - Air velocity inside the grille

Order codes

Type - Blade's angle + **CC** + **A** (Mounting spring) + **FIL** + **OBD** + **ADP IZ/N** / **Dimensions** / **O**

- **LBD/LBDB / LBDD -0°/15°**
- Settlement frame (LBD)
- No. of springs depending on grille dimensions
- Air filter (LBD)
- Air control damper (LBD/LBDB)
- Insulated/ Non-insulated adapter
- On request
- **0** Painted white RAL 9016
  R Other RAL paints
  Natural anodised (LBD)

Circular Ceiling Diffuser RCD/RCD-R/RCDH/RCD-RH

RCD-R

RCDH

Description
Ceiling diffusers with circular or square front panel and adjustable circular opening.

Application
Supply/ exhaust of the air flow.

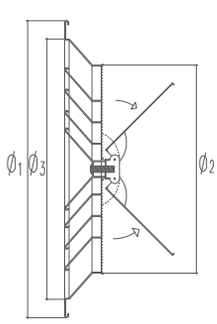
Characteristics
Recommended difference between air temperature and indoor air is ± 10 K .
Conic shape with removable central core.
Depending on the type of roof and front panel, the following alternatives are possible:
RCD - for low ceiling with circular front panel
RCD-R - for low ceiling with square front panel 595 X 595 mm.
RCDH - for high ceiling with circular front panel
RCD-RH - for high ceiling with square front panel 595 X 595 mm.

Accessories
Insulated/Non-insulated adapter **ADP IZ/N**

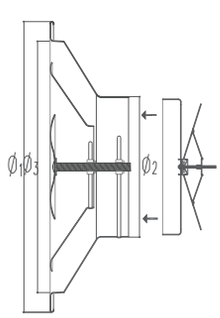
Materials and finishing
Aluminum material electrostatic field painted white RAL 9016.
Other RAL colors available on request.

Technical design

RCD



RCDH



Available dimensions

RCDH / RCD / RCD-RH / RCD-R

Diffuser type	Ø2	Ø1	Ø3
RCDH	100	250	170
	160	335	280
	200	423	360
	250	517	445
	315	640	555
	355	730	640
	400	776	680
	450	825	735
	500	915	820
	630	1045	943
RCD	150	258	208
	200	308	255
	250	358	305
	300	408	355
	350	458	405
RCD-RH	160	□595X595	280
	200		360
	250		445
	315		555
RCD-R	200	□595X595	395
	250		395
	300		395

Selection table RCD/RCD-R/RCDH/RCD-RH

RCDH/RCD-RH - high ceiling

Air flow m³/h	Connection size Ø(mm)	160	200	250	315	355	400	450	500	630
	Ak [m²]	0.015	0.027	0.04	0.056	0.074	0.09	0.011	0.013	0.019
200	X [m] NR [dB(A)] Veff[m/s] ΔPt [pa]	1.4 25.0 3.7 14.0								
250	X [m] NR [dB(A)] Veffm/s] ΔPt [pa]	1.7 30.0 4.4 20.0	1.3 18.0 2.5 6.6							
300	X [m] NR [dB(A)] Veff[m/s] ΔPt [pa]	2.1 36.0 5.6 33.0	1.5 23.0 3.0 10.0	1.3 15.0 2.0 4.2						
400	X [m] NR [dB(A)] Veff[m/s] ΔPt [pa]	2.8 45 7.2 54	2.1 30.0 4.0 17.0	1.7 22.0 2.7 7.7	1.5 3.0 2.0 4.2					
500	X [m] NR [dB(A)] Veff[m/s] ΔPt [pa]		2.6 36.0 5.0 26.0	2.1 27.0 3.4 12.0	1.8 20.0 2.4 6.0					
600	X [m] NR [dB(A)] Veff[m/s] ΔPt [pa]		3.2 42.0 6.0 38.0	2.6 33.0 4.0 17.0	2.2 25.0 3.0 10.0	1.9 20.0 2.3 5.0				
800	X [m] NR [dB(A)] Veff[m/s] ΔPt [pa]			3.5 41.0 5.6 33.0	3.0 34.0 4.0 17.0	2.6 25.0 3.0 9.0	2.3 20.0 2.5 6.5			
1000	X [m] NR [dB(A)] Veff[m/s] ΔPt [pa]			4.5 50.0 7.0 52.0	3.7 40.0 5.0 26.0	3.3 34.0 3.8 16.0	3.0 28.0 3.0 9.0	2.7 22.0 2.5 6.5		
1250	X [m] NR [dB(A)] Veff[m/s] ΔPt [pa]				4.6 47.0 6.2 40.0	4.1 42.0 4.8 23.0	3.8 35.0 3.8 16.0	3.3 26.0 3.0 9.0	3.0 24.0 2.6 6.9	
1500	X [m] NR [dB(A)] Veff[m/s] ΔPt [pa]					5.0 47.0 5.5 32.0	4.5 41.0 4.6 22.0	4.0 35.0 3.8 16.0	3.6 29.0 3.1 10.0	
1750	X [m] NR [dB(A)] Veff[m/s] ΔPt [pa]							4.8 40.0 4.4 20.0	4.3 34.0 3.6 14.0	3.5 25.0 2.6 6.9
2000	X [m] NR [dB(A)] Veff[m/s] ΔPt [pa]								4.9 38.0 4.1 19.0	4.1 30.0 2.9 8.0
2500	X [m] NR [dB(A)] Veff[m/s] ΔPt [pa]								6.0 45.0 5.0 29.0	5.1 37.0 3.6 14.0

RCD/RCD-R - low ceiling

Air flow m³/h	Connection size Ø(mm)	160	200	250	315	355
	Ak [m²]	0.011	0.02	0.031	0.046	0.047
150	X [m] NR [dB(A)] Veff[m/s] ΔPt [pa]	1.8 - 3.9 1.0	1.3 - 2.1 6.0			
200	X [m] NR [dB(A)] Veffm/s] ΔPt [pa]	2.3 27.0 5.0 22.0	1.8 - 2.8 7.8	1.4 - 1.9 -		
300	X [m] NR [dB(A)] Veff[m/s] ΔPt [pa]	3.4 41.0 7.5 50.0	2.6 23.0 4.1 15.0	2.1 - 2.7 7.2	1.7 - 1.9 -	
400	X [m] NR [dB(A)] Veff[m/s] ΔPt [pa]		3.4 33.0 5.7 28.0	2.7 20.0 3.6 12.0	2.3 - 2.5 5.5	1.9 7.0 2.4 3.2
500	X [m] NR [dB(A)] Veff[m/s] ΔPt [pa]		4.3 41.0 7.0 45.0	3.5 27.0 4.5 18.0	2.8 15.0 3.1 8.0	2.0 10.0 2.8 4.8
600	X [m] NR [dB(A)] Veff[m/s] ΔPt [pa]			4.1 34.0 5.5 26.0	3.3 22.0 3.7 14.0	2.8 20.0 3.5 7.0
800	X [m] NR [dB(A)] Veff[m/s] ΔPt [pa]			5.5 45.0 7.5 50.0	4.6 32.0 5.0 22.0	3.5 30.0 4.8 12.0
1000	X [m] NR [dB(A)] Veff[m/s] ΔPt [pa]				5.7 40.0 6.2 35.0	4.5 37.0 6.0 20.0
1200	X [m] NR [dB(A)] Veff[m/s] ΔPt [pa]				6.7 46.0 7.5 50.0	5.4 40.0 7.0 28.0
1400	X [m] NR [dB(A)] Veff[m/s] ΔPt [pa]					7.1 50.0 10.2 39.2
1600	X [m] NR [dB(A)] Veff[m/s] ΔPt [pa]					7.8 57.0 11.6 50.3

Correction coefficients for other velocities inside sitting area.

Vk[m/s]	0.2	0.25	0.375	0.5
x[m]	1.25	1	0.75	0.5

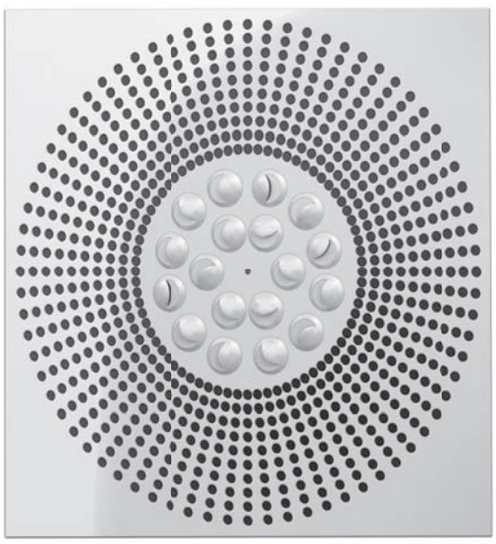
Annotation

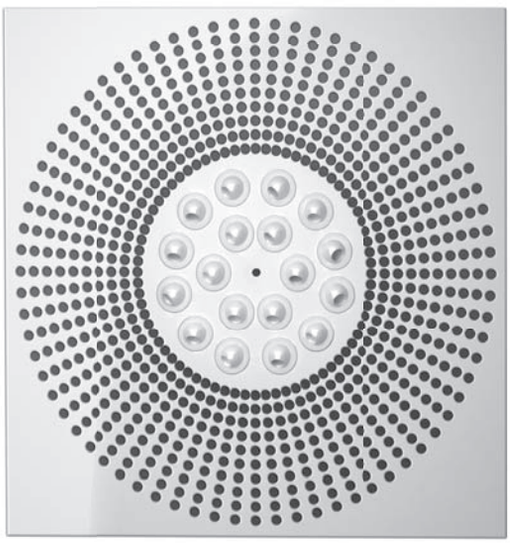
* Ak [m²] - Free surface
* Veff[m/s] - Air velocity inside the diffuser
* ΔPt [pa] - Pressure loss
* X [m] - Air jet length
* NR [dB(A)] - Noise level without the room attenuation
* Vk = 0.25 [m/s] - velocity at which the air jet length was calculated "x" [m]

Order codes

Type	+	ADP IZ/N	/	Dimensions	/	0
				Ø		
RCDH/RCD **RCD-RH/RCD-R**		Insulated/ Non-insulated adapter		high ceiling: 160, 200, 250, 315, 355, 400, 450, 500, 630 low ceiling: 160, 200, 250, 315, 355		**0** Painted white RAL 9016 **R** Other RAL paints

Ceiling Diffuser PGP/PGR





Description
Ceiling diffuser with maneuverable disks/ nozzles and perforated metal sheet, designed for use in crowded areas.

Application
Air supply.

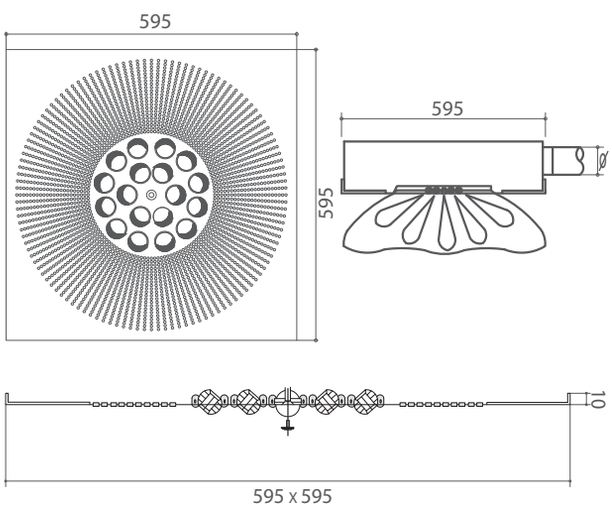
Characteristics
The speaker is equipped with Ø5 mm circular perforations radially arranged and 18 individually maneuverable disks/ nozzles , located in the center of the front.
Specially designed for paneled ceiling size 600 x 600 mm.
The diffuser is supplied with gasket seals, fastening screw and plastic cover for concealment.

Accessories
Insulated/Non-insulated adapter **ADP IZ/N**

Materials and finishing
Pickled steel sheet electrostatic field painted white RAL 9016.
Other RAL colors available on request.
Polypropylene disks/ nozzles.

Technical design

PGP **PGR**



Assembly Methods
Paneled ceiling 600 x 600 mm mounting, fitted with a central screw M6 x 100 mm on mounting traverse of the adapter.

Selection table PGP/PGR

PGP

Q m³/h	595 X 595 Ak [m²]	0.03536
175	Veff (m/s) NR[dB(A)] Pt(Pa) X(m)	1.37 <20 6 0.9
200	Veff (m/s) NR[dB(A)] Pt(Pa) X(m)	1.57 <20 8 1
250	Veff (m/s) NR[dB(A)] Pt(Pa) X(m)	1.96 24 13 1.3
300	Veff (m/s) NR[dB(A)] Pt(Pa) X(m)	2.36 29 19 1.5
350	Veff (m/s) NR[dB(A)] Pt(Pa) X(m)	2.75 34 25 1.8
400	Veff (m/s) NR[dB(A)] Pt(Pa) X(m)	3.14 38 33 2
450	Veff (m/s) NR[dB(A)] Pt(Pa) X(m)	3.54 41 42 2.3
500	Veff (m/s) NR[dB(A)] Pt(Pa) X(m)	3.93 44 52 2.5

ΔT=-10K
H(room height)=2,84 m
Vk=0,2 m/s

PGR

Q m³/h	595 X 595 Ak [m²]	0.0485
400	Veff (m/s) NR[dB(A)] Pt(Pa) X(m)	2.29 31 15 2
600	Veff (m/s) NR[dB(A)] Pt(Pa) X(m)	3.44 44 33 5
800	Veff (m/s) NR[dB(A)] Pt(Pa) X(m)	4.58 53 58 7

ΔT=-10K
Vk=0,25 m/s

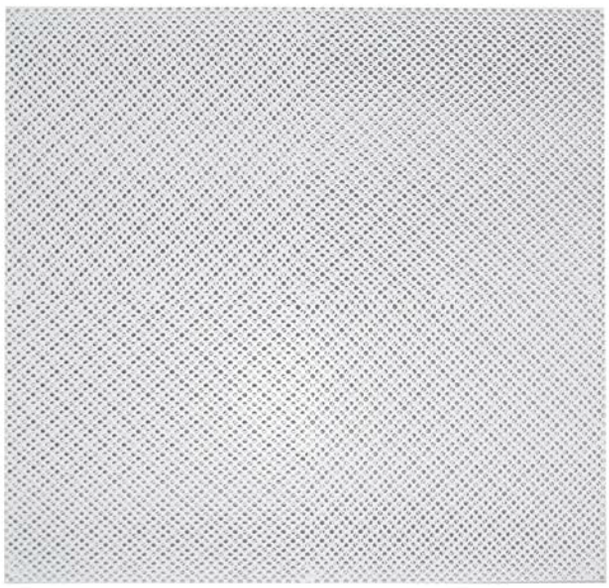
Annotation

* Ak [m²] - Free surface
* Veff[m/s] - Air velocity inside diffuser
* NR [dB(A)] - Noise level without the room attenuation
* ΔPt [pa] - Pressure loss
* X [m] - Air jet length
* Vk = velocity at which the air jet length was calculated "x" [m]

Order codes

Type	+	ADP IZ/N	/	Dimensions	/	Nozzles/ disks color	/	0
PGR / PGP		Insulated/Non-insulated adapter		L X H 595 X 595		PGR - white RAL 9016 PGP - white RAL9016 / black RAL9005		**0** Vopsit alb RAL 9016 **R** Vopsit RAL…

Perforated Ceiling Diffuser PFD/PF



PFD

Description
Rectangular ceiling diffuser with circular perforations for exhaust/ supply of the air flow **PFD/PF.**

Application
For air supply - **PFD**, for air exhaust **PF.**

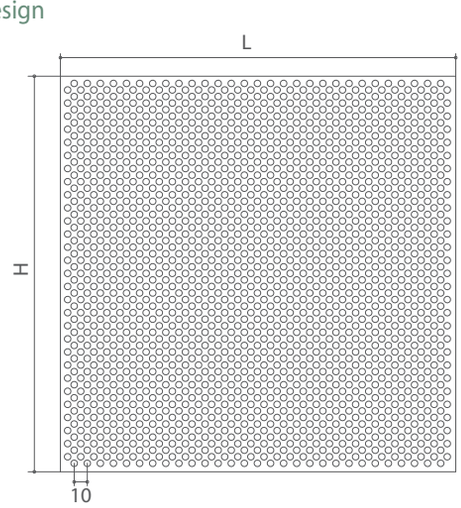
Characteristics
PFD/PF is projected for paneled ceiling mounting (600 x 600 mm).
Circular perforations of 5 mm diameter.
The PFD diffuser is provided with an air jet orientation system in 1, 2, 3 or 4 directions, without air flow deviations or pressure loss.
The diffuser connects to the ventilation duct through the adapter.

Accessories
Insulated/Non-insulated adapter **ADP IZ/N**.

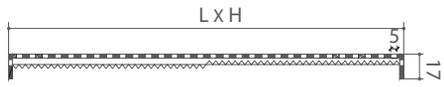
Materials and finishing
Galvanized steel sheet painted in electrostatic field (white RAL 9016).
Other RAL colors available on request.

Technical design

PFD/PF



PFD



PF

Assembly Method
Paneled ceiling mounting 600 x 600 mm.

Selection table PFD/ PF

PFD - supply

Air flow (m^3/h)	LxH	595x595
	Ak [m^2]	0.150
700	Veff(m/s)	1.3
	X [m]	1.8
	NR [dB(A)]	27.0
	ΔPt [pa]	7.0
1000	Veff(m/s)	1.9
	X [m]	2.6
	NR [dB(A)]	39.0
	ΔPt [pa]	15.0
1600	Veff(m/s)	3.0
	X [m]	3.1
	NR [dB(A)]	54.0
	ΔPt [pa]	36.0

PF - exhaust

Air flow (m^3/h)	LxH	595x595
	Ak [m^2]	0.150
700	Veff(m/s)	
	NR [dB(A)]	
	ΔPt [pa]	
1000	Veff(m/s)	1.9
	NR [dB(A)]	20.0
	ΔPt [pa]	33.0
1600	Veff(m/s)	3.0
	NR [dB(A)]	36.0
	ΔPt [pa]	85.0

Note: The values for the supply mode from selection table are calculated for a 4 direction system diffuser and for an air velocity of Vk = 0.2 m/s.

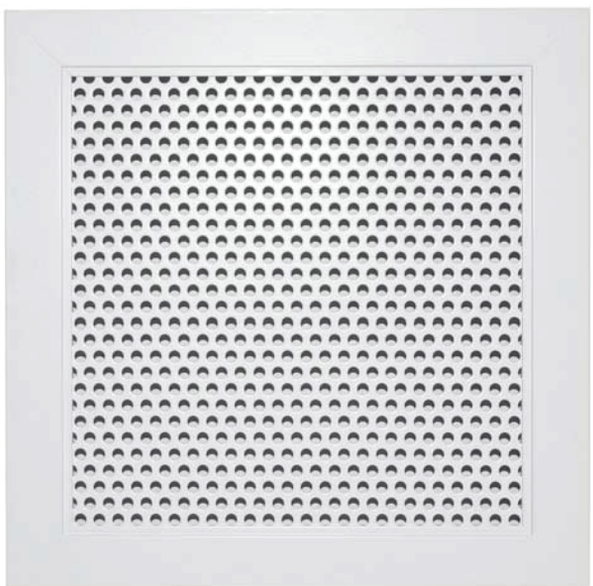
Annotation

* Ak [m²] - Free surface
* Veff[m/s] - Air velocity inside the diffuser
* X [m] - Air jet length
* NR [dB(A)] - Noise level without the room attenuation
* ΔPt [pa] - Pressure loss

Order codes

Type	+	**ADP IZ/N**	/	**Dimensions**	/	**L x H**	/	**0**
PFD/PF		Insulated/ Non-insulated adapter		L X H 595 x 595		Other dimensions on request		**0** Painted white RAL 9016 **R** Other RAL paints

Perforated Diffuser PFF/PFDD



PFF

Description
Rectangular ceiling diffuser for exhaust/ supply of the air flow provided with circular perforations and mounting frame PFF/PFDD. PFDD is provided with access door to ease access to the back of the diffuser.

Application
Exhaust/ supply of the air flow.

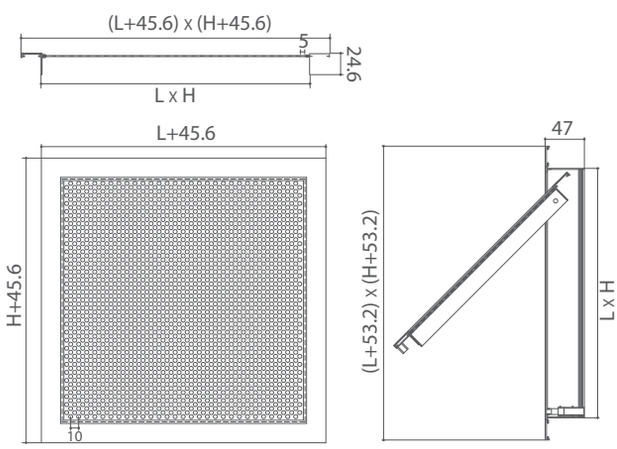
Characteristics
The diffusers are executed with mounting holes on the frame, except those with dimensions 595 x 595 mm for paneled ceiling.
PFDD is provided with access door, in this case the mounting holes will not be executed on the frame, but on the inlet connection of the diffuser.
The access door is opened via a click type system (open/ close by pressing).
Circular perforations of 5 mm diameter.
The air flow can be adjusted by mounting an air control damper **OBD** on the diffuser inlet.
An air filter FIL can be optionally mounted on the inlet connection of the diffuser .
The diffuser connects to the ventilation duct through the adapter.

Accessories
Air filter **FIL**
Air control damper with opposable blades **OBD**
Insulated/Non-insulated adapter **ADP IZ/N**

Materials and finishing
Galvanized steel sheet painted in electrostatic field (white RAL 9016).
Other RAL colors available on request.

Technical design

PFF **PFDD**



Assembly Method
With screws (standard).
PFDD - screws on the inlet connection (optional).

Selection table PFF/PFDD

Supply

Air flow m³/h	LxH	300x300	400x400	500x500	595x595
	Ak [m²]	0.030	0.060	0.100	0.150
160	Veff(m/s) X [m] NR [dB(A)] ΔPt [pa]	1.5 0.9 22.0 9.0			
200	Veff(m/s) X [m] NR [dB(A)] ΔPt [pa]	1.8 1.1 31.0 14.0	0.9 0.8 < 20 4.0		
300	Veff(m/s) X [m] NR [dB(A)] ΔPt [pa]	2.8 1.7 44.0 31.0	1.4 1.2 25.0 8.0		
400	Veff(m/s) X [m] NR [dB(A)] ΔPt [pa]	3.7 2.1 53.0 53.0	1.9 1.6 35.0 15.0	1.1 1.3 20.0 5.0	
500	Veff(m/s) X [m] NR [dB(A)] ΔPt [pa]	4.6 2.5 61.0 81.0	2.3 2.0 42.0 23.0	1.4 1.6 27.0 8.0	
700	Veff(m/s) X [m] NR [dB(A)] ΔPt [pa]		3.2 2.2 52.0 45.0	1.9 2.2 39.0 16.0	1.3 1.8 27.0 7.0
1000	Veff(m/s) X [m] NR [dB(A)] ΔPt [pa]		4.6 3.1 63.0 90.0	2.8 2.4 51.0 30.0	1.9 2.6 39.0 15.0
1600	Veff(m/s) X [m] NR [dB(A)] ΔPt [pa]				3.0 3.1 54.0 36.0

Exhaust

Air flow m³/h	LxH	300x300	400x400	500x500	595x595
	Ak [m²]	0.030	0.060	0.100	0.150
160	Veff(m/s) NR [dB(A)] ΔPt [pa]	1.5 <20 20.0			
200	Veff(m/s) NR [dB(A)] ΔPt [pa]	1.8 <20 31.0			
300	Veff(m/s) NR [dB(A)] ΔPt [pa]	2.8 26.0 76.0			
400	Veff(m/s) NR [dB(A)] ΔPt [pa]	3.7 35.0 125.0	1.9 <20 33.0		
500	Veff(m/s) NR [dB(A)] ΔPt [pa]	4.6 42.0 185.0	2.3 23.0 47.0		
700	Veff(m/s) NR [dB(A)] ΔPt [pa]		3.2 34.0 95.0	1.9 20.0 37.0	
1000	Veff(m/s) NR [dB(A)] ΔPt [pa]			2.8 31.0 73.0	1.9 20.0 33.0
1600	Veff(m/s) NR [dB(A)] ΔPt [pa]				3.0 36.0 85.0

NR < 35 | 35≤NR≤45 | NR>45

Annotation

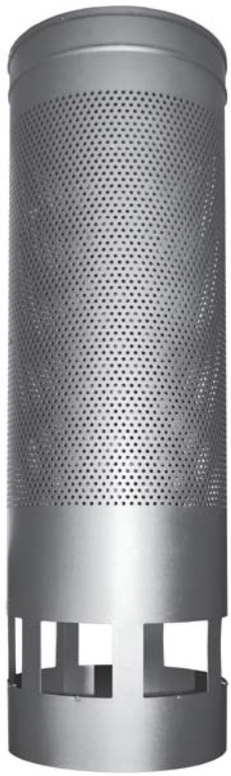
* Ak [m²] - Free surface
* Veff[m/s] - Air velocity inside the diffuser
* X [m] - Air jet length
* Vk = 0.2 m/s - Velocity at which the air jet length was calculated "x" [m]
* NR [dB(A)] - Noise level without the room attenuation
* ΔPt [pa] - Pressure loss

Order codes

Type + **FIL** + **OBD** + **ADP IZ/N** / **Dimensions** / **L x H** / **0**

- **Type**: **PFF / PFDD**
- **FIL**: Air filter
- **OBD**: Air control damper
- **ADP IZ/N**: Insulated/Non-insulated adapter
- **Dimensions**: L X H — 300 x 300, 400 x 400, 500 x 500, 595 x 595
- **L x H**: Other dimensions on request
- **0**: **0** Painted white RAL 9016, **R** Other RAL paints

Perforated Cylinder Diffuser PC



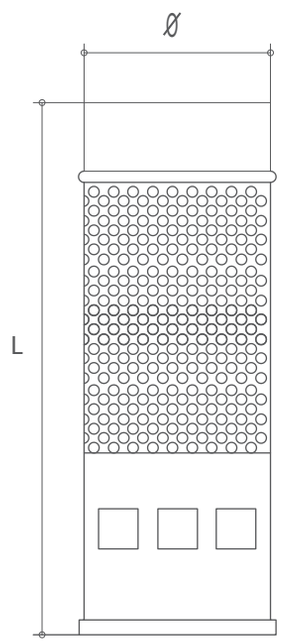
Description
Perforated cylinder for use in industrial areas.

Application
Air supply

Characteristics
Available diameters: D250, D355, D450 and D560 mm.
Length of the cylinder: 825 mm.
Height for mounting is over 3.5 m.
Suitable for use for both heating and cooling.
Horizontal and vertical dispersion of air.
Can function as an air damper, which makes the air supply possible on both horizontal and vertical levels, depending on requirements (heating/ cooling).
The diffuser is provided with a internal adjustable disk manipulated by a chain drive.

Materials and finishing
Perforated cylinder made of galvanized steel sheet
RAL colors available on request.

Technical design



Selection table PC

Air flow (m^3/h)	Ø	250	355	450	560
700	NR [dB(A)]	31.0			
	X(m)	2.2			
	ΔPt [pa]	12.0			
1000	NR [dB(A)]	42.0	26.0		
	X(m)	3.5	2.6		
	ΔPt [pa]	23.0	7.0		
1500	NR [dB(A)]	55.0	38.0	21.0	
	X(m)	5.2	3.5	2.8	
	ΔPt [pa]	50.0	18.0	7.0	
2000	NR [dB(A)]		47.0	31.0	<25
	X(m)		4.6	3.3	2.5
	ΔPt [pa]		31.0	13.0	6.0
3000	NR [dB(A)]		>55	43.0	30.0
	X(m)		7.0	5.2	4.2
	ΔPt [pa]		70.0	28.0	13.0
4000	NR [dB(A)]			53.0	40.0
	X(m)			7.0	5.2
	ΔPt [pa]			50.0	27.0
6000	NR [dB(A)]				53.0
	X(m)				8.0
	ΔPt [pa]				60.0

Annotation

* NR [dB(A)] - Noise level
* ΔPt [pa] - Pressure loss
* X [m] - Air jet length

Note: The values in the selection table are calculated for an air jet discharge set in heating mode and for temperature deviation of ΔT= + 10 K.

Order codes

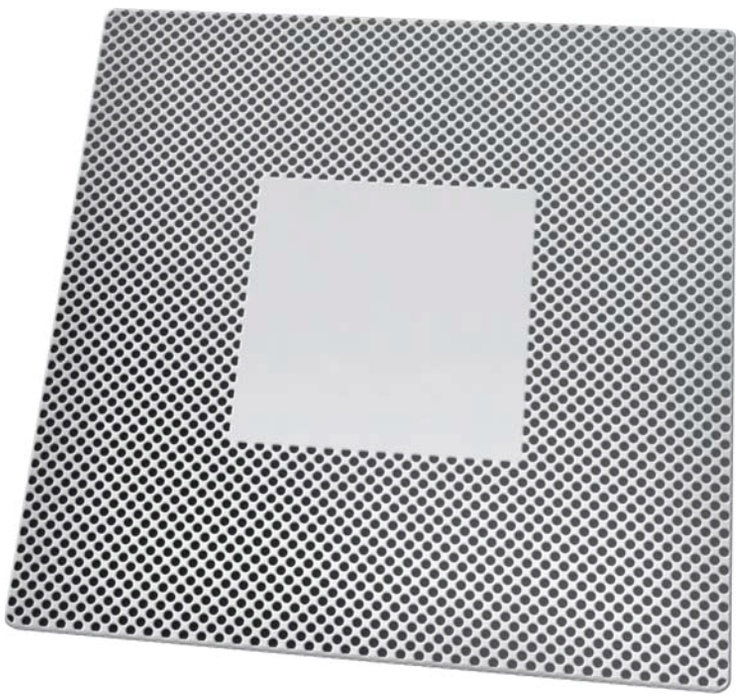
Type / **Ø** / **0**

PC

250
355
450
560

Natural steel material
0 Painted white RAL 9016
R Other RAL paints

Architectural Diffuser PA



Description
Architectural diffuser with circular perforations, for paneled ceiling mounting.

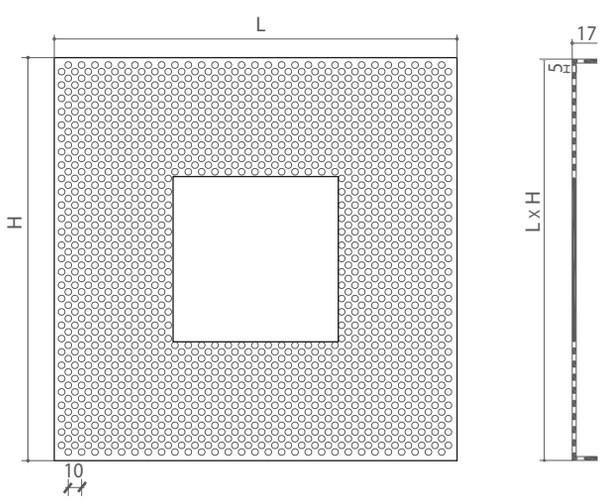
Application
Supply/ exhaust ot the air flow.

Characteristics
Diffuser is provided with circular perforation of Ø 5 mm placed on the front side.
Specially designed for paneled ceiling mounting - size 600 x 600 mm.
Connects to the air ducts through the adapter.

Accessories
Insulated/Non-insulated adapter **ADP IZ/N.**

Materials and finishing
Pickled steel sheet electrostatic field painted white RAL 9016.
Other RAL colors available on request.

Technical design



Assembly Methods
Paneled ceiling 600 x 600 mm

Selection table PA

Air flow (m^3/h)	L x H	595x595
450	X [m] NR [dB(A)] ΔPt [pa]	2.3 <10 100
500	X [m] NR [dB(A)] ΔPt [pa]	2.6 10.0 125.0
600	X [m] NR [dB(A)] ΔPt [pa]	3.1 15.0 150.0
700	X [m] NR [dB(A)] ΔPt [pa]	3.6 18.0 180.0
780	X [m] NR [dB(A)] ΔPt [pa]	4.0 20.0 200.0
890	X [m] NR [dB(A)] ΔPt [pa]	4.6 25.0 250.0
1100	X [m] NR [dB(A)] ΔPt [pa]	5.5 30.0 300.0
1300	X [m] NR [dB(A)] ΔPt [pa]	6.8 35.0 380.0
1450	X [m] NR [dB(A)] ΔPt [pa]	7.5 40.0 450.0

Annotation

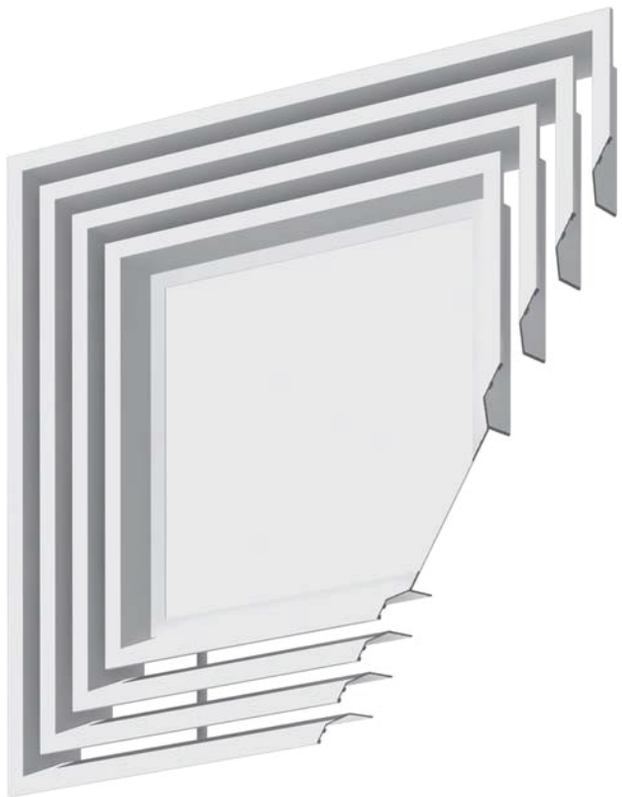
* X [m] - Air jet length at 0.2 m/s velocity
* NR [dB(A)] - Noise level without the room attenuation
* ΔPt [pa] - Pressure loss

Order codes

Type + **ADP IZ/N** / **Dimensions** / **O**

- **Type**: **PA**
- **ADP IZ/N**: Insulated/Non-insulated adapter
- **Dimensions**: L X H, 595 x 595
- **O**: **0** Painted white RAL 9016; **R** Other RAL paints

Linear Ceiling Diffuser LCD



Description
Rectangular diffuser, paneled ceiling type, with four direction air jet diffuser.

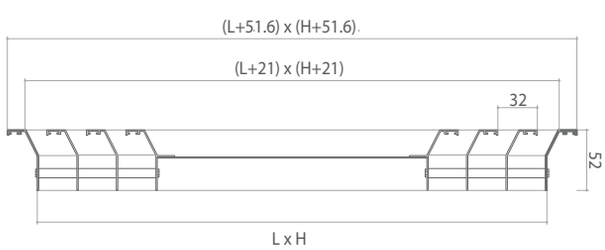
Application
Designed for various air supply or exhaust systems.

Characteristics
Preferred for paneled ceiling, 600 x 600 mm diffuser with 1, 2 or 3 peripheral slots.
The air flow can be adjusted by mounting an air control damper OBD on the diffuser inlet.

Accessories
Air control damper with opposable blades **OBD**
Insulated/non-insulated adapter **ADP IZ/N**

Materials and finishing
Extruded aluminum blades, electrostatic field painted white RAL 9016.
Other RAL colors available on request.

Technical design



Assembly Methods
Paneled ceiling 600 x 600 mm (standard)

Selection table LCD

Air flow (m^3/h)	595X595			
	No. slots	1	2	3
	Ak [m^2]	0.028	0.048	0.068
150	Veff(m/s) X [m] NR [dB(A)] ΔPt [pa]	1.5 0.8 <20 2.1		
200	Veff(m/s) X [m] NR [dB(A)] ΔPt [pa]	2.0 1.0 20.0 3.3	1.2 0.7 <20 1.1	
300	Veff(m/s) X [m] NR [dB(A)] ΔPt [pa]	3.0 1.4 28.0 7.3	1.7 1.1 22.0 2.5	1.2 0.9 <20 1.2
400	Veff(m/s) X [m] NR [dB(A)] ΔPt [pa]	4.0 1.9 34.0 13.0	2.3 1.5 29.0 4.4	1.6 1.2 25.0 2.2
500	Veff(m/s) X [m] NR [dB(A)] ΔPt [pa]	5.0 2.4 39.0 20.4	2.9 1.8 33.0 6.9	2.0 1.5 30 3.4
600	Veff(m/s) X [m] NR [dB(A)] ΔPt [pa]	6.0 2.9 43.0 29.3	3.5 2.2 37.0 9.9	2.5 1.9 33.0 4.9
700	Veff(m/s) X [m] NR [dB(A)] ΔPt [pa]	6.9 3.4 46.0 39.9	4.1 2.6 40.0 13.5	2.9 2.2 37.0 6.7
1000	Veff(m/s) X [m] NR [dB(A)] ΔPt [pa]	9.9 4.8 54.0 81.5	5.8 3.7 48.0 27.6	4.1 3.1 44.0 13.7
1500	Veff(m/s) X [m] NR [dB(A)] ΔPt [pa]		8.7 5.5 56.0 62.1	6.1 4.6 53.0 30.9

NR < 35 | 35≤NR≤45 | NR>45

Annotation

* Ak [m^2] - Free surface of the diffuser
* Veff[m/s] - Air velocity in diffuser
* X [m] - Length airflow at speed Vk [m/s] = 0.2 m/s
* NR [dB(A)] - Noise level without the room attenuation
* ΔPt [pa] - Pressure loss inside diffuser

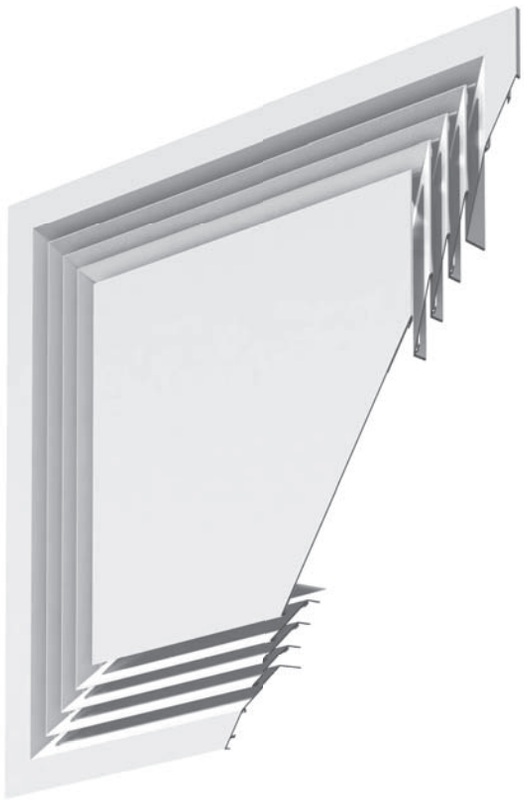
Note: The values in the selection table are calculated for 2.8 m high ceiling.

Order codes

Type - Number of slots + **ADP IZ/N** / **Dimensions** / **L X H** / **O**

- **LCD**
- 1 / 2 / 3
- Insulated/Non-insulated adapter
- **L X H** (595 x 595)
- Other dimensions on request
- **0** Painted white RAL 9016
- **R** Other RAL paints

Decorative Diffuser AC



Description
Rectangular diffuser, paneled ceiling type, with four direction air jet diffuser.

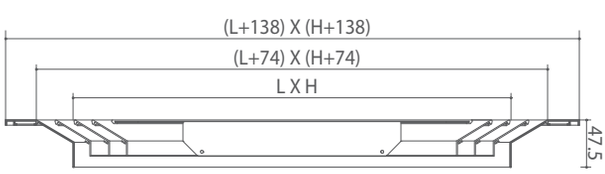
Application
Supply or exhaust of the airflow.

Characteristics
Preferred for paneled ceiling, 600 x 600 mm diffuser with 1, 2, 3 or 4 outlying slots around the square core.
Removable central core.
The air flow can be adjusted by mounting an air control damper **OBD** on the diffuser inlet.
Uniform aspect with adjacent diffusers.

Accessories
Air control damper with opposable blades **OBD.**
Insulated/non-insulated adaptor **ADP IZ/N.**

Materials and finishing
Extruded shaped aluminum blades , electrostatic field painted white RAL 9016
Other RAL colors available on request.

Technical design



Assembly Methods
Paneled ceiling 600 x 600 mm (standard)
Mounting holes for diffuser connection (optional)

Selection table AC

Air flow (m^3/h)	No. slots	595 X 595			
		1	2	3	4
	Ak(mp)	0.022	0.039	0.048	0.055
90	Veff[m/s] NR [dB(A)] ΔPt [pa] X [m]	1.12 22.00 3.00 1.80			
125	Veff[m/s] NR [dB(A)] ΔPt [pa] X [m]	1.56 28.00 6.00 2.10			
180	Veff[m/s] NR [dB(A)] ΔPt [pa] X [m]	2.24 35.00 12.00 2.60			
250	Veff[m/s] NR [dB(A)] ΔPt [pa] X [m]	3.11 42.00 24.00 3.60	1.79 25.00 17.00 2.40		
320	Veff[m/s] NR [dB(A)] ΔPt [pa] X [m]	3.99 47.00 40.00 4.80	2.30 34.00 29.00 3.00	1.84 20.00 8.00 2.40	
430	Veff[m/s] NR [dB(A)] ΔPt[pa] X [m]		3.09 45.00 51.00 4.20	2.48 30.00 14.00 3.20	2.17 26.00 11.00 2.80
540	Veff[m/s] NR [dB(A)] ΔPt [pa] X [m]		3.88 55.00 80.00 5.20	3.11 38.00 21.00 4.00	2.73 33.00 17.00 3.60
650	Veff[m/s] NR [dB(A)] ΔPt [pa] X [m]			3.75 45.00 31.00 4.80	3.28 39.00 24.00 4.20
750	Veff[m/s] NR [dB(A)] ΔPt [pa] X [m]			4.32 52.00 42.00 5.60	3.79 44.00 33.00 5.00
850	Veff[m/s] NR [dB(A)] ΔPt [pa] X [m]				4.29 49.00 43.00 5.60

Annotation

* Ak [m²] - Free surface of the diffuser
* Veff[m/s] - Air velocity in diffuser
* NR [dB(A)] - Noise level without room attenuation
* ΔPt [pa] - Pressure loss inside diffuser
* X [m] - Airflow length
* Vk = 0.2 m/s - Speed at which airflow length was calculated "x" [m]

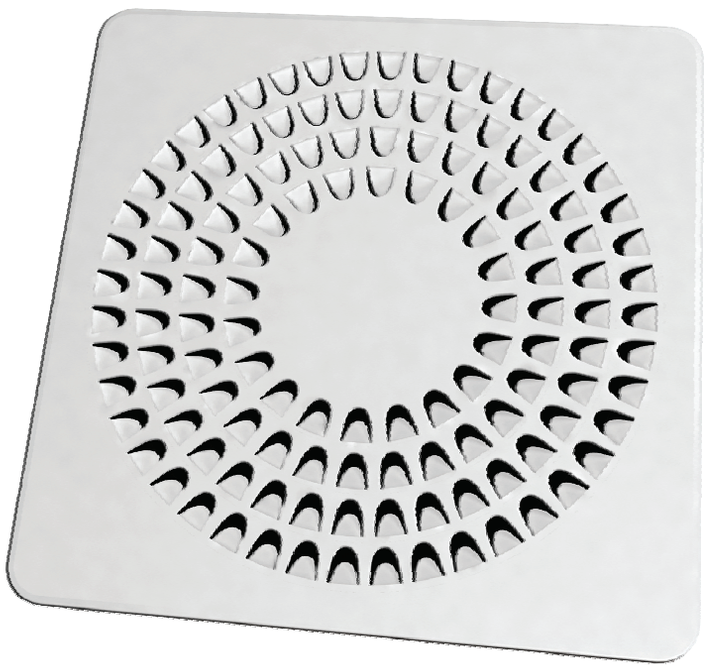
NR < 35 | 35≤NR≤45 | NR>45

Order codes

Type - Number of slots + **OBD** + **ADP IZ/N** / **Dimensions** / **L X H** / **O**

- **AC**
- 1, 2, 3
- Air control damper
- Insulated/non-insulated adapter
- **L X H** 595 x 595
- Other dimensions on request
- **0** Painted white RAL 9016
- **R** Other RAL paints…

Ceiling Diffuser CD-CK



Description
Square diffuser preferred for paneled ceiling mounting.

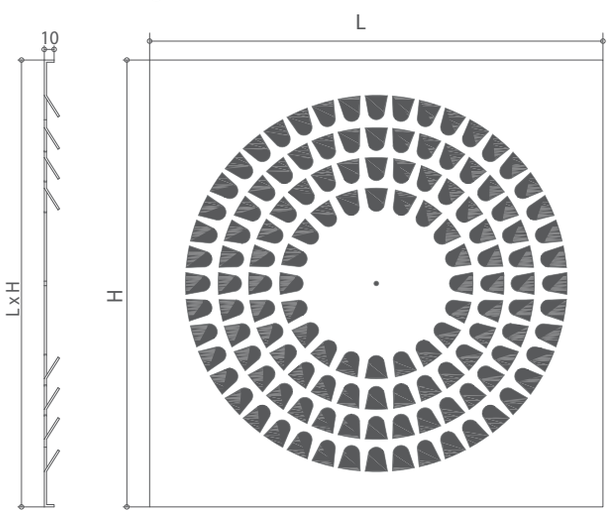
Application
Supply/ exhaust of airflow.

Characteristics
Specifically designed for paneled ceiling mounting - size 600 x 600 mm.
Keeps a uniform aspect with adjacent diffusers.
Connects to the air ducts through the adapter.
The diffuser is delivered with fitting gaskets.

Accessories
Insulated/ Non-insulated adapter **ADP IZ/N**.

Materials and finishing
Pickled steel sheet electrostatic field painted white RAL 9016
Other RAL colors available on request

Technical design



Assembly Methods
Paneled ceiling 600 x 600 mm

Selection table CD-CK

Supply

Air flow (m^3/h)	LxH	595 X 595
480	X [m] NR [dB(A)] ΔPt [pa]	5.7 5.0 7.0
540	X [m] NR [dB(A)] ΔPt [pa]	6.3 10.0 8.0
610	X [m] NR [dB(A)] ΔPt [pa]	7.2 20.0 10.0
720	X [m] NR [dB(A)] ΔPt [pa]	8.5 25.0 15.0
890	X [m] NR [dB(A)] ΔPt [pa]	11.0 32.0 22.0
1010	X [m] NR [dB(A)] ΔPt [pa]	13.0 35.0 28.0
1200	X [m] NR [dB(A)] ΔPt [pa]	14.0 40.0 40.0
1315	X [m] NR [dB(A)] ΔPt [pa]	16.0 45.0 50.0

Exhaust

Air flow (m^3/h)	LxH	595 X 595
480	NR [dB(A)] ΔPt [pa]	15.0 7.5
540	NR [dB(A)] ΔPt [pa]	22.0 9.0
610	NR [dB(A)] ΔPt [pa]	26.0 11.0
720	NR [dB(A)] ΔPt [pa]	32.0 17.0
890	NR [dB(A)] ΔPt [pa]	37.0 24.0
1010	NR [dB(A)] ΔPt [pa]	42.0 30.0
1200	NR [dB(A)] ΔPt [pa]	45.0 40.0
1315	NR [dB(A)] ΔPt [pa]	50.0 50.0

Annotation

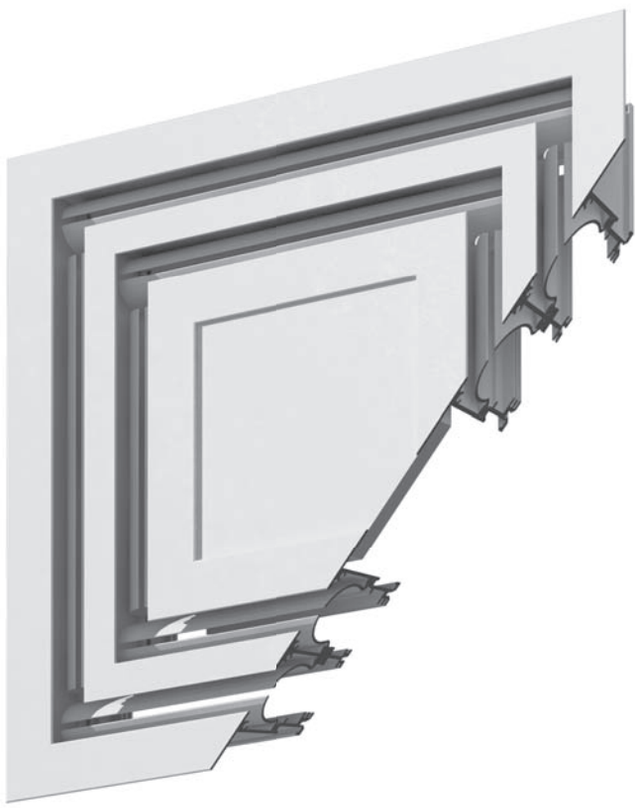
* X [m] - Length air jet at speed Vk = 0.2 m/s
* NR [dB(A)] - Noise level without the room attenuation
* ΔPt [pa] - Pressure loss

Order codes

Type + **ADP IZ/N** / **Dimensions** / **O**

- **CD-CK**
- Insulated/Non-insulated adapter
- **L X H** 595 x 595
- **0** Painted white RAL 9016
  R Other RAL paints

4 Direction Slot Diffuser SL



Description
Air diffuser specially designed for paneled ceiling mounting.

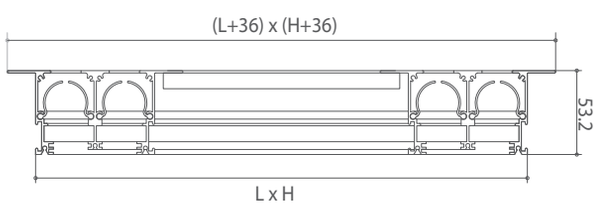
Application
Supply/ exhaust of the airflow

Characteristics
The diffuser is designed with two slots perimetrically located around a square center of plasterboard or other material, keeping a uniform aspect with the rest of the ceiling.
Inside each slot there are individually adjustable twin blades mounted, which allows adjustment of air flow passing through each slot.
The blades inside the slot can be painted differently from the diffuser, thus giving a decorative look.

Accessories
Insulated/Non-insulated adapter **ADP IZ/N.**

Materials and finishing
Extruded aluminum blades, electrostatic field painted white RAL 9016.
Other RAL colors available on request.

Technical design



Assembly Methods
Paneled ceiling 600 x 600 mm (standard)

Selection table SL

Air flow (m^3/h)	595 x 595	
	Ak (mp)	0.076
86	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] ΔPt [pa]	4.00 3.00 --- 4.02
176	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] ΔPt [pa]	5.00 4.00 18.00 7.94
219	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] ΔPt [pa]	8.00 7.00 30.00 22.16
262	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] ΔPt [pa]	9.00 7.00 34.00 31.00
305	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] ΔPt [pa]	10.00 8.00 38.00 43.00
352	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] ΔPt [pa]	10.00 9.00 41.00 56.00
395	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] ΔPt [pa]	11.00 9.00 44.00 72.00
485	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] ΔPt [pa]	12.00 10.00 47.00 73.00
552	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] ΔPt [pa]	12.50 10.50 48.00 73.50
625	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] ΔPt [pa]	12.80 11.00 50.00 74.00
850	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] ΔPt [pa]	13.50 12.00 52.00 92.00

Annotation

* Ak [m²] - Free surface of the diffuser
* X [m] - Length airflow
* NR [dB(A)] - Noise level without the room attenuation
* ΔPt [pa] - Pressure loss inside diffuser

Order codes

Type + **ADP IZ/N** / **Dimensions** / **L X H** / **O**

- **Type**: **SL**
- **ADP IZ/N**: Insulated/Non-insulated adapter
- **Dimensions**: **L X H** — 595 x 595
- **L X H**: Other dimensions on request
- **O**: **0** Painted white RAL 9016; **R** Other RAL paints

Ceiling Diffuser FD

Description
Aluminum rectangular diffuser with individual adjustable blades.

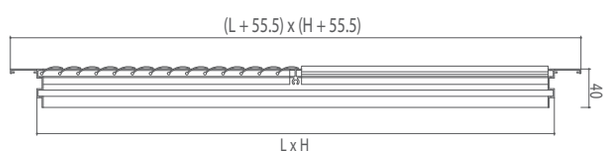
Application
Designed for air exhaust from ventilation and air conditioning systems.

Characteristics
The diffuser is equipped with adjustable blades arranged in four directions, causing rotation of the air with Coanda effect, obtaining a high rate of induction and reduced stratification.
On request the diffuser can be provided with 1, 2 or 3 discharge directions.
Recommended for rooms with high ceilings (eg. factories, airports, sports halls, etc.)
Minimum recommended height is 3.8 m.
Designed specifically for integration into the paneled ceiling box (dimensions 600 x 600 mm), the speaker is provided without mounting holes.
On request, the diffuser can be produced in custom sizes, in which case it is provided with mounting holes.

Accessories
Insulated/Non-insulated adapter **ADP IZ/N**.

Materials and finishing
Extruded aluminum material, electrostatic field painted white RAL 9016.
Other RAL colors available on request.

Technical design



Assembly Methods
Paneled ceiling 600 x 600 mm (standard)
With screws, visible on the diffuser's frame.

Selection table FD

Air flow (m^3/h)	Dimensions	595x595
	Ak [m^2]	0.1819
960	Veff(m/s) X [m] NR [dB(A)] ΔPt [pa]	1.5 3.0 23.0 13.0
1150	Veff(m/s) X [m] NR [dB(A)] ΔPt [pa]	1.8 3.8 30.0 18.0
1380	Veff(m/s) X [m] NR [dB(A)] ΔPt [pa]	2.1 4.5 35.0 25.0
1610	Veff(m/s) X [m] NR [dB(A)] ΔPt [pa]	2.5 5.5 38.0 33.0
1860	Veff(m/s) X [m] NR [dB(A)] ΔPt [pa]	2.8 6.2 42.0 47.0
2300	Veff(m/s) X [m] NR [dB(A)] ΔPt [pa]	3.4 7.5 50.0 75.0
2700	Veff(m/s) X [m] NR [dB(A)] ΔPt [pa]	4.1 8.0 55.0 100.0

Annotation

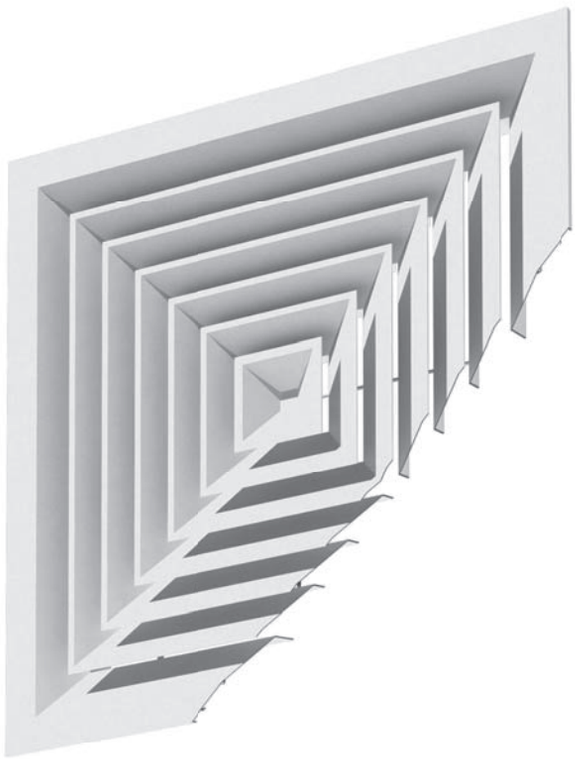
* Ak [m^2] - Free surface of the diffuser
* Veff[m/s] - Air velocity in diffuser
* X [m] - Length air jet at speed Vk = 0.2 m/s
* NR [dB(A)] - Noise level without the room attenuation
* ΔPt [pa] - Pressure loss

Note: Table values are for maximum blade opening

Order codes

Type	+	**ADP IZ/N**	/	**Dimensions**	/	**L X H**	/	**O**
FD		Insulated/ Non-insulated adapter		**L X H** 595 x 595		Other dimensions on request		**0** Painted white RAL 9016 **R** Other RAL paints

Ceiling Diffuser CD



Description
4 direction aluminum ceiling diffuser, for paneled ceiling mounting.

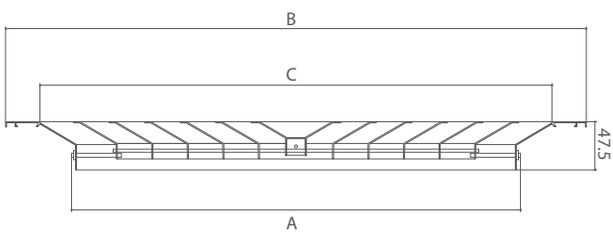
Application
Supply or exhaust of the airflow

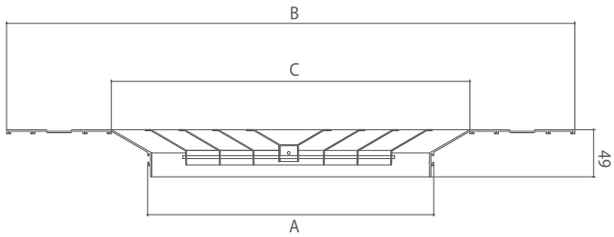
Characteristics
Recommended for rooms with a maximum height of 4 meters .
Removable central cone for easy installation and cleaning.
The difference between the recommended supply air temperature and the room air temperature is ± 10 K.
The air flow can be adjusted by mounting an air control damper **OBD** on the diffuser inlet.

Accessories
Air control damper with opposable blades **OBD**
Insulated/Non-insulated adapter **ADP IZ/N**

Materials and finishing
Extruded aluminum material, electrostatic field painted white RAL 9016.
Other RAL colors available on request.
Natural anodised aluminum.

Technical design
CD narrow/large frame





Assembly Methods
Paneled ceiling 600 x 600 mm (standard)
Mounting holes on diffuser connection (optional)

Selection table CD

Air flow (m^3/h)	LXH	150x150	225x225	300x300	375x375	455x455	525x525	600x600
	Ak [m^2]	0.0109	0.0244	0.0435	0.0679	0.0978	0.1331	0.1739
100	Veff [m/s] X (m) NR [dB(A)] ΔPt [pa]	2.6 0,6 18.0 5.0						
160	Veff [m/s] X (m) NR [dB(A)] ΔPt [pa]	4.1 1,0 29.0 12.0	1.8 0,7 13.0 2.0					
200	Veff [m/s] X (m) NR [dB(A)] ΔPt [pa]	5.1 1,2 35.0 18.0	2.3 0,8 18.0 4.0					
300	Veff [m/s] X (m) NR [dB(A)] ΔPt [pa]	7.7 1,9 45.0 41.0	3.4 1,3 28.0 8.0	1.9 0,9 16.0 3.0				
400	Veff [m/s] X (m) NR [dB(A)] ΔPt [pa]		4.6 1,7 35.0 15.0	2.6 1,3 24.0 5.0	1.6 1,0 15.0 2.0			
500	Veff [m/s] X (m) NR [dB(A)] ΔPt [pa]		5.7 2,1 41.0 23.0	3.2 1,6 29.0 7.0	2.1 1,3 20.0 3.0			
600	Veff [m/s] X (m) NR [dB(A)] ΔPt [pa]		6.8 2,5 45.0 33.0	3.8 1,9 34.0 10.0	2.5 1,5 25.0 4.0	1.7 1,3 17.0 2.0		
700	Veff [m/s] X (m) NR [dB(A)] ΔPt [pa]		8.0 2,9 49.0 44.0	4.5 2,2 37.0 14.0	2.9 1,8 28.0 6.0	2.0 1,5 21.0 3.0		
800	Veff [m/s] X (m) NR [dB(A)] ΔPt [pa]		9.1 3,3 53.0 58.0	5.1 2,5 41.0 18.0	3.3 2,0 32.0 7.0	2.3 1,7 24.0 4.0	1.7 1,4 18.0 2.0	
900	Veff [m/s] X (m) NR [dB(A)] ΔPt [pa]			5.8 2,8 44.0 23.0	3.7 2,3 35.0 9.0	2.6 1,9 27.0 5.0	1.9 1,6 21.0 2.0	
1000	Veff [m/s] X (m) NR [dB(A)] ΔPt [pa]			6.4 3,1 46.0 29.0	4.1 2,5 37.0 12.0	2.8 2,1 30.0 6.0	2.1 1,8 23.0 3.0	1.6 1,6 18.0 2.0
1200	Veff [m/s] X (m) NR [dB(A)] ΔPt [pa]			7.7 3,8 51.0 41.0	4.9 3,0 42.0 17.0	3.4 2,5 34.0 8.0	2.5 2,1 28.0 4.0	1.9 1,9 23.0 3.0
1600	Veff [m/s] X (m) NR [dB(A)] ΔPt [pa]				6.6 4,0 49.0 30.0	4.5 3,3 41.0 14.0	3.3 2,9 35.0 8.0	2.6 2,5 30.0 5.0
2000	Veff [m/s] X (m) NR [dB(A)] ΔPt [pa]				8.2 5,0 54.0 47.0	5.7 4,2 47.0 23.0	4.2 3,6 41.0 12.0	3.2 3,1 35.0 7.0
3000	Veff [m/s] X (m) NR [dB(A)] ΔPt [pa]					8.5 6,3 57.0 51.0	6.3 5,4 51.0 27.0	4.8 4,7 45.0 16.0

NR < 35 | 35≤NR≤45 | NR>45

Annotation

* Ak [m²] - Free surface of the diffuser
* Veff[m/s] - Air velocity in diffuser
* X [m] - Air jet length
* NR [dB(A)] - Noise level without the room attenuation
* ΔPt [pa] - Pressure loss inside diffuser
* Vk = 0.2 [m/s] - Speed at which airflow length was calculated "x" [m]

Standard dimensions

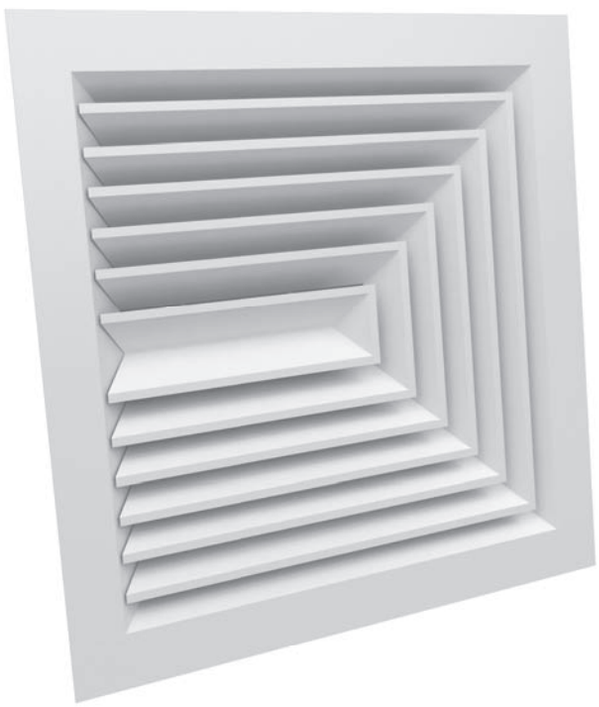
CD			
neck / external size	**A**	**B**	**C**
150 / 290	150 x 150	290 x 290	230 x 230
150 / 595	150 x 150	595 x 595	235 x 235
225 / 365	225 x 225	365 x 365	305 x 305
225 / 595	225 x 225	595 x 595	315 x 315
300 / 440	300 x 300	440 x 440	380 x 380
300 / 595	300 x 300	595 x 595	385 x 385
375 / 515	375 x 375	515 x 515	455 x 455
375 / 595	375 x 375	595 x 595	465 x 465
455 / 595	455 x 455	595 x 595	535 x 535
480 / 620	480 x 480	620 x 620	560 x 560
525 / 665	525 x 525	665 x 665	605 x 605
600 / 740	600 x 600	740 x 740	680 x 680

Order codes

Type + **OBD** + **ADP IZ/N** / **Dimensions** / **L X H** / **O**

- **Type**: **CD**
- **OBD**: Air control damper
- **ADP IZ/N**: Insulated/ Non-insulated adapter
- **Dimensions** — **neck / external size**: 150 / 290; 150 / 595; 225 / 365; 225 / 595; 300 / 440; 300 / 595; 375 / 515; 375 / 595; 455 / 595; 480 / 620; 525 / 665; 600 / 740
- **L X H**: Other dimensions on request
- **O**: **0** Painted white RAL 9016; **R** Other RAL paints; Natural anodised

Ceiling Diffuser CD-S



CD-S3

Description
1, 2 or 3 direction square diffuser preferred for paneled ceiling mounting.

Application
Supply/ exhaust of airflow.

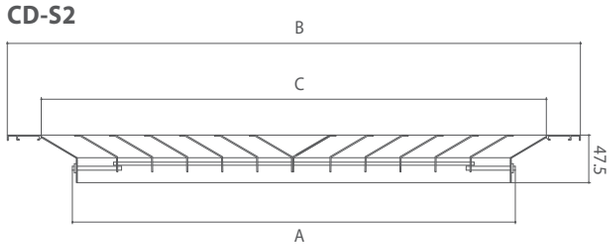
Characteristics
Recommended for rooms of a maximum height of 4 meters .
Removable central cone for easy installation and cleaning.
Suitable for horizontal air supply.
The difference between the recommended supply air temperature and the room air temperature is ± 10 K.
The air flow can be adjusted by mounting a air control damper **OBD** on the diffuser inlet.

Accessories
Air control damper with opposable blades **OBD**
Insulated/Non-insulated adapter **ADP IZ/N**

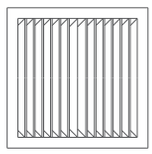
Materials and finishing
Extruded aluminum material, electrostatic field painted white RAL 9016
Other RAL colors available on request.
Natural anodised aluminum.

Technical design

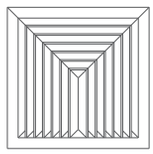
CD-S2



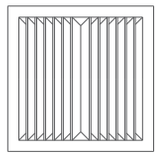
CD-S1



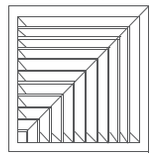
CD-S3



CD-S2



CD-S2-90



Assembly Methods
Paneled ceiling 600 x 600 mm (standard)
Mounting holes on diffuser connection (optional)

Ceiling Diffuser CD-S

Selection table CD-S

Air flow (m^3/h)	LxH	150 x 150			225 x 225			300 x 300			375 x 375			455 x 455			525 x 525			600 x 600		
	Model	S1	S2	S3	S1	S2	S3	S1	S2	S3	S1	S2	S3	S1	S2	S3	S1	S2	S3	S1	S2	S3
	Ak(m^2)	0.0096	0.0093	0.0088	0.0215	0.0209	0.02	0.0383	0.0373	0.0355	0.0598	0.0582	0.0554	0.0863	0.0838	0.0798	0.1174	0.1141	0.1086	0.1534	0.149	0.1418
100	Veff [m/s]	2.9	3.0																			
	X (m)	2.2	0.8																			
	NR [dB(A)]	22	22																			
	ΔPt(Pa)	5.9	6.7																			
160	Veff [m/s]	4.6	4.8	5.1																		
	X (m)	3.5	1.2	1.2																		
	NR [dB(A)]	33	33	29																		
	ΔPt(Pa)	15	17.1	22																		
200	Veff [m/s]	5.8	6.0	6.3	2.6	2.7	2.8															
	X (m)	4.4	1.5	1.5	2.9	1	1															
	NR [dB(A)]	39	39	34	22	22	<20															
	ΔPt(Pa)	23.4	26.8	34	4.7	5.3	7															
300	Veff [m/s]	8.7	9.0	9.5	3.9	4.0	4.2	2.2	2.2													
	X (m)	6.6	2.3	2.3	4.4	1.5	1.5	3.3	1.1													
	NR [dB(A)]	49	49	44	32	32	28	20	21													
	ΔPt(Pa)	52.7	60.2	77	10.5	11.9	15	3.3	3.7													
400	Veff [m/s]				5.2	5.3	5.6	2.9	3.0	3.1	1.9	1.9										
	X (m)				5.8	2	2	4.4	1.5	1.5	3.5	1.2										
	NR [dB(A)]				39	40	35	27	28	23	<20	<20										
	ΔPt(Pa)				18.7	21.2	27	5.9	6.7	9	2.4	2.7										
600	Veff [m/s]				7.8	8.0	8.3	4.4	4.5	4.7	2.8	2.9	3.0	1.9	2.0							
	X (m)				8.8	3	3	6.6	2.3	2.3	5.3	1.8	1.8	4.4	1.5							
	NR [dB(A)]				49	50	45	37	38	33	28	29	24	20	21							
	ΔPt(Pa)				42.1	47.7	61	13.3	15	19	5.4	6.2	8	2.6	3							
800	Veff [m/s]							5.8	6.0	6.3	3.7	3.8	4.0	2.6	2.7	2.8	1.9	1.9				
	X (m)							8.8	3	3	7	2.4	2.4	5.8	2	2	5	1.7				
	NR [dB(A)]							44	45	40	35	36	31	27	28	24	21	22				
	ΔPt(Pa)							23.6	26.6	34	9.7	10.9	14	4.6	5.3	7	2.5	2.8				
1000	Veff [m/s]							7.3			4.6	4.8	5.0	3.2	3.3	3.5	2.4	2.4	2.6	1.8	1.9	
	X (m)							11			8.8	3	3	7.3	2.5	2.5	6.3	2.2	2.1	5.5	1.9	
	NR [dB(A)]							50			41	41	37	33	34	29	26	28	23	21	22	
	ΔPt(Pa)							36.8			15.1	17.1	22	7.3	8.2	11	3.9	4.4	6	2.3	2.6	
1200	Veff [m/s]										5.6	5.7	6.0	3.9	4.0	4.2	2.8	2.9	3.1	2.2	2.2	2.4
	X (m)										10.5	3.7	3.6	8.8	3	3	7.5	2.6	2.6	6.6	2.3	2.3
	NR [dB(A)]										45	46	41	37	38	34	31	32	28	25	27	22
	ΔPt(Pa)										21.7	24.6	32	10.4	11.9	15	5.6	6.4	8	3.3	3.8	5
1600	Veff [m/s]										7.4		8.0	5.1	5.3	5.6	3.8	3.9	4.1	2.9	3.0	3.1
	X (m)										14		4.8	11.7	4.1	4	10	3.5	3.4	8.8	3	3
	NR [dB(A)]										52		48	44	46	41	38	39	35	33	34	29
	ΔPt(Pa)										38.7		56	18.6	21.2	27	10	11.4	15	5.9	6.7	9
2000	Veff [m/s]																	4.9	5.1		3.7	3.9
	X (m)																	4.3	4.3		3.8	3.8
	NR [dB(A)]																	45	40		39	35
	ΔPt(Pa)																	17.8	23		10.4	13
3000	Veff [m/s]																		7.7		5.6	5.9
	X (m)																		6.4		5.7	5.6
	NR [dB(A)]																		50		49	45
	ΔPt(Pa)																		51		23.5	30

NR < 35 | 35≤NR≤45 | NR>45

Annotation

- Ak [m²] - Free surface of the diffuser
- Veff[m/s] - Air velocity in diffuser
- X [m] - Air jet length
- NR [dB(A)] - Noise level without the room attenuation
- ΔPt [pa] - Pressure loss
- Vk = 0.25 [m/s] - Speed at which air jet length was calculated "x" [m]

Standard dimensions

CD-S 1,2,3			
neck / external size	**A**	**B**	**C**
150 / 290	150 x 150	290 x 290	230 x 230
150 / 595	150 x 150	595 x 595	235 x 235
225 / 365	225 x 225	365 x 365	305 x 305
225 / 595	225 x 225	595 x 595	315 x 315
300 / 440	300 x 300	440 x 440	380 x 380
300 / 595	300 x 300	595 x 595	385 x 385
375 / 515	375 x 375	515 x 515	455 x 455
375 / 595	375 x 375	595 x 595	465 x 465
455 / 595	455 x 455	595 x 595	535 x 535
480 / 620	480 x 480	620 x 620	560 x 560
525 / 665	525 x 525	665 x 665	605 x 605
600 / 740	600 x 600	740 x 740	680 x 680

Order codes

Type	–	**No. of directions**	+	**OBD**	+	**ADP IZ/N**	/	**Dimensions**	/	**L X H**	/	**O**
CD-S		1, 2, 3		Air control damper		Insulated/ Non-insulated adapter		**neck / external** 150 / 290 150 / 595 225 / 365 225 / 595 300 / 440 300 / 595 375 / 515 375 / 595 455 / 595 480 / 620 525 / 665		Other dimensions on request		**0** Painted white RAL 9016 **R** Other RAL paints Natural anodised

Ceiling Diffuser With Double Deflection CD-DD

Description
Aluminum rectangular diffuser designed for paneled ceiling mounting.

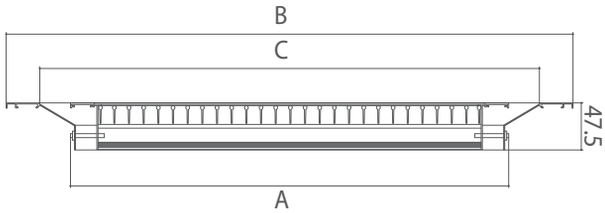
Application
Suitable for supply and exhaust air systems.

Characteristics
The diffuser is provided with a central core that consists of a **DD** -grille (double deflection) with two rows of individually adjustable blades.
The adjustable baldes allow airflow distribution to desired areas.
Removable central cone for easy installation and cleaning.
The air flow can be adjusted by mounting an air control damper **OBD** on the diffuser inlet connection.

Accessories
Air control damper with opposable blades **OBD**
Insulated/Non-insulated adapter **ADP IZ/N**

Materials and finishing
Extruded aluminum material, electrostatic field painted white RAL 9016
Other RAL colors available on request
Natural anodised aluminum

Technical design



Assembly Methods
Paneled ceiling 600x600mm (standard)
Mounting holes on the diffuser inlet connection (optional)

Selection table CD-DD

Air flow m³/h	LXH	150x150	225x225	300x300	375x375	455x455	480x480	525x525	600x600
	Ak [m²]	0.014	0.028	0.048	0.075	0.110	0.123	0.147	0.192
100	Veff [m/s] X (m) NR [dB(A)] ΔPt [pa]	2.0 3.0 20.0 3.0							
160	Veff [m/s] X (m) NR [dB(A)] ΔPt [pa]	3.2 3.7 23.0 7.0	1.6 2.4 20.0 2.0						
200	Veff [m/s] X (m) NR [dB(A)] ΔPt [pa]	4.0 5.0 25.0 12.0	2.0 3.5 20.0 3.0						
300	Veff [m/s] X (m) NR [dB(A)] ΔPt [pa]	6.0 7.5 30.0 30.0	3.0 5.3 25.0 7.0	1.7 2.5 20.0 3.0					
400	Veff [m/s] X (m) NR [dB(A)] ΔPt [pa]		4.0 7.0 30.0 12.0	2.3 4.0 23.0 5.0	1.5 2.0 15.0 2.0				
500	Veff [m/s] X (m) NR [dB(A)] ΔPt [pa]		4.9 9.0 30.0 20.0	2.9 5.0 28.0 7.0	1.9 1.3 19.0 3.0				
600	Veff [m/s] X (m) NR [dB(A)] ΔPt [pa]		5.9 9.0 35.0 33.0	3.4 7.0 33.0 9.0	2.2 4.0 25.0 4.0	1.5 3.0 17.0 2.0			
700	Veff [m/s] X (m) NR [dB(A)] ΔPt [pa]		6.9 12.0 40.0 40.0	4.0 8.0 35.0 15.0	2.6 7.5 30.0 5.0	1.8 5.0 21.0 3.0			
800	Veff [m/s] X (m) NR [dB(A)] ΔPt [pa]		7.9 18.0 45.0 50.0	4.6 11.0 40.0 20.0	3.0 8.0 31.0 7.0	2.0 6.0 23.0 4.0	1.8 3.0 20.0 3.0	1.5 2.0 17.0 2.0	
900	Veff [m/s] X (m) NR [dB(A)] ΔPt [pa]			5.2 12.0 43.0 25.0	3.3 9.0 34.0 15.0	2.3 7.0 26.0 5.0	2.0 4.0 23.0 3.0	1.7 2.5 20.0 2.0	
1000	Veff [m/s] X (m) NR [dB(A)] ΔPt [pa]			5.7 15.0 46.0 32.0	3.7 12.0 36.0 18.0	2.5 8.0 29.0 6.0	2.3 7.0 25.0 4.0	1.9 4.0 23.0 3.0	1.4 2.0 18.0 2.0
1200	Veff [m/s] X (m) NR [dB(A)] ΔPt [pa]			6.9 20.0 50.0 41.0	4.4 15.0 41.0 20.0	3.0 11.0 33.0 8.0	2.7 9.0 30.0 6.0	2.3 7.0 27.0 4.0	1.7 3.0 22.0 3.0
1600	Veff [m/s] X (m) NR [dB(A)] ΔPt [pa]				5.9 17.0 48.0 29.0	4.0 13.0 40.0 14.0	3.6 12.0 37.0 10.0	3.0 10.0 33.0 8.0	2.3 7.0 29.0 5.0
2000	Veff [m/s] X (m) NR [dB(A)] ΔPt [pa]				7.4 20.0 53.0 45.0	5.0 17.0 45.0 23.0	4.5 15.0 42.0 18.0	3.8 11.0 40.0 12.0	2.9 8.5 34.0 7.0
3000	Veff [m/s] X (m) NR [dB(A)] ΔPt [pa]					7.6 20.0 55.0 50.0	6.8 18.0 53.0 40.0	5.7 15.0 50.0 25.0	4.3 12.0 44.0 16.0

Annotation

- Ak [m²] - Free surface of the diffuser
- Veff[m/s] - Air velocity in diffuser
- X [m] - Air jet length
- ΔPt [pa] - Pressure loss

* NR [dB(A)] - Noise level without the room attenuation

Standard dimensions

CD - DD			
neck/external size	**A**	**B**	**C**
150 / 290	150 x 150	290 x 290	230 x 230
150 / 595	150 x 150	595 x 595	235 x 235
225 / 365	225 x 225	365 x 365	305 x 305
225 / 595	225 x 225	595 x 595	315 x 315
300 / 440	300 x 300	440 x 440	380 x 380
300 / 595	300 x 300	595 x 595	385 x 385
375 / 515	375 x 375	515 x 515	455 x 455
375 / 595	375 x 375	595 x 595	465 x 465
455 / 595	455 x 455	595 x 595	535 x 535
480 / 620	480 x 480	620 x 620	560 x 560
525 / 665	525 x 525	665 x 665	605 x 605
600 / 740	600 x 600	740 x 740	680 x 680

Order codes

Type + **OBD** + **ADP IZ/N** / **Dimensions** / **L X H** / **O**

- **Type**: **CD-DD**
- **OBD**: Air control damper
- **ADP IZ/N**: Insulated/Non-insulated adapter
- **Dimensions**: **neck/external size**
  - 150 / 290
  - 150 / 595
  - 225 / 365
  - 225 / 595
  - 300 / 440
  - 375 / 515
  - 375 / 595
  - 455 / 595
  - 480 / 620
  - 525 / 665
  - 600 / 740
- **L X H**: Other dimensions on request
- **O**:
  - **0** Painted white RAL 9016
  - **R** Other RAL paints
  - Natural anodised

Linear Slot Diffuser SLOT

Description
Linear aluminum diffuser with adjustable blades.

Application
Supply/ exhaust of the airflow

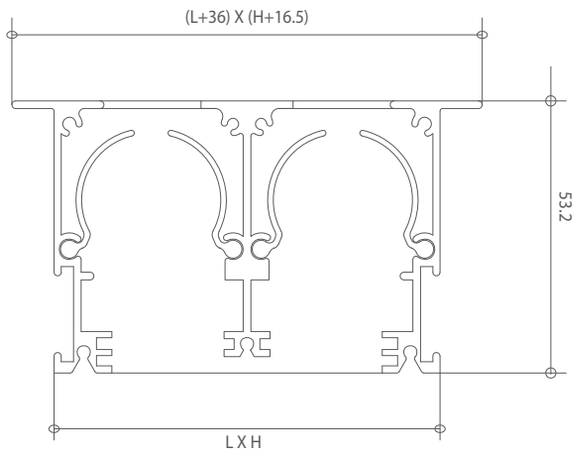
Characteristics
Recommended for rooms between 2.60 to 4.00 meters in height.
The recommended temperature difference between supplied air and indoor air is ± 10 K.
Long run diffusers are made of assembled modules (maximum 3 meters full length). In this case the product is delivered with joining elements between modules.
Blades are used for air flow control.
On request, the diffuser can be made perimetral.

Accessories
Insulated/Non-insulated adapter **ADP IZ/N**
Attachment spring **A**

Materials and finishing
Extruded aluminum, electrostatic field painted white RAL 9016.
Other RAL colors available on request.
Anodised aluminium

Assembly Methods
Attachment spring.

Technical design



No. of slots	H interior (height) slot = 19 mm	H interior (height) slot = 25 mm
1	40	45
2	76	88
3	112	130
4	149	174
5	186	216
6	223	260
7	260	304
8	297	348

Selection table SLOT 19 mm

Air flow (m^3/h)	No. of slots Ak (mp)	1 0.019
Length = 1000 mm		
33	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	1.00 1.00 - 0.98
67	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	2.00 2.00 - 4.02
106	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	4.00 3.00 15.00 9.22
140	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	5.00 4.00 21.00 16.18
173	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	5.00 4.00 27.00 25.40
206	X [m] (Horizontal X [m] (Vertical) NR [dB(A)] △Pt [pa]	6.00 5.00 31.00 36.38
240	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	7.00 5.00 35.00 49.52
279	X [m] (Horizontal) X[m] (Vertical) NR [dB(A)] △Pt [pa]	7.00 5.00 38.00 64.72
312	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	8.00 5.00 41.00 82.18

Air flow (m^3/h)	No. of slots Ak (mp)	2 0.038
Length = 1000 mm		
67	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	1.00 1.00 - 0.98
140	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	3.00 3.00 - 4.02
206	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	5.00 4.00 18.00 9.22
279	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	7.00 5.00 24.00 16.18
346	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	8.00 6.00 30.00 25.40
413	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	9.00 6.00 34.00 36.38
485	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	9.00 7.00 38.00 49.52
552	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	10.00 7.00 41.00 64.72
625	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	11.00 8.00 44.00 82.18

Air flow (m^3/h)	No. of slots Ak (mp)	3 0.057
Length = 1000 mm		
106	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	2.00 2.00 - 0.98
206	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	4.00 3.00 - 4.02
312	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	6.00 5.00 19.00 9.22
413	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	8.00 6.00 26.00 16.18
519	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	10.00 7.00 32.00 25.40
625	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	11.00 8.00 36.00 36.38
725	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	12.00 9.00 40.00 49.52
831	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	12.00 9.00 43.00 64.72
932	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	13.00 10.00 46.00 82.18

Air flow (m^3/h)	No. of slots Ak (mp)	4 0.076
Length = 1000 mm		
140	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	2.00 2.00 - 0.98
279	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	5.00 4.00 - 4.02
413	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	7.00 6.00 21.00 9.22
552	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	10.00 7.00 27.00 16.18
692	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	11.00 8.00 33.00 25.40
831	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	12.00 9.00 37.00 36.38
971	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	13.00 10.00 41.00 49.52
1105	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	14.00 10.00 44.00 64.72
1244	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	15.00 11.00 47.00 82.18

Air flow (m^3/h)	No. of slots Ak (mp)	5 0.095
Length = 1000 mm		
173	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	2.00 2.00 - 0.98
346	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	5.00 4.00 - 4.02
519	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	8.00 6.00 22.00 9.22
692	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	11.00 8.00 28.00 16.18
865	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	12.00 9.00 34.00 25.40
1038	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	14.00 10.00 38.00 36.38
1211	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	15.00 11.00 42.00 49.52
1384	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	16.00 12.00 45.00 64.72
1557	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	17.00 12.00 48.00 82.18

Air flow (m^3/h)	No. of slots Ak (mp)	6 0.114
Length = 1000 mm		
206	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	3.00 2.00 - 0.98
413	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	6.00 5.00 - 4.02
625	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	9.00 7.00 22.00 9.22
831	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	12.00 9.00 29.00 16.18
1038	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	14.00 10.00 35.00 25.40
1244	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	15.00 11.00 39.00 36.38
1451	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	16.00 12.00 43.00 49.52
1663	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	17.00 13.00 46.00 64.72
1869	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	18.00 14.00 49.00 82.18

Selection table SLOT 25 mm

Air flow (m^3/h)	No. of slots Ak (mp)	1 0.025
Length = 1000 mm		
39	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	1.00 1.00 - 0.98
73	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	2.00 2.00 1.00 3.53
112	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	4.00 4.00 14.00 7.94
145	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	5.00 4.00 21.00 14.22
184	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	6.00 5.00 27.00 22.16
223	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	6.00 5.00 31.00 31.87
257	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	7.00 5.00 35.00 43.34
296	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	7.00 6.00 38.00 56.78
329	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	8.00 6.00 41.00 71.78

Air flow (m^3/h)	No. of slots Ak (mp)	2 0.05
Length = 1000 mm		
73	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	1.00 2.00 - 0.98
145	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	4.00 3.00 - 3.53
223	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	5.00 5.00 17.00 7.94
296	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	7.00 6.00 24.00 14.22
368	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	8.00 7.00 30.00 22.16
441	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	9.00 7.00 34.00 31.87
513	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	10.00 8.00 38.00 43.34
591	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	10.00 9.00 41.00 56.78
664	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	11.00 9.00 44.00 71.78

Air flow (m^3/h)	No. of slots Ak (mp)	3 0.075
Length= 1000 mm		
112	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	2.00 2.00 - 0.98
223	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	4.00 4.00 - 3.53
329	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	6.00 6.00 19.00 7.94
441	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	9.00 7.00 26.00 14.22
552	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	10.00 8.00 31.00 22.16
664	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	11.00 9.00 36.00 31.87
776	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	12.00 10.00 40.00 43.34
882	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	13.00 10.00 43.00 56.78
993	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	13.00 11.00 46.00 71.78

Air flow (m^3/h)	No. of slots Ak (mp)	4 0.1
Length = 1000 mm		
145	X[m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	2.00 2.00 - 0.98
296	X[m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	5.00 5.00 - 3.53
441	X[m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	7.00 7.00 20.00 7.94
591	X[m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	10.00 9.00 27.00 14.22
737	X[m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	12.00 9.00 33.00 22.16
882	X[m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	13.00 10.00 37.00 31.87
1032	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	14.00 11.00 41.00 43.34
1177	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	15.00 12.00 44.00 56.78
1328	X [m] (Orizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	16.00 13.00 47.00 71.78

Air flow (m^3/h)	No. of slots Ak (mp)	5 0.125
Length = 1000 mm		
184	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	2.00 3.00 - 0.98
368	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	5.00 5.00 - 3.53
552	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	8.00 8.00 21.00 7.94
737	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	11.00 9.00 28.00 14.22
921	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	13.00 10.00 34.00 22.16
1105	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	14.00 12.00 38.00 31.87
1289	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	15.00 12.00 42.00 43.34
1473	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	16.00 13.00 45.00 56.78
1657	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	17.00 14.00 48.00 71.78

Air flow (m^3/h)	No. of slots Ak (mp)	6 0.15
Length = 1000 mm		
223	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	3.00 3.00 - 0.98
441	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	6.00 6.00 - 3.53
664	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	9.00 9.00 22.00 7.94
882	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	12.00 10.00 29.00 14.22
1105	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	14.00 12.00 34.00 22.16
1328	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	16.00 12.00 39.00 31.87
1546	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	17.00 14.00 43.00 43.34
1769	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	18.00 15.00 46.00 56.78
1986	X [m] (Horizontal) X [m] (Vertical) NR [dB(A)] △Pt [pa]	19.00 16.00 48.00 71.78

Annotation

* Ak [m²] - Free surface
* X [m] - Air jet length
* NR [dB(A)] - Noise level without the room attenuation
* ΔPt [pa] - Pressure loss

Order codes

Type - No. of slots + **ADP IZ/N** + **A** (Attachment spring) / **Dimensions** / **Blades color** / **O**

- **Type - No. of slots**
  - SLOT
  - 1…..8 slots
- **ADP IZ/N**
  - Insulated/Non-insulated adapter
- **A** (Attachment spring)
  - No. of springs depending on length and no. of slots
- **Dimensions**
  - L (length) on request
- **Blades color**
  - Same as the frame
  - Black RAL 9005
- **O**
  - **0** Painted white RAL 9016
  - **R** Other RAL paints
  - Natural anodised

Linear Diffuser With Nozzles BD1/BD2/BDV

BDV

Description
Aluminum linear diffuser with individually maneuvarable nozzles. Recommended for use in traffic areas, classrooms, laboratories, etc.

Application
Airflow supply.

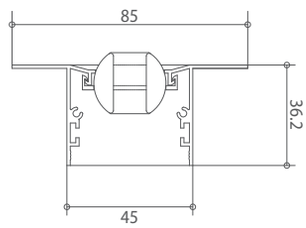
Characteristics
Recommended for rooms between 2.40 and 4.00 m in height.
Long run diffusers are made of assembled modules (maximum 3 m full length). In this case the product is delivered with joining elements between modules.
The spherical design of the nozzles enables air flow redirection in the desired areas.
The spherical nozzles are available in two colors: white RAL 9016 and black RAL 9005.
Depending on the technical and architectural requirements, the diffuser is made with a single row of nozzles - **BD1** or two parallel rows of nozzles - **BD2/BDV**.

Accessories
Attachment spring **A**
Insulated/Non-insulated adapter **ADP IZ/N**

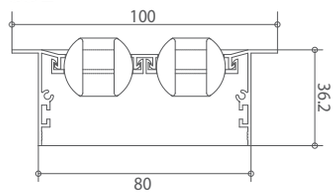
Materials and finishing
Extruded aluminum material, electrostatic field painted white RAL 9016.
Other RAL colors available on request.
Natural/black anodised aluminum.
Polypropylene nozzles.

Technical design

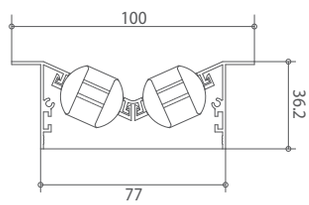
BD1



BD2



BDV



Assembly Methods
Attachment spring.

Selection table BD1, BD2, BDV

Air flow m³/h	Model	BD1		BD2		BDV
		1	2	1	2	3
	Ak(m²)	0.0034	0.0028	0.0067	0.0057	0.0063
30	X [m] NR [dB(A)] ΔPt [pa] Veff [m/s]	- <20 11.0 2.45	- <20 13.0 2.98			
40	X [m] NR [dB(A)] ΔPt [pa] Veff [m/s]	1.5 21.0 19.0 3.27	1.5 25.0 23.0 3.97			
50	X [m] NR [dB(A)] ΔPt [pa] Veff [m/s]	1.6 26.0 29.0 4.08	1.6 30.0 36.0 4.96	1.7 <20 9.0 2.07	1.6 <20 10.0 2.44	- <20 9.0 2.20
60	X [m] NR [dB(A)] ΔPt [pa] Veff [m/s]	1.8 31.0 41.0 4.90	1.7 35.0 52.0 5.95	1.9 <20 12.0 2.49	1.7 21.0 14.0 2.92	1.8 <20 13.0 2.65
70	X [m] NR [dB(A)] ΔPt [pa] Veff [m/s]	2.0 35.0 55.0 5.72	1.9 39.0 71.0 6.94	2.1 23.0 16.0 2.90	1.8 25.0 19.0 3.41	1.9 23.0 17.0 3.09
80	X [m] NR [dB(A)] ΔPt [pa] Veff [m/s]	2.3 38.0 71.0 6.54	2.1 42.0 93.0 7.94	2.3 26.0 21.0 3.32	1.9 29.0 25.0 3.90	2.0 26.0 22.0 3.53
90	X [m] NR [dB(A)] ΔPt [pa] Veff [m/s]	2.6 41.0 90.0 7.35	2.3 45.0 118.0 8.93	2.5 29.0 27.0 3.73	2.1 32.0 32.0 4.39	2.2 29.0 28.0 3.97
100	X [m] NR [dB(A)] ΔPt [pa] Veff [m/s]			2.7 32.0 33.0 4.15	2.2 34.0 39.0 4.87	2.4 31.0 34.0 4.41
110	X [m] NR [dB(A)] ΔPt [pa] Veff [m/s]			3.0 34.0 40.0 4.56	2.4 37.0 48.0 5.36	2.6 33.0 41.0 4.85
120	X [m] NR [dB(A)] ΔPt [pa] Veff [m/s]			3.2 36.0 47.0 4.98	2.6 39.0 57.0 5.85	2.9 35.0 49.0 5.29

Note: The values of the selection table are given for a length of 1m.

Annotation

- Ak [m²] - Free surface of the diffuser
- X [m] - Length airflow at speed Vk [m/s] = 0.3 m/s
- NR [dB(A)] - Noise level without the room attenuation
- ΔPt [pa] - Pressure loss
- Veff[m/s] - Air velocity in diffuser

COLUMN 1 - nozzles directed toward the residence area
COLUMN 2 - nozzles directed 50% toward the residence area and 50% toward the ceiling
COLUMN 3 - nozzles directed toward the ceiling

Order codes

Type	+	**A** (Attachment spring)	+	**ADP IZ/N**	/	**Dimensions**	/	**Nozzles color**	/	**O**
BD1 **BD2** **BDV**		No. of springs depending on the size of the diffuser		Insulated/ Non-insulated adapter		**L** (lenght) on request		White RAL 9016 Black RAL 9005		**0** Painted white RAL 9016 **R** Other RAL paints Black/natural anodised

Linear Disc Diffuser SP 2 / SP 3

SP3

Description
Aluminium linear diffuser provided with aerodynamic shaped discs.

Application
Airflow supply.

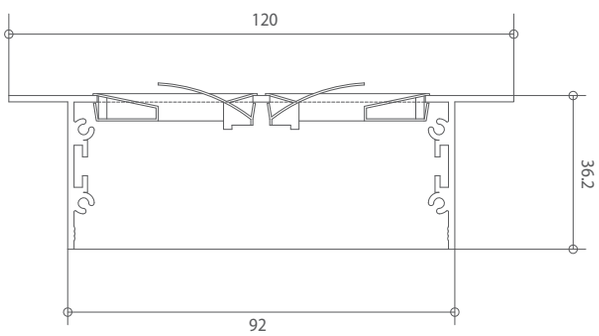
Characteristics
Depending on the technical and architectural requirements, the diffuser is provided with 2 or 3 parallel rows of discs **SP2/ SP3**
Each disk can be rotated at 360 degrees, providing unlimited possibilities to diffuse the air, without affecting the air flow, pressure loss or noise level.
Discs are available in two colors: white RAL 9016 and black RAL 9005
Long run diffusers are made out of assembled modules (maximum 3 m full length). In this case the product is delivered with joining elements between modules.
Diffuser recommended for constant or variable air flow.

Accessories
Attachment spring **A**
Insulated/ Non-insulated adapter **ADP IZ/N**

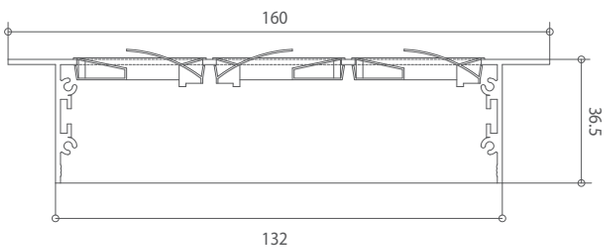
Materials and finishing
Extruded aluminum material, electrostatic field painted white RAL 9016
Other RAL colors available on request
Natural/ black anodised aluminum
Polypropylene discs.

Technical design

SP 2



SP 3



Assembly Method
Attachment spring

Selection table SP 2, SP 3

SP2-900

Air flow (m^3/h)	Length Vk[m/s]	900 0.2
105	X [m] NR [dB(A)] △Pt [pa]	6.5 20.0 50.0
125	X [m] NR [dB(A)] △Pt [pa]	7.5 25.0 75.0
150	X [m] NR [dB(A)] △Pt [pa]	9.5 30.0 100.0
180	X [m] NR [dB(A)] △Pt [pa]	12.0 35.0 150.0
210	X [m] NR [dB(A)] △Pt [pa]	14.0 40.0 200.0

SP2-1500

Air flow (m^3/h)	Length Vk[m/s]	1500 0.2
190	X [m] NR [dB(A)] △Pt [pa]	8.0 20.0 40.0
240	X [m] NR [dB(A)] △Pt [pa]	10.0 27.0 75.0
270	X [m] NR [dB(A)] △Pt [pa]	12.0 30.0 90.0
320	X [m] NR [dB(A)] △Pt [pa]	14.0 35.0 130.0
380	X [m] NR [dB(A)] △Pt [pa]	17.0 40.0 180.0

SP3-900

Air flow (m^3/h)	Length Vk[m/s]	900 0.2
110	X [m] NR [dB(A)] △Pt [pa]	5.5 22 50
130	X [m] NR [dB(A)] △Pt [pa]	6.5 27 75
160	X [m] NR [dB(A)] △Pt [pa]	8 32 110
190	X [m] NR [dB(A)] △Pt [pa]	9.5 37 175
230	X [m] NR [dB(A)] △Pt [pa]	12 42 225

SP3-1500

Air flow (m^3/h)	Length Vk[m/s]	1500 0.2
270	X [m] NR [dB(A)] △Pt [pa]	9.5 20.0 40.0
340	X [m] NR [dB(A)] △Pt [pa]	12 27 70
400	X [m] NR [dB(A)] △Pt [pa]	14 32 85
470	X [m] NR [dB(A)] △Pt [pa]	16 37 120
520	X [m] NR [dB(A)] △Pt [pa]	18 38 150

Annotation

* Vk [m/s] - Speed at which airflow length was calculated "x" [m]
* X [m] - Airflow length
* NR [dB(A)] - Noise level without room attenuation
* ΔPt [pa] - Pressure loss inside diffuser

Order codes

Type + **ADP IZ/N** + **A** (Attachment spring) / **Dimensions** / **Discs colour** / **O**

- **Type**: **SP2**, **SP3**
- **ADP IZ/N**: Insulated/non-insulated adapter
- **A** (Attachment spring): No of springs depending on grille dimensions
- **Dimensions**: L (length) on request
- **Discs colour**: White RAL 9016, Black RAL 9005
- **O**: **0** Painted white RAL 9016; **R** Painted RAL…; Natural/black anodised

Swirl Diffuser SWR

Description
Swirl jet diffuser with radial fixed blades **SWR**.

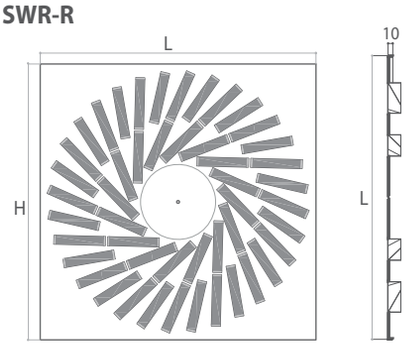
Application
Supply/ exhaust of the air flow

Characteristics
Recommended for rooms between 2.60 to 4.00 meters in height.
Depending on the technical and architectural requirements, the diffuser is produced with round or rectangular front **R/C** .
Blade design, radial layout and swirl action, result in a high level induction rate.
The diffuser should be embedded into the ceiling to ensure a stable horizontal discharge of air.
The diffuser is supplied with seals, screw fastening and masking plastic lid.
Blades are available in two colors: white RAL 9016 and black RAL9005 .

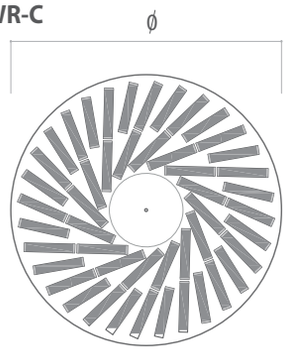
Accessories
Insulated/Non-insulated adapter **ADP IZ/N.**

Materials and finishing
Pickled steel sheet electrostatic field painted white RAL 9016.
Other RAL colors available on request.

Technical design
SWR-R



SWR-C



Assembly Method
Paneled ceiling 600 x 600 mm mounting, fitted with a central screw M6 x 100 mm

Swirl Diffuser SWR

Selection table SWR

SWR - 8 slots

Air flow (m^3/h)	Ak [m^2]	300 x 300 / D300	
		0.007	
		Exhaust	Supply
100	Veff[m/s] NR [dB(A)] ΔPt [pa]	4.0 <20 9.0	4.0 <20 10.0
125	Veff[m/s] NR [dB(A)] ΔPt [pa]	5.0 <20 14.0	5.0 <20 12.0
150	Veff[m/s] NR [dB(A)] ΔPt [pa]	6.0 22.0 18.0	6.0 <20 14.0
175	Veff[m/s] NR [dB(A)] ΔPt [pa]	6.9 28.0 28.0	6.9 <20 16.0
200	Veff[m/s] NR [dB(A)] ΔPt [pa]	7.9 30.0 30.0	7.9 22.0 17.0
250	Veff[m/s] NR [dB(A)] ΔPt [pa]	9.9 38.0 52.0	9.9 26.0 22.0
300	Veff[m/s] NR [dB(A)] ΔPt [pa]		11.9 35.0 35.0
400	Veff[m/s] NR [dB(A)] ΔPt [pa]		15.9 42.0 60.0

SWR - 16 slots

Air flow (m^3/h)	Ak [m^2]	400 x 400/D400	
		0.014	
		Exhaust	Supply
200	Veff[m/s] NR [dB(A)] ΔPt [pa]	4.0 15.0 10.0	4.0 <20 5.0
250	Veff[m/s] NR [dB(A)] ΔPt [pa]	5.0 25.0 15.0	5.0 <20 6.0
300	Veff[m/s] NR [dB(A)] ΔPt [pa]	6.0 30.0 25.0	6.0 <20.0 10.0
400	Veff[m/s] NR [dB(A)] ΔPt [pa]	7.9 38.0 40.0	7.9 26.0 15.0
500	Veff[m/s] NR [dB(A)] ΔPt [pa]	9.9 46.0 60.0	9.9 33.0 25.0
600	Veff[m/s] NR [dB(A)] ΔPt [pa]	11.9 48.0 75.0	11.9 40.0 36.0
700	Veff[m/s] NR [dB(A)] ΔPt [pa]		13.9 45.0 50.0
800	Veff[m/s] NR [dB(A)] ΔPt [pa]		15.9 50.0 65.0

SWR - 24 slots

Air flow (m^3/h)	Ak [m^2]	500 x 500 / D500	
		0.021	
		Exhaust	Supply
200	Veff[m/s] NR [dB(A)] ΔPt [pa]	2.6 <25 7.0	2.6 <20 3.0
300	Veff[m/s] NR [dB(A)] ΔPt [pa]	4.0 28.0 13.0	4.0 <20 4.0
400	Veff[m/s] NR [dB(A)] ΔPt [pa]	5.3 35.0 25.0	5.3 <20 7.0
600	Veff[m/s] NR [dB(A)] ΔPt [pa]	7.9 45.0 52.0	7.9 30.0 17.0
700	Veff[m/s] NR [dB(A)] ΔPt [pa]	9.3 50.0 70.0	9.3 33.0 20.0
800	Veff[m/s] NR [dB(A)] ΔPt [pa]	10.6 55.0 80.0	10.6 42.0 32.0
1000	Veff[m/s] NR [dB(A)] ΔPt [pa]		13.2 50.0 52.0
1200	Veff[m/s] NR [dB(A)] ΔPt [pa]		15.9 60.0 65.0

SWR - 24 slots

Air flow (m^3/h)	Ak [m^2]	595 x 595 / D595	
		0.0295	
		Exhaust	Supply
250	Veff[m/s] NR [dB(A)] ΔPt [pa]	2.4 <20 4.0	2.4 <20 3.0
400	Veff[m/s] NR [dB(A)] ΔPt [pa]	3.8 24.0 12.0	3.8 <20 4.0
500	Veff[m/s] NR [dB(A)] ΔPt [pa]	4.7 30.0 15.0	4.7 <20 6.0
600	Veff[m/s] NR [dB(A)] ΔPt [pa]	5.6 33.0 20.0	5.6 <20 8.0
700	Veff[m/s] NR [dB(A)] ΔPt [pa]	6.6 36.0 25.0	6.6 23.0 9.0
800	Veff[m/s] NR [dB(A)] ΔPt [pa]	7.5 43.0 35.0	7.5 30.0 15.0
1000	Veff[m/s] NR [dB(A)] ΔPt [pa]		9.4 35.0 20.0
1200	Veff[m/s] NR [dB(A)] ΔPt [pa]		11.3 40.0 30.0

Selection table SWR

SWR - 48 slots

Air flow (m^3/h)	Ak [m^2]	595 x 595 / D595	
		0.0390	
		Exhaust	Supply
350	Veff[m/s] NR [dB(A)] ΔPt [pa]	2.5 <20 6.0	2.5 <20 3.0
400	Veff[m/s] NR [dB(A)] ΔPt [pa]	2.8 23.0 12.0	2.8 <20 4.0
500	Veff[m/s] NR [dB(A)] ΔPt [pa]	3.6 27.0 14.0	3.6 <20 5.0
600	Veff[m/s] NR [dB(A)] ΔPt [pa]	4.3 30.0 17.0	4.3 <20 6.0
700	Veff[m/s] NR [dB(A)] ΔPt [pa]	5.0 33.0 20.0	5.0 23.0 7.5
800	Veff[m/s] NR [dB(A)] ΔPt [pa]	5.7 36.0 28.0	5.7 25.0 8.0
1000	Veff[m/s] NR [dB(A)] ΔPt [pa]	7.1 43.0 45.0	7.1 35.0 15.0
1400	Veff[m/s] NR [dB(A)] ΔPt [pa]		10.0 45.0 30.0

SWR - 24 slots

Air flow (m^3/h)	Ak [m^2]	623 x 623 / D623	
		0.0295	
		Exhaust	Supply
300	Veff[m/s] NR [dB(A)] ΔPt [pa]		2.8 <20 6.0
400	Veff[m/s] NR [dB(A)] ΔPt [pa]	3.8 <20 4.0	3.8 27.0 12.0
600	Veff[m/s] NR [dB(A)] ΔPt [pa]	5.6 <20 8.0	5.6 33.0 18.0
700	Veff[m/s] NR [dB(A)] ΔPt [pa]	6.6 25.0 10.0	6.6 35.0 23.0
800	Veff[m/s] NR [dB(A)] ΔPt [pa]	7.5 30.0 15.0	7.5 42.0 33.0
1000	Veff[m/s] NR [dB(A)] ΔPt [pa]	9.4 38.0 28.0	9.4 46.0 45.0
1200	Veff[m/s] NR [dB(A)] ΔPt [pa]	11.3 42.0 32.0	11.3 53.0 70.0
1400	Veff[m/s] NR [dB(A)] ΔPt [pa]	13.2 50.0 40.0	

SWR - 54 slots

Air flow (m^3/h)	Ak [m^2]	623 x 623 / D623	
		0.047	
		Exhaust	Supply
300	Veff[m/s] NR [dB(A)] ΔPt [pa]		1.8 <20 3.0
400	Veff[m/s] NR [dB(A)] ΔPt [pa]		2.4 <20 6.0
600	Veff[m/s] NR [dB(A)] ΔPt [pa]	3.5 <20 3.0	3.5 27.0 13.0
800	Veff[m/s] NR [dB(A)] ΔPt [pa]	4.7 25.0 6.0	4.7 35.0 23.0
1000	Veff[m/s] NR [dB(A)] ΔPt [pa]	5.9 30.0 8.5	5.9 43.0 40.0
1200	Veff[m/s] NR [dB(A)] ΔPt [pa]	7.1 35.0 12.0	7.1 46.0 50.0
1500	Veff[m/s] NR [dB(A)] ΔPt [pa]	8.9 42.0 20.0	8.9 52.0 73.0
1800	Veff[m/s] NR [dB(A)] ΔPt [pa]	10.6 50.0 30.0	

Annotation

- Ak [m^2] - Free surface
- Veff[m/s] - Air velocity inside the diffuser
- NR [dB(A)] - Noise level without the room attenuation
- ΔPt [pa] - Pressure loss

Order codes

Type	-	R/C	+	ADP IZ/N	/	Dimensions - No. of slots	/	Blades color	/	O
SWR		Rectangular / Circular		Insulated/Non-insulated adapter		300 x 300 / D300 - 8 400 x 400 / D400 - 16 500 x 500 / D500 - 24 595 x 595 / D595 - 24 595 x 595 / D595 - 48 623 x 623 / D623 - 24 623 x 623 / D623 - 54		White RAL9016 Black RAL9005		**0** Painted white RAL 9016 **R** Other RAL paints

Swirl Jet Diffuser STD/STD1

Description
Swirl jet diffuser with radial fixed blades **STD/STD1**.

Application
Supply/ exhaust of the air flow

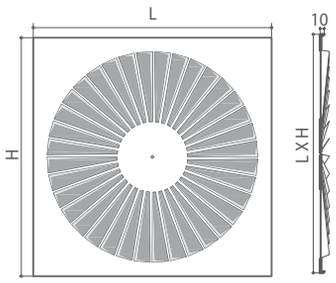
Characteristics
Recommended for rooms between 2.60 to 4.00 meters in height.
Depending on the technical and architectural requirements, the diffuser is produced with round or rectangular front **R/C.**
Blade design, radial layout and swirl action, result in a high level induction rate.
The diffuser should be embedded into the ceiling to ensure a stable horizontal discharge of air.
The diffuser is supplied with seals, screw fastening and masking plastic lid.

Accessories
Insulated/Non-insulated adapter **ADP IZ/N**.

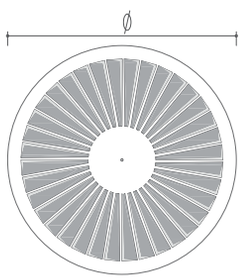
Materials and finishing
Pickled steel sheet electrostatic field painted white RAL 9016.
Other RAL colors available on request.

Technical design

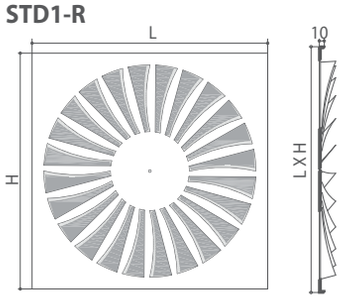
STD-R



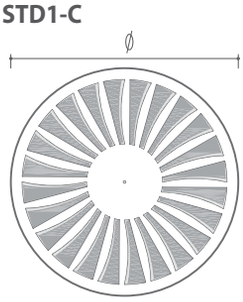
STD-C



STD1-R



STD1-C



Assembly method
Paneled ceiling 600 x 600 mm mounting, fitted with a central screw M6 x 100 mm.

Selection table STD/STD1

STD-R / STD-C

Air flow (m^3/h)	595 x 595 / D595	
	Ak [m^2]	0.0367
300	Veff[m/s] X [m] NR [dB(A)] △Pt [pa]	2.27 1.80 22.00 22.00
450	Veff[m/s] X [m] NR [dB(A)] △Pt [pa]	3.41 3.30 25.00 30.00
500	Veff[m/s] X [m] NR [dB(A)] △Pt [pa]	3.78 3.90 35.00 33.00
600	Veff[m/s] X [m] NR [dB(A)] △Pt [pa]	4.54 4.80 45.00 38.00
700	Veff[m/s] X [m] NR [dB(A)] △Pt [pa]	5.30 5.70 64.00 43.00
800	Veff[m/s] X [m] NR [dB(A)] △Pt [pa]	6.06 6.10 79.00 49.00

STD1-R / STD1-C

Air flow (m^3/h)	595 x 595 /D595	
	Ak [m^2]	0.058
300	Veff[m/s] X [m] NR [dB(A)] △Pt [pa]	1.44 2.60 - 8.00
450	Veff[m/s] X [m] NR [dB(A)] △Pt [pa]	2.16 3.80 15.00 18.00
500	Veff[m/s] X [m] NR [dB(A)] △Pt [pa]	2.39 4.20 18.00 20.00
600	Veff[m/s] X [m] NR [dB(A)] △Pt [pa]	2.87 5.00 26.10 26.00
700	Veff[m/s] X [m] NR [dB(A)] △Pt [pa]	3.35 5.70 35.00 32.00
800	Veff[m/s] X [m] NR [dB(A)] △Pt [pa]	3.83 6.70 44.70 36.00

Annotation

* Ak [m²] - Free surface
* Veff[m/s] - Air velocity inside the diffuser
* X [m] - Air jet length
* NR [dB(A)] - Noise level without the room attenuation
* ΔPt [pa] - Pressure loss

Order codes

Type	-	R/C	+	ADP IZ/N	/	O
STD **STD1**		Rectangular \ Circular		Insulated/ Non-insulated adapter		**0** Painted white RAL 9016 **R** Other RAL paints

Swirl Jet Diffuser SS

Description
Rectangular swirl jet diffuser suitable for paneled ceiling mounting.

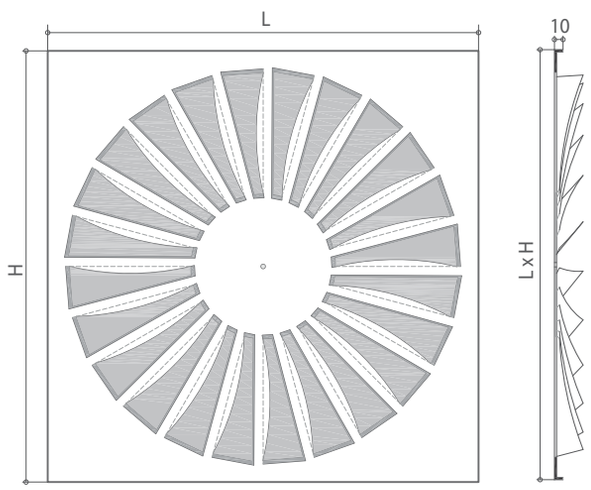
Application
Supply /exhaust of the air flow

Characteristics
Recommended for rooms between 2.60 to 4.00 m. in height
The design, radial layout and vortex action of the blades leads to a high level of induction rate.
The recommended temperature difference between supplied air and indoor air is ± 10 K.
The blades that guide the air jet are plastic white RAL 9016 or black RAL 9005.
The diffuser is supplied with gasket seals, fastening screw and plastic cover for concealment.

Accessories
Insulated/Non-insulated adapter **ADP IZ/N.**

Materials and finishing
Pickled steel sheet electrostatic field painted white RAL 9016.
Other RAL colors available on request.
Polypropylene blades.

Technical design



Assembly Methods
Paneled ceiling 600 x 600 mm mounting, fitted with a central screw M6 x 100 mm on the mounting traverse of the adapter.

Selection table SS

Air flow (m^3/h)	Dimensions	595x595	
	Ak [m^2]	0.0440	
	Vk[m/s]	0.3	0.5
360	Veff [m/s]	2.27	2.27
	X [m]	1.9	-
	NR [dB(A)]	<20	<20
	△Pt [pa]	8.0	8.0
450	Veff [m/s]	2.84	2.84
	X [m]	2.3	1.4
	NR [dB(A)]	22.0	22.0
	△Pt [pa]	12.0	12.0
540	Veff [m/s]	3.41	3.41
	X [m]	2.8	1.7
	NR [dB(A)]	30.0	30.0
	△Pt [pa]	17.0	17.0
648	Veff [m/s]	4.09	4.09
	X [m]	3.3	2.0
	NR [dB(A)]	36.0	36.0
	△Pt [pa]	24.0	24.0
720	Veff [m/s]	4.55	4.55
	X [m]	3.7	2.2
	NR [dB(A)]	39.0	39.0
	△Pt [pa]	30.0	30.0
900	Veff [m/s]	5.68	5.68
	X [m]	4.6	2.8
	NR [dB(A)]	46.0	46.0
	△Pt [pa]	47.0	47.0

Annotation

* Ak [m^2] - Free surface
* Vk - Velocity at which the air jet length was calculated "x" [m]
* Veff[m/s] - Air velocity inside diffuser
* X [m] - Air jet length
* NR [dB(A)] - Noise level without the room attenuation
* ΔPt [pa] - Pressure loss

Order codes

Type	+	ADP IZ/N	/	Dimensions	/	Control blades	/	0
SS		Insulated/ Non-insulated adapter		L x H 595*595		White RAL 9016 Black RAL 9005		**0** Painted white RAL **R** Other RAL paints

Swirl Jet Diffuser VS

Description
Swirl jet diffuser with fixed and radial arranged blades.

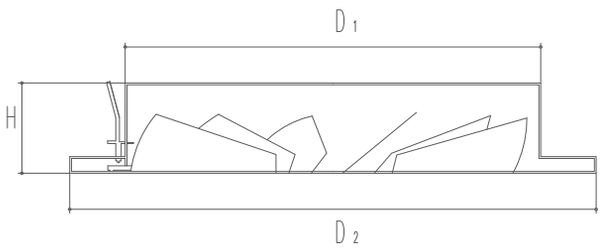
Application
Supply/ exhaust of the air.

Characteristics
Recommended for rooms of 2.60 to 4.00 m in height.
The design, radial layout and vortex action of the blades leads to a high level of induction rate.
Can be used in systems with both constant or variable air volumes.

Accessories
Insulated/Non-insulated adapter **ADP IZ/N.**

Materials and finishing
Pickled steel sheet electrostatic field painted white RAL 9016
Other RAL colors available on request

Technical design



Ø	D1	H	D2
125	123	50	158
160	158	53	197
200	198	53	241
250	248	50	295
315	313	53	364
400	398	53	450

Selection table VS

Air flow (m³/h)	Ø Ak [m²]	125 0.0025
35	Veff(m/s)	3.9
	NR [dB(A)]	17.0
	△ Pt [pa]	8.2
43	Veff(m/s)	4.8
	NR [dB(A)]	22.0
	△ Pt [pa]	13.5
50	Veff(m/s)	5.6
	NR [dB(A)]	27.0
	△ Pt [pa]	18.0
61	Veff(m/s)	6.8
	NR [dB(A)]	32.0
	△ Pt [pa]	28.0
76	Veff(m/s)	8.4
	NR [dB(A)]	37.0
	△ Pt [pa]	39.0
96	Veff(m/s)	10.7
	NR [dB(A)]	42.0
	△ Pt [pa]	60.0

Air flow (m³/h)	Ø Ak [m²]	160 0.004
43	Veff(m/s)	3.0
	NR [dB(A)]	17.0
	△ Pt [pa]	7.5
52	Veff(m/s)	3.6
	NR [dB(A)]	22.0
	△ Pt [pa]	9.0
65	Veff(m/s)	4.5
	NR [dB(A)]	27.0
	△ Pt [pa]	15.0
81	Veff(m/s)	5.6
	NR [dB(A)]	32.0
	△ Pt [pa]	23.0
100	Veff(m/s)	6.9
	NR [dB(A)]	37.0
	△ Pt [pa]	35.0
120	Veff(m/s)	8.3
	NR [dB(A)]	42.0
	△ Pt [pa]	50.0

Air flow (m³/h)	Ø Ak [m²]	200 0.007
90	Veff(m/s)	3.6
	NR [dB(A)]	17.0
	△ Pt [pa]	12.0
120	Veff(m/s)	4.8
	NR [dB(A)]	22.0
	△ Pt [pa]	18.0
140	Veff(m/s)	5.6
	NR [dB(A)]	27.0
	△ Pt [pa]	25.0
160	Veff(m/s)	6.3
	NR [dB(A)]	32.0
	△ Pt [pa]	35.0
185	Veff(m/s)	7.3
	NR [dB(A)]	37.0
	△ Pt [pa]	48.0
230	Veff(m/s)	9.1
	NR [dB(A)]	42.0
	△ Pt [pa]	70.0

Air flow (m³/h)	Ø Ak [m²]	250 0.011
190	Veff(m/s)	4.8
	NR [dB(A)]	17.0
	△ Pt [pa]	15.0
240	Veff(m/s)	6.1
	NR [dB(A)]	22.0
	△ Pt [pa]	20.0
280	Veff(m/s)	7.1
	NR [dB(A)]	27.0
	△ Pt [pa]	30.0
320	Veff(m/s)	8.1
	NR [dB(A)]	32.0
	△ Pt [pa]	40.0
410	Veff(m/s)	10.4
	NR [dB(A)]	37.0
	△ Pt [pa]	58.0
520	Veff(m/s)	13.1
	NR [dB(A)]	42.0
	△ Pt [pa]	60.0

Selection table VS

Air flow (m³/h)	Ø	315
	Ak [m²]	0.021
240	Veff(m/s)	3.2
	NR [dB(A)]	17.0
	△ Pt [pa]	9.0
270	Veff(m/s)	3.6
	NR [dB(A)]	22.0
	△ Pt [pa]	13.5
320	Veff(m/s)	4.2
	NR [dB(A)]	27.0
	△ Pt [pa]	18.0
380	Veff(m/s)	5.0
	NR [dB(A)]	32.0
	△ Pt [pa]	25.0
470	Veff(m/s)	6.2
	NR [dB(A)]	37.0
	△ Pt [pa]	35.0
530	Veff(m/s)	7.0
	NR [dB(A)]	42.0
	△ Pt [pa]	48.0

Air flow (m³/h)	Ø	400
	Ak [m²]	0.028
290	Veff(m/s)	2.9
	NR [dB(A)]	17.0
	△ Pt [pa]	10.0
330	Veff(m/s)	3.3
	NR [dB(A)]	22.0
	△ Pt [pa]	15.0
350	Veff(m/s)	3.5
	NR [dB(A)]	27.0
	△ Pt [pa]	20.0
500	Veff(m/s)	5.0
	NR [dB(A)]	32.0
	△ Pt [pa]	30.0
600	Veff(m/s)	6.0
	NR [dB(A)]	37.0
	△ Pt [pa]	42.0
720	Veff(m/s)	7.1
	NR [dB(A)]	42.0
	△ Pt [pa]	65.0

Annotation

* Ak [m²] - Free surface
* Veff[m/s] - Air velocity inside diffuser
* ΔPt [pa] - Pressure loss
* NR [dB(A)] - Noise level without the room attenuation

Order codes

Type	+	ADP IZ/N	/	Ø	/	0
VS		Insulated/ Non-insulated adapter		125 160 200 250 315 400		**0** Painted white RAL **R** Other RAL paints

Swirl Jet Diffuser VSR

Description
Swirl jet diffuser with adjustable and radial arranged blades.

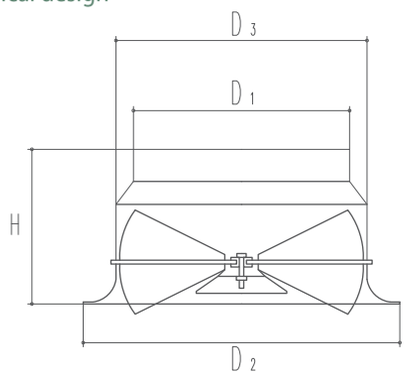
Application
Supply/ exhaust of the air.

Characteristics
Recommended for rooms over 3 m in height.
Circular and helical design promotes airflow swirling diffusion obtaining a high level of induction rate and reduces air stratification.

Accessories
Insulated/Non-insulated adapter **ADP IZ/N.**

Materials and finishing
Aluminum diffuser electrostatic field painted white RAL 9016.
Other RAL colors available on request.

Technical design



D1	D2	D3	H
250	400	300	205
315	500	400	230
400	615	465	270
500	780	565	320
630	935	665	390

Swirl Jet Diffuser VSR

Selection table VSR

Air flow (m^3/h)	Ø	250
	Ak [m^2]	0.050
250	Veff(m/s)	1.4
	NR [dB(A)]	<20
	X(m)	1.2
	△Pt [pa]	10.0
300	Veff(m/s)	1.7
	NR [dB(A)]	<20
	X(m)	1.5
	△Pt [pa]	15.0
350	Veff(m/s)	1.9
	NR [dB(A)]	22.0
	X(m)	1.6
	△Pt [pa]	20.0
400	Veff(m/s)	2.2
	NR [dB(A)]	26.0
	X(m)	1.8
	△Pt [pa]	28.0
500	Veff(m/s)	2.8
	NR [dB(A)]	33.0
	X(m)	2.8
	△Pt [pa]	45.0
600	Veff(m/s)	3.3
	NR [dB(A)]	38.0
	X(m)	3.6
	△Pt [pa]	70.0
700	Veff(m/s)	3.9
	NR [dB(A)]	44.0
	X(m)	4.6
	△Pt [pa]	91.0

Air flow (m^3/h)	Ø	315
	Ak [m^2]	0.078
500	Veff(m/s)	1.8
	NR [dB(A)]	<20
	X(m)	1.6
	△Pt [pa]	17.0
600	Veff(m/s)	2.1
	NR [dB(A)]	24.0
	X(m)	2.0
	△Pt [pa]	23.0
700	Veff(m/s)	2.5
	NR [dB(A)]	27.0
	X(m)	2.6
	△Pt [pa]	31.0
800	Veff(m/s)	2.8
	NR [dB(A)]	33.0
	X(m)	3.0
	△Pt [pa]	41.0
900	Veff(m/s)	3.2
	NR [dB(A)]	38.0
	X(m)	3.8
	△Pt [pa]	51.0
1000	Veff(m/s)	3.6
	NR [dB(A)]	43.0
	X(m)	4.2
	△Pt [pa]	69.0
1100	Veff(m/s)	3.9
	NR [dB(A)]	46.0
	X(m)	5.0
	△Pt [pa]	77.0

Air flow (m^3/h)	Ø	400
	Ak [m^2]	0.125
900	Veff(m/s)	2
	NR [dB(A)]	<20
	X(m)	1.8
	△Pt [pa]	13.0
1000	Veff(m/s)	2.2
	NR [dB(A)]	23.0
	X(m)	2.1
	△Pt [pa]	17.0
1100	Veff(m/s)	2.4
	NR [dB(A)]	26.0
	X(m)	2.5
	△Pt [pa]	21.0
1200	Veff(m/s)	2.7
	NR [dB(A)]	28.0
	X(m)	3.0
	△Pt [pa]	25.0
1400	Veff(m/s)	3.1
	NR [dB(A)]	32.0
	X(m)	3.8
	△Pt [pa]	33.0
1500	Veff(m/s)	3.3
	NR [dB(A)]	35.0
	X(m)	4.0
	△Pt [pa]	38.0
1600	Veff(m/s)	3.6
	NR [dB(A)]	38.0
	X(m)	4.5
	△Pt [pa]	44.0

Selection table VSR

Air flow (m^3/h)	Ø / Ak [m^2]	500 / 0.195
1200	Veff(m/s)	1.7
	NR [dB(A)]	30.0
	X(m)	1.8
	△Pt [pa]	17.0
1400	Veff(m/s)	2.0
	NR [dB(A)]	33.0
	X(m)	2.5
	△Pt [pa]	22.0
1600	Veff(m/s)	2.3
	NR [dB(A)]	36.0
	X(m)	3.0
	△Pt [pa]	28.0
1800	Veff(m/s)	2.6
	NR [dB(A)]	38.0
	X(m)	3.5
	△Pt [pa]	35.0
2000	Veff(m/s)	2.8
	NR [dB(A)]	41.5
	X(m)	4.0
	△Pt [pa]	42.0
2250	Veff(m/s)	3.2
	NR [dB(A)]	44.0
	X(m)	4.5
	△Pt [pa]	51.0
2500	Veff(m/s)	3.6
	NR [dB(A)]	47.0
	X(m)	5.2
	△Pt [pa]	70.0

Air flow (m^3/h)	Ø / Ak [m^2]	630 / 0.310
2100	Veff(m/s)	1.9
	NR [dB(A)]	25.0
	X(m)	3.5
	△Pt [pa]	18.0
2500	Veff(m/s)	2.2
	NR [dB(A)]	28.0
	X(m)	3.8
	△Pt [pa]	31.0
3000	Veff(m/s)	2.7
	NR [dB(A)]	35.0
	X(m)	4.8
	△Pt [pa]	38.0
3500	Veff(m/s)	3.1
	NR [dB(A)]	40.0
	X(m)	5.0
	△Pt [pa]	53.0
4000	Veff(m/s)	3.6
	NR [dB(A)]	44.0
	X(m)	5.5
	△Pt [pa]	72.0
4500	Veff(m/s)	4.0
	NR [dB(A)]	47.0
	X(m)	6.0
	△Pt [pa]	86.0
5000	Veff(m/s)	4.5
	NR [dB(A)]	50.0
	X(m)	7.0
	△Pt [pa]	110.0

Note:
The values from selection table are calculated for 45° blades inclination and 5 degrees temperature deviation.

Annotation
* Ak [m^2] - Free surface
* Veff[m/s] - Air velocity inside diffuser
* ΔPt [pa] - Pressure loss
* NR [dB(A)] - Noise level without the room attenuation
* X [m] - Air jet length

Order codes

Type	+	ADP IZ/N	/	Ø	/	0
VSR		Insulated/ Non-insulated adapter		250 315 400 500 630		**0** Painted white RAL **R** Other RAL paints

Access Grille AD

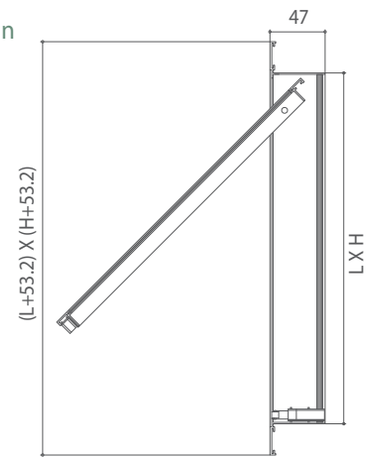
Description
Aluminum access door is used in areas where regular access is required for installation works maintenance.

Application
To access the desired maintenance area.

Characteristics
Suitable for wall or ceiling mounting.
The access door is provided with countersunk holes for mounting the visible frame .
Size 595 x 595 mm (for paneled ceiling) is provided without mounting holes.
Opening/closing the door is done by pushing a click type system.

Materials and finishing
Extruded aluminum material, electrostatic field painted white RAL 9016.
Pickled steel sheet.
Other RAL colors available on request.

Technical design



Assembly Methods
Mounting countersunk holes on the visible frame (standard)
Mounting holes on the connection frame inlet (optional)

Order codes

Type	/	Dimensions	/	0
AD		On request		0 Painted white RAL 9016 R Other RAL paints

Access Door AG

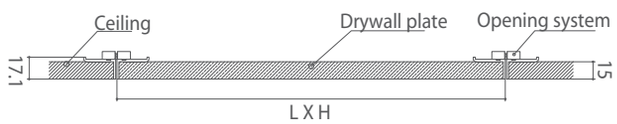
Description
Access doors are used in areas where regular access is required for work maintenance and installations.

Application
To access the desired maintenance area.

Characteristics
Access door is made of an aluminum profile with frontal drywall plate.
Opening/closing of the roof is carried out via a click type system.
Perfect incorporation with the wall or ceiling.

Materials and finishing
Natural aluminum frame
Gypsum plate

Technical design



Order codes

Type	/	Dimensions	/	L x H
AG		L X H 300 x 300 400 x 400 500 x 500 600 x 600		Other dimensions on request

Air Duct With Nozzles JR

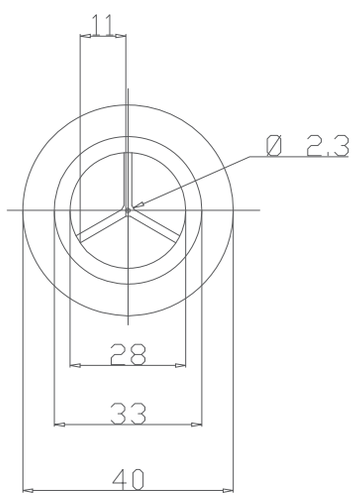
Description
Air duct with nozzles.

Application
Air supply

Characteristics
Available lenghts: 0.5m, 1.0m, 1.5m and 2.0m.
The nozzles are 32 mm in diameter, available in two colors: white RAL 9016 and black RAL 9005
The ideal temperature difference between supplied air and room air is between 2 and 4K and the velocity between 2.5 and 4.5 m/s.
Uniform and permanent air distribution.
Low pressure loss.

Materials and finishing
Galvanized steel sheet
Other RAL colors available on request.

Technical design



Selection table JR

Q (m³/h)	Length = 2000 (mm)		
	D (mm)	NR [dB(A)]	ΔPt [pa]
128	160	27	5.0
	200	26	5.0
	250	26	5.0
	315	26	5.0
No. of rows 1	400	26	5.0
	500	26	5.0
249	160	34	7.0
	200	30	7.0
	250	29	7.0
	315	29	7.0
No. of rows 2	400	29	7.0
	500	29	7.0
337	160	41	9.0
	200	32	9.0
	250	29	9.0
	315	29	9.0
No. of rows 3	400	29	9.0
	500	29	9.0
413	160	>35	10.0
	200	35	10.0
	250	30	10.0
	315	29	10.0
No. of rowsi 4	400	29	10.0
	500	29	10.0
483	160	>35	11.0
	200	>35	11.0
	250	31	11.0
	315	29	11.0
No. of rows 5	400	29	11.0
	500	29	11.0
559	160	>35	12.0
	200	>35	12.0
	250	33	12.0
	315	29	12.0
No. of rows 6	400	29	12.0
	500	29	12.0

Q (m³/h)	Length = 3000 (mm)		
	D (mm)	NR [dB(A)]	ΔPt [pa]
199	160	31	6.0
	200	29	6.0
	250	29	6.0
	315	29	6.0
No. of rows 1	400	29	6.0
	500	29	6.0
381	160	-	8.0
	200	35	8.0
	250	32	8.0
	315	31	8.0
No. of rows 2	400	31	8.0
	500	31	8.0
518	160	-	10.0
	200	40	10.0
	250	44	10.0
	315	31	10.0
No. of rows 3	400	31	10.0
	500	31	10.0
637	160	-	11.0
	200	-	11.0
	250	36	11.0
	315	32	11.0
No. of rows 4	400	31	11.0
	500	31	11.0
746	160	-	12.0
	200	-	12.0
	250	39	12.0
	315	32	12.0
No. of rows 5	400	31	12.0
	500	31	12.0
859	160	-	13.0
	200	-	13.0
	250	-	13.0
	315	33	13.0
No. of rows 6	400	31	13.0
	500	31	13.0

Q (m³/h)	Length= 4000 (mm)		
	D (mm)	NR [dB(A)]	ΔPt [pa]
270	160	36	9.0
	200	31	9.0
	250	30	9.0
	315	30	9.0
No. of rows 1	400	30	9.0
	500	30	9.0
512	160	-	11.0
	200	40	11.0
	250	34	11.0
	315	33	11.0
No. of rows 2	400	32	11.0
	500	32	11.0
698	160	-	16.0
	200	-	16.0
	250	38	16.0
	315	38	16.0
No. of rows 3	400	32	16.0
	500	32	16.0
860	160	-	20.0
	200	-	20.0
	250	42	20.0
	315	33	20.0
No. of rows 4	400	32	20.0
	500	32	20.0
1008	160	-	23.0
	200	-	23.0
	250	-	23.0
	315	36	23.0
No. of rows 5	400	33	23.0
	500	32	23.0
1159	160	-	25.0
	200	-	25.0
	250	-	25.0
	315	38	25.0
No. of rows 6	400	33	25.0
	500	32	25.0

Note

Described values are for:

* Velocity in sitting area <0.2 m/s (measured at 1.7 meters when mounting)
* Assembly height - 3m (coefficient value F1=1)
* Temperature difference - 3K-COOLING (coefficient value F2=1)

THE CALCULATED AIR FLOW WILL BE ADJUSTED BY MULTIPLYING WITH F1 AND F2 COEFFICIENTS, DEPENDING ON APPLICATION Qeff= Q *F1*F2.

Correction factor for temperature	
Temperature difference (K)	Correction coefficient F2
0	1.055
1	1.044
2	1.026
3	1
4	0.967
5	0.925
6	0.874

Correction factor for the mounting height	
Mounting height (m)	Correction coefficient F1
2.6	0.856
2.8	0.932
3	1
3.2	1.065
3.4	1.125
3.6	1.183
3.8	1.238
4	1.3

Annotation

* Dn (mm) - Duct diameter
* NR [dB(A)] - Noise level without the room attenuation
* ΔPt [pa] - Pressure loss

Order codes

Type	+	Ø	+	No. of rows	/	L	/	Nozzles color	/	0
JR		160 200 250 315 400 500		1 2 3 4 5 6		500 1000 1500 2000		WHITE RAL 9016 BLACK RAL 9005		Natural **0** Painted white RAL 9016 **R** Other RAL paints

Jet Diffuser JD

Description
Diffuser with long range air supply capacity.
Ideal for use in traffic areas, theaters, discos, museums, concert halls, etc.

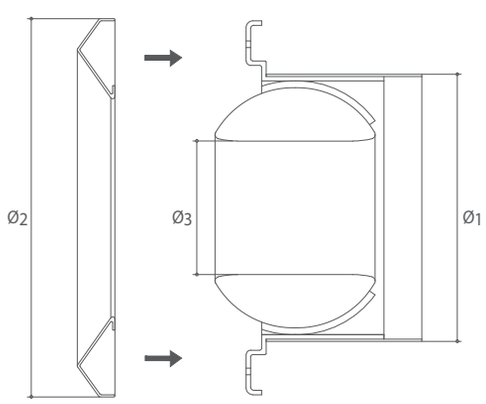
Application
Air supply

Characteristics
Low noise level.
Spherical nozzle bordered by a ring with mounting holes, allows airflow direction in the desired area.
Decorative ring for masked mounting.
Circular inlet connection for mounting to air ducts.

Accessories
Insulated/ Non-insulated adapter **ADP IZ/N**.

Materials and finishing
Aluminum nozzle and decorative ring, electrostatic field painted white RAL 9016.
Other RAL colors available on request.
Polypropylene ring (mounting)
Galvanized steel sheet connection (for air ducts)

Technical design



Ø1	Ø2	Ø3
125	170	60
160	210	80
200	255	105
250	300	135
315	380	185
400	470	230
500	595	285

Assembly Methods
Standard mounting, with screws.

Selection table JD

Air flow (m³/h)	Ø	125	160	200	250	315	400
	Ak(m²)	0.0031	0.005	0.0085	0.0135	0.0225	0.0385
100	Veff [m/s] NR [dB(A)] ΔPt [pa] X [m]	8.96 20 48 5	5.56 <20 20 4				
150	Veff [m/s] NR [dB(A)] ΔPt [pa] X [m]	13.44 32 100 7	8.33 20 40 5.50	4.90 <20 8.00 4.30			
200	Veff [m/s] NR [dB(A)] ΔPt [pa] X [m]	17.92 40 180 8.5	11.11 28 60 7	6.54 <20 25 5.5			
300	Veff [m/s] NR [dB(A)] ΔPt [pa] X [m]		16.67 40 150 8.5	9.80 28 60 6.3	6.17 <20 22 5.2		
500	Veff [m/s] NR [dB(A)] ΔPt [pa] X [m]			16.34 45 140 9	10.29 37 60 8	6.17 20 25 6	
700	Veff [m/s] NR [dB(A)] ΔPt [pa] X [m]				14.40 40 100 10	8.64 28 48 8	5.05 <20 12 6.5
1000	Veff [m/s] NR [dB(A)] ΔPt [pa] X [m]				20.58 52 230 13	12.35 40 95 12	7.22 25 28 10
1500	Veff [m/s] NR [dB(A)] ΔPt [pa] X [m]					18.52 52 200 18	10.82 35 62 14
2000	Veff [m/s] NR [dB(A)] ΔPt [pa] X [m]					24.7 >50 320 20	14.4 45 120 18

Annotation

* Ak [m²] - Free surface of the diffuser
* Veff[m/s] - Air velocity in diffuser
* X [m] - Air jet length
* ΔPt [pa] - Pressure loss
* NR [dB(A)] - Noise level without the room attenuation

Note: The values are for ΔT = + 10 K.

Order codes

Type + **ADP IZ/N** / **Dimensions** / **O**

- **Type**: **JD**
- **ADP IZ/N**: Insulated/Non-insulated adapter
- **Dimensions**: 125, 160, 200, 250, 315, 400, 500
- **O**: **0** Painted white RAL 9016; **R** Other RAL paints

Jet Diffuser with Nozzles JB

Description
Aluminum linear diffuser with individually steerable nozzles. Recommended for use in traffic areas, cinemas, museums, theaters, etc.

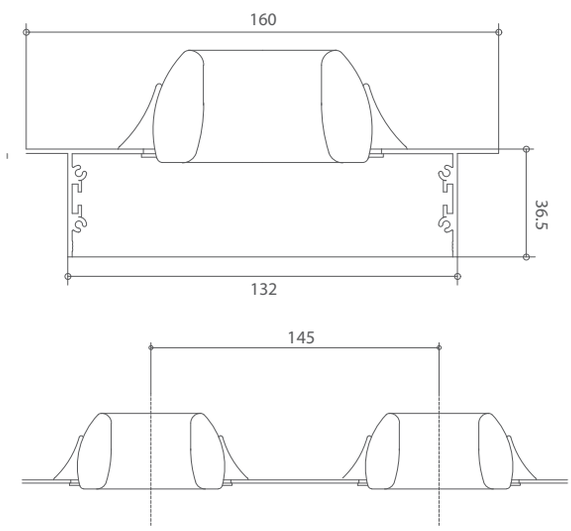
Application
Various air supply systems

Characteristics
The diffuser is produced, depending on technical and architectural requirements, with one or two rows of nozzles.
Nozzles have a diameter of 40 mm for air passage and are available in two colors: white RAL 9016 and black RAL 9005.
Due to the spherical design of the nozzle, the air flow can be directed into desired areas.
Allows unlimited airflow direction settings.

Accessories
Attachment spring **A**.
Insulated/Non-insulated adapter **ADP IZ/N**.

Materials and finishing
Extruded aluminum material, electrostatic field painted white RAL 9016
Other RAL colors available on request
Polypropylene nozzles

Technical design



Number of nozzles	Standard height
2	1row of nozzles = 132 mm 2 rows of nozzles = 294 mm
3	
4	
5	
6	
7	
8	
9	
10	

Assembly Methods
Attachment spring.

Selection table JB

Air flow m³/h	Number of nozzles	3	5	6	8	10
	Ak(m²)	0.0038	0.0063	0.0076	0.01	0.0126
120	$X_{0.3}$ $X_{0.5}$ $X_{1.0}$ [m] Veff [m/s] NR [dB(A)] ΔPt [pa]	9.5 5.7 2.8 8.77 < 20 43.0				
150	$X_{0.3}$ $X_{0.5}$ $X_{1.0}$ [m] Veff [m/s] NR [dB(A)] ΔPt [pa]	11.8 7.1 3.5 10.96 22.0 68.0				
200	$X_{0.3}$ $X_{0.5}$ $X_{1.0}$ [m] Veff [m/s] NR [dB(A)] ΔPt [pa]	15.8 9.5 4.7 14.62 29.0 121.0	12.2 7.3 3.7 8.82 < 20 43.0	11.2 6.7 3.3 7.31 < 20 30.0		
250	$X_{0.3}$ $X_{0.5}$ $X_{1.0}$ [m] Veff [m/s] NR [dB(A)] ΔPt [pa]	19.7 11.8 5.9 18.27 35.0 189.0	15.3 9.2 4.6 11.02 24.0 68.0	13.9 8.4 4.2 9.14 < 20 47.0	12.1 7.2 3.6 6.94 < 20 27.0	
300	$X_{0.3}$ $X_{0.5}$ $X_{1.0}$ [m] Veff [m/s] NR [dB(A)] ΔPt [pa]	23.7 14.2 7.1 21.93 40.0 272.0	18.3 11.0 5.5 13.23 29.0 98.0	16.7 10.0 5.0 10.96 25.0 68.0	14.5 8.7 4.3 8.33 < 20 38.0	
350	$X_{0.3}$ $X_{0.5}$ $X_{1.0}$ [m] Veff [m/s] NR [dB(A)] ΔPt [pa]	27.6 16.6 8.3 25.58 44.0 370.0	21.4 12.8 6.4 15.43 33.0 133.0	19.5 11.7 5.9 12.79 29.0 92.0	16.9 10.1 5.1 9.72 22.0 52.0	15.1 9.1 4.5 7.72 < 20 33.0
400	$X_{0.3}$ $X_{0.5}$ $X_{1.0}$ [m] Veff [m/s] NR [dB(A)] ΔPt [pa]		24.4 14.7 7.3 17.64 36.0 174.0	22.3 13.4 6.7 14.62 32.0 121.0	19.3 11.6 5.8 11.11 26.0 68.0	17.3 10.4 5.2 8.82 21.0 43.0
500	$X_{0.3}$ $X_{0.5}$ $X_{1.0}$ [m] Veff [m/s] NR [dB(A)] ΔPt [pa]		> 30 18.3 9.2 22.05 42.0 272.0	27.9 16.7 8.4 18.27 38.0 189.0	24.1 14.5 7.2 13.89 32.0 106.0	21.6 13.0 6.5 11.02 27.0 68.0
600	$X_{0.3}$ $X_{0.5}$ $X_{1.0}$ [m] Veff [m/s] NR [dB(A)] ΔPt [pa]		> 30 22.0 11.0 26.46 47.0 391.0	> 30 20.1 10.0 21.93 43.0 272.0	29.0 17.4 8.7 16.67 37.0 153.0	25.9 15.6 7.8 13.23 32.0 98.0
700	$X_{0.3}$ $X_{0.5}$ $X_{1.0}$ [m] Veff [m/s] NR [dB(A)] ΔPt [pa]			> 30 23.4 11.7 25.58 47.0 370.0	> 30 20.3 10.1 19.44 41.0 208.0	> 30 18.1 9.1 15.43 36.0 133.0
800	$X_{0.3}$ $X_{0.5}$ $X_{1.0}$ [m] Veff [m/s] NR [dB(A)] ΔPt [pa]				> 30 23.2 11.6 22.22 44.0 272.0	> 30 20.7 10.4 17.64 39.0 174.0
1000	$X_{0.3}$ $X_{0.5}$ $X_{1.0}$ [m] Veff [m/s] NR [dB(A)] ΔPt [pa]					> 30 25.9 13.0 22.05 45.0 272.0

Correction table for more rows

Number of rows	1	2
FNR	0	3
Fx	1	1.4

The values "X" will be corrected by multiplying with the coefficients from the table and "NR" through addition.

Annotation

*Ak [m²] - Free surface of the diffuser
*Veff[m/s] - Air velocity in diffuser
*NR [dB(A)] - Noise level without the room attenuation
*ΔPt [pa] - Pressure loss

X0.3 - Air jet length at speed Vk = 0.3 m/s
X0.5 - Air jet length at speed Vk = 0.5 m/s
X1.0 - Air jet length at speed Vk = 1.0 m/s

Order codes

Type - **Number of rows** + **ADP IZ/N** / **Number of nozzles** / **Color of nozzles** / **O**

- **JB**
- 1, 2
- Insulated/Non-insulated adapter
- On request
- White RAL 9016, Black RAL 9005
- **0** Painted white RAL 9016, **R** Other RAL paints

Circular Valve EV

Description
Circular steel valve with adjustable central cone for exhaust of the air, provided with mounting frame **EV**.

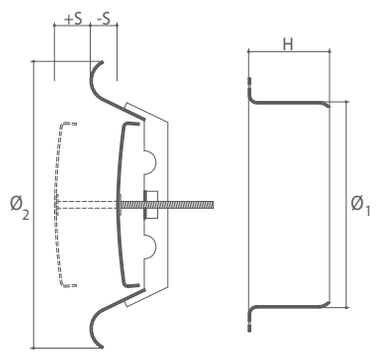
Application
Exhaust the airflow

Characteristics
Valve is provided with gasket seal and mounting frame
The adjustable center cone allows adjustment of air flow
The mounting frame is provided with holes for fixing

Materials and finishing
Steel electrostatic field painted in RAL9016 white
Other RAL colors available on request.
Mounting frame made of galvanized metal sheet

Assembly Methods
Screws on the mounting frame.

Technical design



Tip	Ø1	Ø2	H
EV	100	140	45
	125	165	47
	150	200	47
	160	210	50
	200	250	50

Selection table EV

Air flow	Dimension (mm)	100					125							160						200							
(m^3/h)	S(mm)	-14	-9	-5	0	5	-15	-12	-9	-6	-3	0	5	-18	-15	-10	-5	0	6	-23	-20	-15	-10	-5	0	10	20
20	NR [dB(A)]	24.0	<25				<25	<25																			
	△ Pt [pa]	200.0	25.0				23.0	12.0																			
30	NR [dB(A)]		<25	<25			<25	<25	<25																		
	△ Pt [pa]		60.0	20.0			50.0	25.0	15.0																		
50	NR [dB(A)]		40.0	<25	<25	<25	30.0	<25	<25	<25	<25	<25		<30	<25	<30				<30	<25	<30					
	△ Pt [pa]		150.0	60.0	25.0	12.0	145.0	70.0	40.0	25.0	12.0	10.0		70.0	30.0	12.0				50.0	28.0	12.0					
70	NR [dB(A)]		>40	35.0	<30	<30	40.0	30.0	<25	<25	<25	<25	<25	35.0	<30	<30	<30			30.0	<30	<30	<30				
	△ Pt [pa]		300.0	110.0	50.0	28.0	260.0	110.0	70.0	50.0	30.0	23.0	12.0	125.0	70.0	28.0	15.0			100.0	52.0	22.0	12.0				
100	NR [dB(A)]			>40	37.0	30.0		42.0	37.0	30.0	<35	<30	<25	>40	35.0	<30	<30	<30		37.0	33.0	<30	<30	<30			
	△ Pt [pa]			300.0	100.0	55.0		220.0	150.0	90.0	65.0	40.0	28.0	280.0	125.0	52.0	30.0	18.0		175.0	110.0	40.0	25.0	15.0			
200	NR [dB(A)]					>40					50.0	45.0	40.0			>40	38.0	33.0	28.0			40.0	35.0	30.0	<30	<30	<30
	△ Pt [pa]					200.0					250.0	170.0	110.0			270.0	140.0	70.0	40.0			150.0	90.0	55.0	20.0	20.0	10.0
250	NR [dB(A)]																>40	40.0	34.0			>40	40.0	36.0	31.0	<30	<30
	△ Pt [pa]																200.0	120.0	70.0			250.0	140.0	80.0	52.0	30.0	15.0
300	NR [dB(A)]																>40	>40	40.0				>40	41.0	36.0	40.0	<30
	△ Pt [pa]																300.0	200.0	110.0				200.0	130.0	75.0	45.0	25.0
500	NR [dB(A)]																								>40	42.0	37.0
	△ Pt [pa]																								200.0	110.0	55.0
700	NR [dB(A)]																									>40	>40
	△ Pt [pa]																									210.0	110.0

Annotation

* S [mm] - Disk opening
* ΔPt [pa] - Pressure loss
* NR [dB(A)] - Noise level without the room attenuation

Order codes

Type	/	Diameter	/	O
EV		100 125 150 160 200		**0** Painted white RAL 9016 **R** Other RAL paints

Circular Valve SV

Description
Circular steel valve with adjustable central cone for supply of the air, provided with mounting frame **SV**.

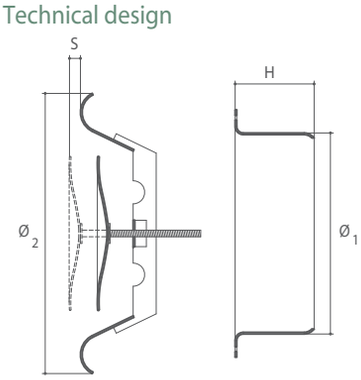
Application
Supply the airflow

Characteristics
Valve is provided with gasket seal and mounting frame
The adjustable center cone allows adjustment of air flow
The mounting frame is provided with holes for fixing.

Materials and finishing
Steel electrostatic field painted in RAL9016 white
Other RAL colors available on request.
Mounting frame made of galvanized metal sheet

Assembly Methods
Screws on the mounting frame

Technical design



Tip	Ø1	Ø2	H
SV	100	140	48
	125	165	48
	150	200	44
	160	210	50
	200	250	45

Selection table SV

Air flow (m^3/h)	Dimension (mm)	100			125				160					200			
	S(mm)	4	9	12	3	7	12	15	4	9	12	15	20	9	12	15	20
20	X [m]	0.65			0.6												
	NR [dB(A)]	<10			<25												
	△Pt [pa]	15.0			20.0												
30	X [m]	1.0			0.8												
	NR [dB(A)]	<10			<25												
	△Pt [pa]	28.0			37.0												
50	X [m]	2.0	0.8	0.8	1.5	1.3			1.4								
	NR [dB(A)]	32.0	<30	<25	30.0	<25			<25								
	△Pt [pa]	83.0	20.0	5.0	105.0	25.0			38.0								
70	X [m]	2.8	1.7	1.3	2.2	1.7	1.1		2.0	1.3							
	NR [dB(A)]	>35	30.0	<30	>35	<25	<25		25.0	<25							
	△Pt [pa]	130.0	32.0	20.0	190.0	40.0	15.0		70.0	15.0							
100	X [m]		2.5	1.8		2.5	1.6	1.2	3.0	1.9	1.5	1.4		2.2	1.5		
	NR [dB(A)]		>35	40.0		30.0	<25	<25	32.0	<25	<25	<25		<25	<25		
	△Pt [pa]		75.0	40.0		80.0	30.0	22.0	145.0	35.0	20.0	13.0		20.0	12.0		
150	X [m]		>3	>3		>3	2.5	2.0		3.0	2.2	2.0	1.5	3.1	2.2	1.8	1.2
	NR [dB(A)]		>35	>40		>35	37.0	35.0		33.0	<25	<25	<25	<25	23.0	<25	<25
	△Pt [pa]		160.0	100.0		180.0	68.0	48.0		72.0	43.0	28.0	18.0	43.0	25.0	15.0	10.0
200	X [m]		>3	>3		>3	>3	2.9		>4	3.5	2.9	2.2	4.7	3.1	2.6	2.2
	NR [dB(A)]		>35	>40		>35	>35	40.0		40.0	35.0	30.0	<30	35.0	30.0	<25	<25
	△Pt [pa]		280.0	155.0		300.0	120.0	70.0		125.0	73.0	48.0	30.0	70.0	48.0	27.0	18.0
250	X [m]									>4	>4	3.9	3.0	>5	4.0	3.6	3.0
	NR [dB(A)]									>40	40.0	35.0	33.0	40.0	37.0	30.0	<25
	△Pt [pa]									195.0	125.0	68.0	45.0	100.0	78.0	40.0	25.0
300	X [m]											>4	3.8	>5	4.8	4.2	3.5
	NR [dB(A)]											40.0	38.0	>40	45.0	36.0	30.0
	△Pt [pa]											100.0	65.0	150.0	120.0	60.0	35.0
500	X [m]															>5	>5
	NR [dB(A)]															>40	42.0
	△Pt [pa]															180.0	100.0
700	X [m]																>5
	NR [dB(A)]																>40
	△Pt [pa]																160.0

Annotation

* S [mm] - Disk opening
* X [m] - Air jet length at the velocity Vk = 0.2 m/s
* ΔPt [pa] - Pressure loss
* NR [dB(A)] - Noise level without the room attenuation

Order codes

Type	/	Diameter	/	O
SV		100 125 150 160 200		**0** Painted white RAL 9016 **R** Other RAL paints

Circular Valve VN/VLB/VLA

Description
Stainless steel circular valve for exhaust/supply of the air flow.

Application
Supply/ exhaust of the airflow

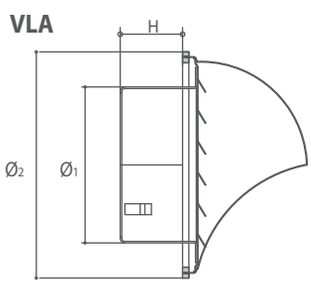
Characteristics
VN - Circular valve with adjustable central cone for air flow adjustments.
- Mounting frame with mounting holes and clips provided
VLA - Exterior valve with blades, 1.5 x 1.5 mm mesh and sealing gasket.
- Clips on the connector and mounting holes on the outer frame
VLB - Exterior valve with 10 x 10 mm mesh and sealing gasket.
- Clips on the connector and mounting holes on the outer frame.

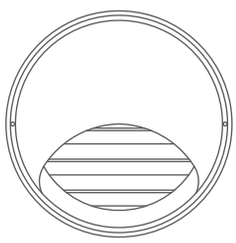
Materials and finishing
Stainless steel, natural finish.

Assembly Methods
Mounting with screw and/or using the clamp located at the connection/framework.

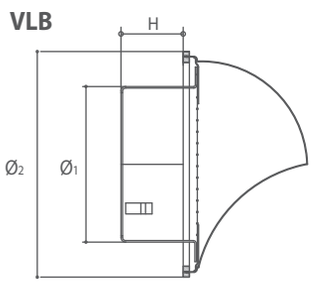
Technical design

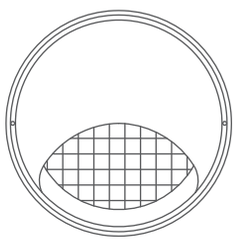
VLA





VLB





Type	Ø1	Ø2	H
VN	100	140	50
	125	170	50
	150	218	63
	160	218	63
	200	298	80
VLA	100	147	50
	125	187	50
	150	210	61
	160	210	64
	200	270	73
VLB	100	147	50
	125	187	50
	150	210	61
	160	210	64
	200	270	73

Selection table VN, VLA, VLB

VN						
Air flow (m^3/h)	Dimensions (mm)	100	125	150	160	200
30	X [m]	0.21				
	ΔPt [pa]	14.00				
40	X [m]	0.42	0.32			
	ΔPt [pa]	27.00	12.00			
60	X [m]	0.75	0.45	0.30	0.20	
	ΔPt [pa]	50.00	25.00	3.00	2.00	
80	X [m]	1.10	0.70	0.50	0.43	
	ΔPt [pa]	95.00	55.00	1.00	9.00	
130	X [m]		1.50	1.00	0.95	1.2
	ΔPt [pa]		125.00	25.00	25.00	45.00
190	X [m]			1.80	1.70	1.7
	ΔPt [pa]			56.00	50.00	60.00
220	X [m]					2.10
	ΔPt [pa]					120.00

VLA				
Air flow (m^3/h)	Dimensions (mm)	100	125	150
30	Veff[m/s]	1.00	0.80	
	ΔPt [pa]	6.50	5.00	
60	Veff[m/s]	2.00	1.40	1.00
	ΔPt [pa]	23.00	18.00	7.00
80	Veff[m/s]	3.00	1.90	1.30
	ΔPt [pa]	45.00	27.00	13.00
110	Veff[m/s]	4.00	2.50	1.80
	ΔPt [pa]	90.00	47.00	23.00
140	Veff[m/s]	5.00	3.20	2.20
	ΔPt [pa]	145.00	72.00	34.00
170	Veff[m/s]		3.90	2.70
	ΔPt [pa]		125.00	50.00
180	Veff[m/s]		4.00	2.90
	ΔPt [pa]		137.00	58.00
190	Veff[m/s]		4.20	3.00
	ΔPt [pa]		150.00	67.00
220	Veff[m/s]		5.00	3.50
	ΔPt [pa]		210.00	85.00
250	Veff[m/s]		5.70	4.00
	ΔPt [pa]		240.00	105.00
270	Veff[m/s]		6.00	4.30
	ΔPt [pa]		270.00	115.00
320	Veff[m/s]			5.00
	ΔPt [pa]			170.00
380	Veff[m/s]			6.00
	ΔPt [pa]			250.00

VLB				
Air flow (m^3/h)	Dimensions (mm)	100	125	150
30	Veff[m/s]	1.00	0.80	
	ΔPt [pa]	4.5	5.00	
60	Veff[m/s]	2.00	1.40	1.00
	ΔPt [pa]	15.00	4.00	3.00
80	Veff[m/s]	3.00	1.90	1.30
	ΔPt [pa]	29.00	8.50	5.00
110	Veff[m/s]	4.00	2.50	1.70
	ΔPt [pa]	55.00	15.00	16.00
140	Veff[m/s]	5.00	3.20	2.20
	ΔPt [pa]	90.00	25.00	24.00
170	Veff[m/s]	6.00	3.90	2.70
	ΔPt [pa]	135.00	37.00	37.00
180	Veff[m/s]		4.00	2.90
	ΔPt [pa]		40.00	43.00
190	Veff[m/s]		4.20	3.00
	ΔPt [pa]		45.00	48.00
220	Veff[m/s]		5.00	3.50
	ΔPt [pa]		65.00	63.00
250	Veff[m/s]		5.70	4.00
	ΔPt [pa]		70.00	73.00
270	Veff[m/s]		6.00	4.30
	ΔPt [pa]		85.00	80.00
320	Veff[m/s]			5.00
	ΔPt [pa]			125.00
380	Veff[m/s]			6.00
	ΔPt [pa]			180.00

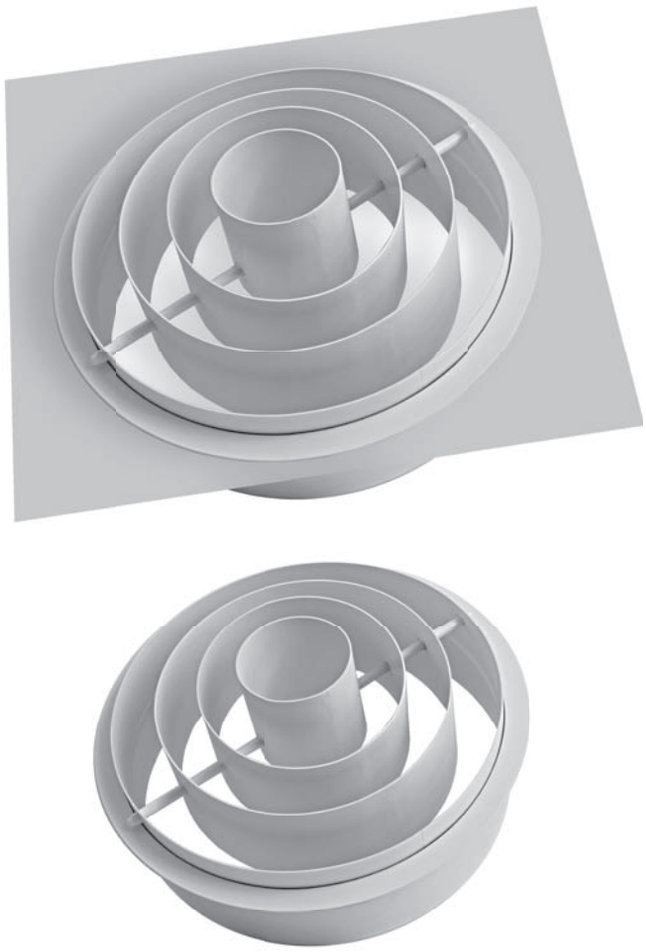
Annotation

* Veff[m/s] - Air velocity in valve
* ΔPt [pa] - Pressure loss
* X [m] - Length air jet at speed Vk [m/s] = 0.5 m/s

Order codes

Type	/	Dimensions
		Ø
VN VLA VLB		100 125 150 160 200

Special Jet Diffuser SJD



Description
Diffuser with long range air supply capacity. Ideal for use in traffic areas, theaters, discos, museums, concert halls, etc.

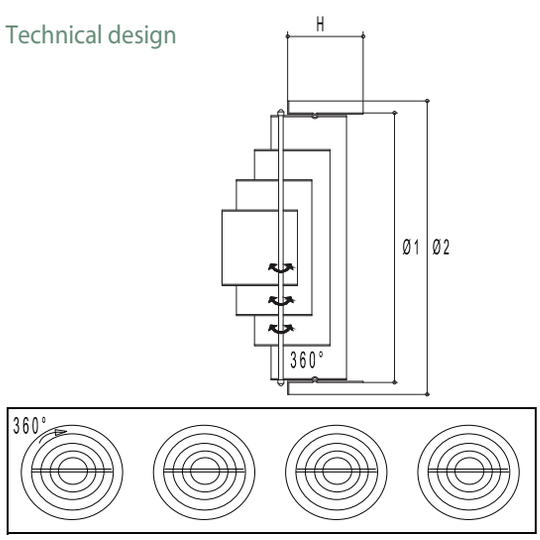
Application
Air supply

Characteristics
Concentric nozzles that allow airflow direction in the desired areas.
Rectangular or circular air duct mounting.
Mounting holes on flange.
Horizontal and vertical 360° rotation.
Produced in an assembly of 1, 2, 3, 4 pieces.
Manually adjustable.

Accessories
Insulated/ Non-insulated adapter **ADP IZ/N.**

Materials and finishing
Aluminum nozzles and galvanized steel sheet connection, electrostatic field painted white RAL 9016.
Other RAL colors available on request.

Technical design



Ø1	Ø2	H
200	240	100
250	290	
300	340	
350	390	

Assembly Methods
Standard assembly with screws.

Selection table SJD

Air flow	Ø	200				250				300				350			
(m^3/h)	Elements no.	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4
342	NR [dB(A)]	—															
	△Pt [pa]	15.0															
	X [m]	8															
504	NR [dB(A)]	25															
	△Pt [pa]	35.0															
	X [m]	9															
684	NR [dB(A)]	31	—														
	△Pt [pa]	60	15														
	X [m]	12	8														
828	NR [dB(A)]	39	22			—				—							
	△Pt [pa]	90	25			30				15							
	X [m]	15	9			12				11							
1008	NR [dB(A)]	43	23			29				—							
	△Pt [pa]	125	35			45				25							
	X [m]	18	11			15				14							
1512	NR [dB(A)]		40	29	22	41	—			34	22						
	△Pt [pa]		75	35	20	95	30			50	15						
	X [m]		15	12	9	21	12			18	11						
2052	NR [dB(A)]		45	37	31	48	32	—		42	24	—		35	—		
	△Pt [pa]		125	60	35	190	50	20		90	25	10		45	10		
	X [m]		21	15	12	28	15	12		24	14	11		20	12		
3060	NR [dB(A)]			48	43		46	31	—	52	38	25		46	29	—	
	△Pt [pa]			125	75		95	45	30	200	50	25		115	25	19	
	X [m]			23	17		24	17	14	37	21	15		30	18	12	
3744	NR [dB(A)]			53	47		52	40	31		44	31	27	52	38	24	
	△Pt [pa]			190	105		145	65	40		75	35	20	150	40	20	
	X [m]			28	21		29	21	17		26	18	14	37	21	15	
4428	NR [dB(A)]				52		57	45	38		47	39	30	57	43	29	—
	△Pt [pa]				150		200	90	50		105	50	25	200	55	25	15
	X [m]				24		34	24	18		29	21	17	43	26	18	15
5940	NR [dB(A)]				58			54	46		55	46	39		49	40	32
	△Pt [pa]				275			155	90		190	80	40		100	45	25
	X [m]				32			32	24		37	28	21		34	24	18
7632	NR [dB(A)]							61	53			51	48		55	47	40
	△Pt [pa]							250	150			140	80		155	60	40
	X [m]							40	32			35	28		40	32	24
10260	NR [dB(A)]								62			57	54		65	53	49
	△Pt [pa]								250			250	145		250	125	70
	X [m]								43			49	37		58	41	32
13608	NR [dB(A)]												61			63	56
	△Pt [pa]												250			210	125
	X [m]												55			55	43

Annotation

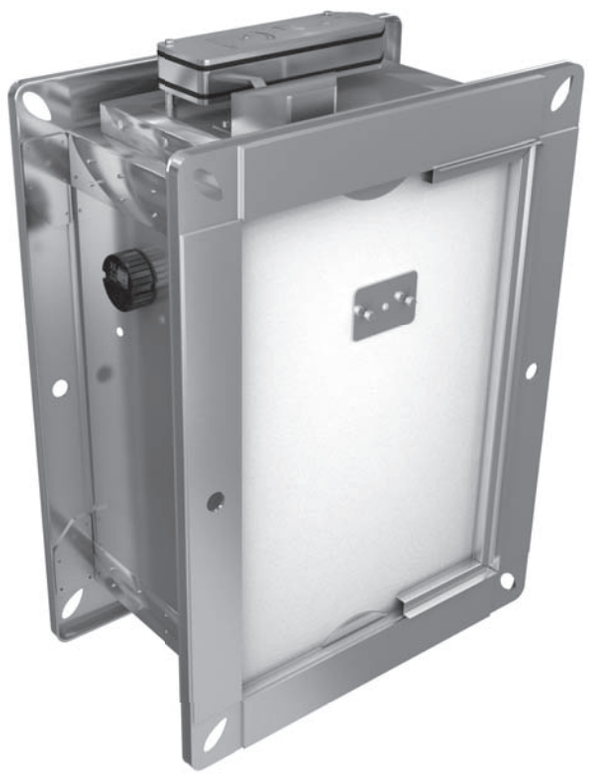
* X [m] - Air flow length for Vk=0.25m/s speed
* ΔPt [pa] - Pressure loss
* NR [dB(A)] - Noise level for 8dB room attenuation

Note: The values in the table are for ΔT = + 10 K.

Order codes

Type	/	Dimensions	/	O
SJD		200 250 300 350		**0** Painted white RAL 9016 **R** Other RAL paints

Rectangular Fire Damper SFD



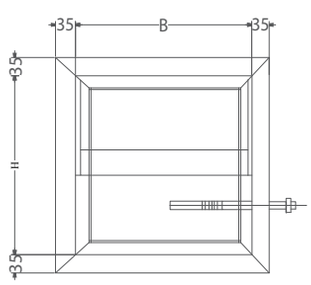
Description
Rectangular fireproof damper designed for automatic isolation of ventilation and air conditioning systems compartments.
Suitable for mounting into concrete walls and ceilings.

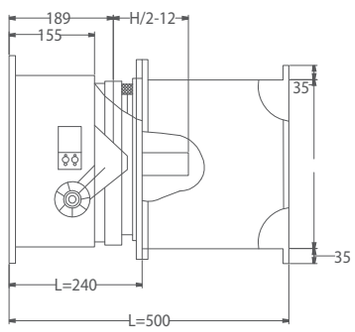
Application
For closing and sealing the ventilation ducts in case of fire.

Characteristics
Drive system with fusible element.
Automatic closing fire damper fusible element is determined to be active at temperatures> 72 ° C.
Various classes of fire resistance (RF60, RF90, RF120).
Optional contact signal for start/ end of the cycle.
Alternatively, an electric actuator can be used.

Materials and finishing
Galvanized sheet steel case
Special insulated damper blade

Technical design





Selection table SFD

Veff (m/s)	H (mm) / B=400 (mm)	200	250	300	350	400	450	500	600	700	800
2.0	NR [db(A)]	33.00	31.00	28.00	27.00	26.00	25.00	25.50	25.00	24.50	24.00
	ΔPt[pa]	18.00	8.50	6.00	4.50	3.80	3.10	2.80	2.20	1.90	1.70
2.5	NR [db(A)]	38.00	34.00	33.00	32.50	31.50	31.00	30.50	30.00	29.50	29.00
	ΔPt[pa]	28.00	14.00	9.20	7.00	5.80	4.80	4.20	3.50	2.90	2.70
3.0	NR [db(A)]	42.50	40.00	38.00	36.80	36.00	35.80	35.50	35.00	34.50	34.00
	ΔPt[pa]	40.00	19.00	14.00	10.00	8.00	7.00	6.00	5.00	4.30	3.80
5.0	NR [db(A)]	53.00	51.80	50.70	49.00	48.00	47.80	47.20	46.80	46.70	46.50
	ΔPt[pa]	110.00	55.00	38.00	28.00	22.00	19.50	17.00	14.00	12.00	11.00
7.0	NR [db(A)]	61.00	58.00	57.20	56.70	56.50	56.10	55.90	55.70	55.20	55.00
	ΔPt[pa]	150.00	110.00	70.00	55.00	45.00	38.00	32.00	28.00	22.00	20.00

Aerodynamic and acoustic corrections based on other lengths of multispot B												
B(mm)	200	250	300	350	400/450	500	600	700/800	900	1000/1100	1200-1400	1500
ΔPt	1.7	1.5	1.3	1.1	1.1	0.9	0.8	0.7	0.6	0.6	0.6	0.5
NR	-1	-1	0	0	0	0	1	2	2	3	4	5

ΔPt [pa] = Correction coefficient x Δ Pt (B=400 mm)
NR[dB(A)] = Correction coefficient + NR (B=400 mm)

Annotation

* Veff[m/s] - Air velocity in damper
* ΔPt [pa] - Pressure loss
* NR [dB(A)] - Noise level without the room attenuation

Order codes

Type	/	Dimensions	
		B	H
SFD		200	200
		250	250
		300	300
		350	350
		400	400
		450	450
		500	500
		600	600
		700	700
		800	800
		900	
		1000	
		1100	
		1200	
		1300	
		1400	
		1500	

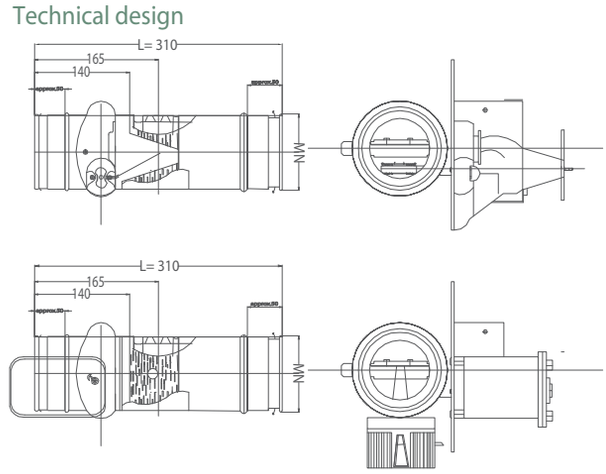
Round Fire Damper RFD

Description
Round fireproof damper designed for automatic isolation of ventilation and air conditioning systems compartments. Suitable for mounting into concrete walls and ceilings.

Application
For closing and sealing the ventilation ducts in case of fire.

Characteristics
Drive system with fusible element.
Automatic closing fire damper fusible element is determined to be active at temperatures> 72 ° C.
Various classes of fire resistance (RF60, RF90, RF120).
Optional contact signal for start/ end of the cycle.
Alternatively, an electric actuator can be used.

Materials and finishing
Galvanized sheet steel case
Special insulated damper blade

Technical design


Selection table RFD

Veff (m/s)	Diameter (mm)	100	125	160	200
4	NR [dB(A)] ΔPt [pa]	34.00 18.20	28.00 10.60	25.00 8.60	20.00 5.80
6	NR [dB(A)] ΔPt [pa]	41.00 41.00	38.00 23.80	34.00 19.40	30.00 13.00
8	NR [dB(A)] ΔPt [pa]	48.00 73.00	45.00 42.20	42.00 34.60	38.00 23.00
10	NR [dB(A)] ΔPt [pa]	53.00 114.00	51.00 66.00	47.00 34.60	45.00 36.00
12	NR [dB(A)] ΔPt [pa]	57.00 164.20	55.00 95.00	52.00 77.80	50.00 51.80
15	NR [dB(A)] ΔPt [pa]	63.00 256.50	60.00 148.50	58.00 121.50	55.00 81.00

Annotation

* Veff[m/s] - Air velocity in damper
* ΔPt [pa] - Pressure loss
* NR [dB(A)] - Noise level without the room attenuation

Order codes

Type / **Dimensions**

RFD

100
125
160
200

Air Flow Control Damper VCD

Description
Aluminum rectangular air flow control damper with opposable blades.

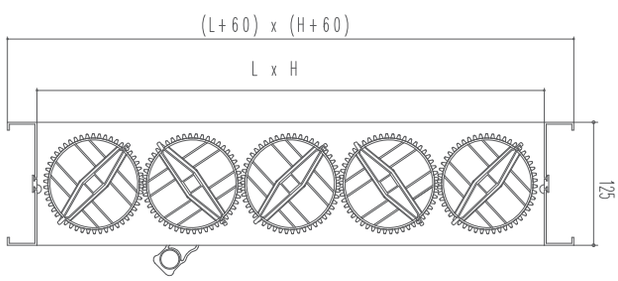
Application
Air flow adjustment in ventilation systems.

Characteristics
The standard control damper is provided with a drive lever and blocking system, an electric drive (actuator) can be delivered on request.
Actuator will be selected according to the size of the damper, voltage and other technical details mentioned in the project (return spring, 0-10V control, etc.)
Control dampers can be made, on request, without drive system - in this case it will be considered the diameter of the blade's drive shaft for coupling the desired actuator. (eg. Control damper without drive lever, prepared for actuator)
Drive lever / actuator operates the opening / closing of the blades through 12 X 12mm and 150mm length shaft .
Aerodynamic blades are a main characteristic of the control damper.
The control damper is provided with gasket seals on both the blades and the frame.
Blade is 100mm height and cannot be adjusted, having the following two cases:
* Case 1: 200 x 200 mm - the control damper consists of two adjustable blades and the final measurements will become 200 x 212 mm [W (H + 12 mm)].
* Case 2: 200 x 150 mm - the control damper consists of one adjustable blade and the difference will be covered with a fixed blade (final measurements - 200 x 150 mm).

Accessories
Drive actuator

Materials and finishing
Natural aluminum material for frame and blades
Polypropylene sprocket wheels

Technical design



Assembly Methods
Screws on the mounting flange.

Selection table VCD

Air flow	Size [LxH]	800x500		1000x500		1500x500		800x800		1000x800		1200x800		1500x800		1200x1200		1500x1200		1800x1200		2000x1200		2200x1200	
(m^3/h)	α [°]	0	45	0	45	0	45	0	45	0	45	0	45	0	45	0	45	0	45	0	45	0	45	0	45
1000	Veff [m/s]	1.3	3.7																						
	△Pt [pa]	1.3	8.8																						
1500	Veff [m/s]	1.9	5.5																						
	△Pt [pa]	2.5	19.4																						
2000	Veff [m/s]	2.5	7.4																						
	△Pt [pa]	4.2	34.3																						
2500	Veff [m/s]	3.1	9.2																						
	△Pt [pa]	6.5	53.5																						
3000	Veff [m/s]	3.8	11.1	1.9	5.5																				
	△Pt [pa]	9.2	76.8	2.5	19.4																				
3500	Veff [m/s]	4.4	12.9	2.2	6.5																				
	△Pt [pa]	12.4	104.5	3.3	26.3																				
4000	Veff [m/s]	5.0	14.8	2.5	7.4																				
	△Pt [pa]	16.1	136.4	4.2	34.3																				
4500	Veff [m/s]	5.7	16.6	2.8	8.3	1.9	5.5																		
	△Pt [pa]	20.3	172.5	5.3	43.4	2.5	19.4																		
5000	Veff [m/s]	6.3	18.4	3.1	9.2	2.1	6.2																		
	△Pt [pa]	25.0	212.9	6.5	53.5	3.0	23.9																		
5500	Veff [m/s]			3.5	10.1	2.3	6.8																		
	△Pt [pa]			7.8	64.6	3.6	28.9																		
6000	Veff [m/s]			3.8	11.1	2.5	7.4																		
	△Pt [pa]			9.2	76.8	4.2	34.3																		
6500	Veff [m/s]			4.1	12.0	2.7	8.0																		
	△Pt [pa]			10.7	90.1	4.9	40.2																		
7000	Veff [m/s]			4.4	12.9	2.9	8.6																		
	△Pt [pa]			12.4	104.5	5.7	46.6																		
7500	Veff [m/s]					3.1	9.2																		
	△Pt [pa]					6.5	53.5																		
8000	Veff [m/s]					3.4	9.8																		
	△Pt [pa]					7.3	60.8																		
8500	Veff [m/s]					3.6	10.5																		
	△Pt [pa]					8.2	68.6																		
10000	Veff [m/s]							5.0	14.6	4.0	11.7	3.3	9.7	2.7	7.8	2.2	6.5								
	△Pt [pa]							15.7	132.8	1.0	85.1	0.7	59.2	4.7	38.0	3.4	26.8								
11000	Veff [m/s]							5.5	16.0	4.4	12.8	3.6	10.7	2.9	8.5	2.4	7.2								
	△Pt [pa]							18.9	160.6	12.2	102.9	8.6	71.5	5.6	45.9	4.0	32.3								
15000	Veff [m/s]									6.0	17.5	5.0	14.6	4.0	11.7	3.3	9.8	2.7	7.8	2.2	6.5	2.0	5.9		
	△Pt [pa]									22.4	191.1	15.7	132.8	10.1	85.1	7.2	59.9	4.7	38.4	3.4	26.8	2.8	21.7		
16000	Veff [m/s]									6.4	18.6	5.3	15.5	4.2	12.4	3.6	10.4	2.8	8.3	2.4	6.9	2.1	6.3	1.9	5.6
	△Pt [pa]									25.5	217.3	17.8	151.1	11.5	96.8	8.2	68.1	5.3	43.7	3.8	30.4	3.1	24.7	2.5	20.1
17000	Veff [m/s]									6.8	19.8	5.6	16.5	4.5	13.2	3.8	11.1	3.0	8.9	2.5	7.4	2.3	6.6	2.0	6.0
	△Pt [pa]									28.7	245.3	20.0	170.4	12.9	109.2	9.2	76.8	6.0	49.3	4.2	34.3	3.5	27.8	2.9	22.5
18000	Veff [m/s]											6.0	17.5	4.8	14.0	4.0	11.7	3.2	9.4	2.7	7.8	2.4	7.0	2.1	6.3
	△Pt [pa]											22.4	191.1	14.5	122.4	10.3	86.1	6.7	55.2	4.7	38.4	3.9	31.2	3.2	25.4
20000	Veff [m/s]											6.6	19.4	5.3	15.5	4.4	13.0	3.6	10.4	3.0	8.7	2.7	7.8	2.4	6.2
	△Pt [pa]											27.6	235.8	17.8	151.0	12.6	106.2	8.2	68.1	5.8	47.4	4.7	38.4	3.8	31.1
30000	Veff [m/s]															6.7	19.5	5.3	15.6	4.4	13.0	4.0	11.7	3.6	10.5
	△Pt [pa]															28.0	238.5	18.0	152.8	12.6	106.2	10.3	86.1	8.4	69.8
35000	Veff [m/s]																	6.2	18.2	5.2	15.2	4.7	13.7	4.2	12.3
	△Pt [pa]																	24.4	207.8	17.0	144.4	13.8	117.0	11.2	94.8
40000	Veff [m/s]																			5.9	17.4	5.3	15.6	4.8	14.1
	△Pt [pa]																			22.1	188.5	18.0	152.8	14.7	123.9
45000	Veff [m/s]																			6.7	19.5	6.0	17.6	5.4	15.8
	△Pt [pa]																			28.0	238.5	22.7	193.3	18.4	156.7

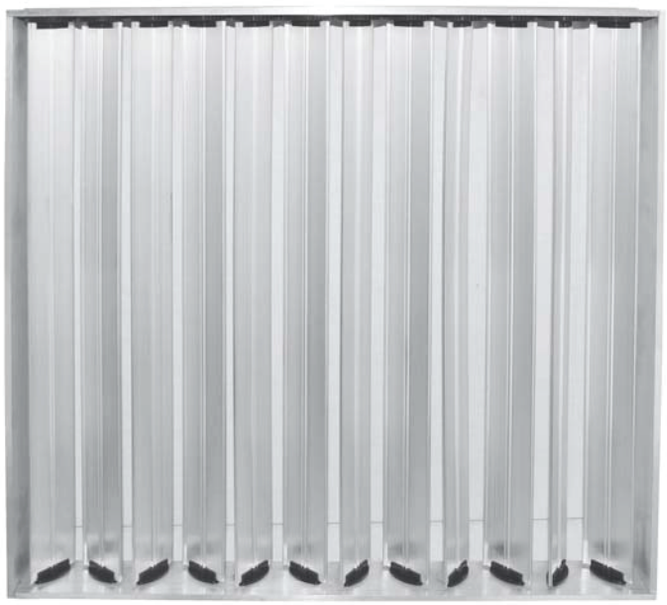
Annotation

* α [°] - Closing angle of the blades
* Veff[m/s] - Air velocity through the damper
* ΔPt [pa] - Pressure loss
* *Veff [m/s] = Air flow [mc/h] / 3600[s] / Ak[m^2]

Order codes

Type	/	S	/	L x H
VCD		Actuator		On request

Opposable Blades Damper OBD



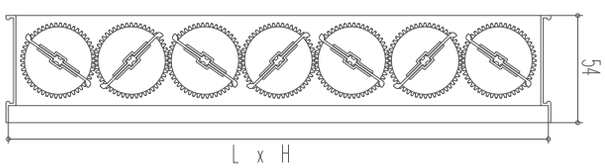
Description
Air control register is mounted on connection of the grille and it is used to adjust the air flow.

Application
Air flow adjustment in ventilation systems

Characteristics
Built with aluminium opposable blades and sprocket wheels.
Manual operation.

Materials and finishing
Natural aluminium blades and frame
Polypropylene sprocket wheels

Technical design



Assembly Methods
With pop rivets on the inlet connection of the grille.

Order codes

Type	/	Dimensions
OBD		On request

Non Return Circular Damper RSD

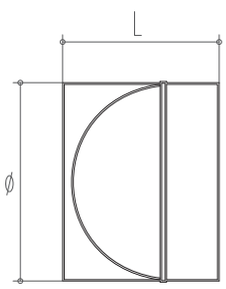
Description
Non return circular damper for mounting between two circular sections of tubing.

Application
It is used to prevent the return of the air when the fan stops.

Characteristics
The damper blades will open when exposed to excess pressure.
The damper is provided with a return spring , which determines closing of the blades when pressure drops.
By closing the blades return of the air is prevented when the fan stops.

Materials and finishing
Galvanized steel cylinder
Aluminium blades
Natural galvanized finish

Technical design



Dimensions	L
100	100
125	100
150	150
160	150
200	150
250	200
315	200
400	250

Order codes

Type	/	Ø
RSD		100 200 125 250 150 315 160 400

Circular Air Control Damper RD

Description
Circular air control damper with driving system.

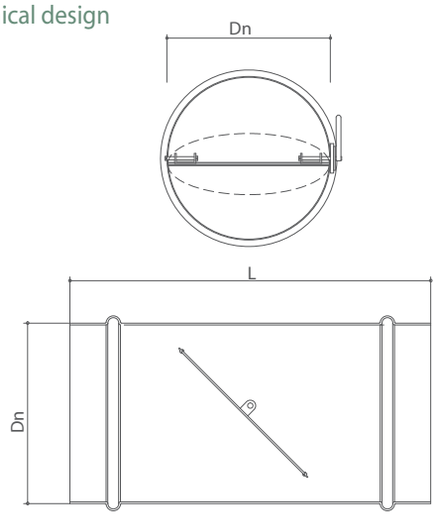
Application
To adjust the air flow in ventilation/air conditioning ducts systems.

Characteristics
The standard damper is produced with manual driving device (lever), unless electric drive (actuator) is requested.
The actuator will be selected according to the size of the damper, voltage and other technical details mentioned in the project (return spring, 0-10V control, etc.).
Actuator operates the opening/ closing of the blade through Ø8 mm shaft.
The damper is provided with a sealing gasket.

Accessories
Electric drive (actuator).

Materials and finishing
Galvanized steel sheet

Technical design



Ø	L	Dn
100	200	95
125	200	120
160	200	155
200	220	195
250	270	245
315	335	310
400	420	395

Selection table RD

Ø [mm]	Ak [mp]
100	0.0077
125	0.0121
160	0.0198
200	0.0311
250	0.0487
315	0.0774
400	0.1250

Correction factor values				
opening angle(˚)	0	15	30	45
Correction coeff.	1	1,8	8	20

ΔPt Corrected = ΔPt * Correction coeff.

Ø (mm)	Air flow m³/h	100	200	300	400	500	600	700	800	900	1000	1500	2000	3000	4000
100	NR [dB(A)] ΔPt [pa] Veff[m/s]	18 7 3.6	33 25 7.2	46 55 10.8											
125	NR [dB(A)] ΔPt [pa] Veff[m/s]		18 6 4.6	30 13 6.9	35 20 9.2										
160	NR [dB(A)] ΔPt [pa] Veff[m/s]			8 4 4.2	20 6 5.6	27 8 7.0	30 14 8.4								
200	NR [dB(A)] ΔPt [pa] Veff[m/s]				8 2 3.6	10 4 4.5	20 5 5.4	24 6.5 6.3	27 8 7.1	33 12 8.0					
250	NR [dB(A)] ΔPt [pa] Veff[m/s]						4 1 3.4	6 1.8 4.0	8 2.5 4.6	18 4 5.1	21 4.5 5.7	30 8 8.6	40 18 11.4		
315	NR [dB(A)] ΔPt [pa] Veff[m/s]											25 2.5 5.4	33 4.5 7.2	44 9 10.8	
400	NR [dB(A)] ΔPt [pa] Veff[m/s]													37 3 6.7	45 55 8.9

Annotation

* Ak [m²] - Free surface
* NR [dB(A)] - Noise level
* ΔPt [pa] - Pressure loss
* Veff[m/s] - Air velocity through the damper

Order codes

Type	+	S	/	Ø
RD		Actuator		On request

Air Volume Control Damper IRIS

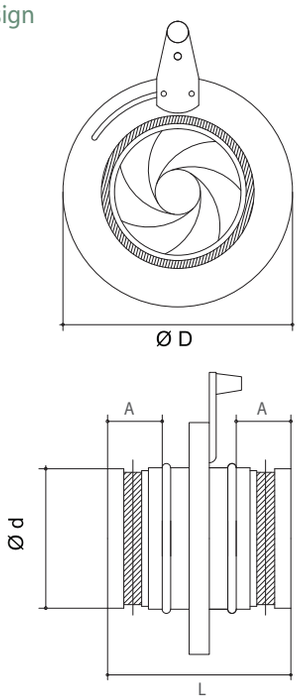
Description
Iris diaphragm damper allows adjustment of the air flow by measuring pressure at two points. .

Application
Adjustment of the air flow in ventilation and air conditioning systems.

Characteristics
The damper is provided with an iris diaphragm, pressure measuring points, operating lever and fitting gaskets.
Various levels of blade opening.
The iris is adjusted by operating lever.
The damper is delivered along with a selection chart (pressure loss/ air flow).

Materials and finishing
Galvanized steel sheet
Plastic adjustment device

Technical design



Selection table IRIS

IRIS			
Ø d	Ø D	L	A
100	165	110	30
160	230	110	30
200	285	110	30
250	335	135	35
315	410	135	35

Diameter	Air flow	
	min (m^3/h) 35 dB(A)	max (m^3/h) 55 dB(A)
100	40	270
160	60	400
200	120	600
250	270	1100
315	400	2500

Order codes

Type	/	Diameter
IRIS		100 160 200 250 315

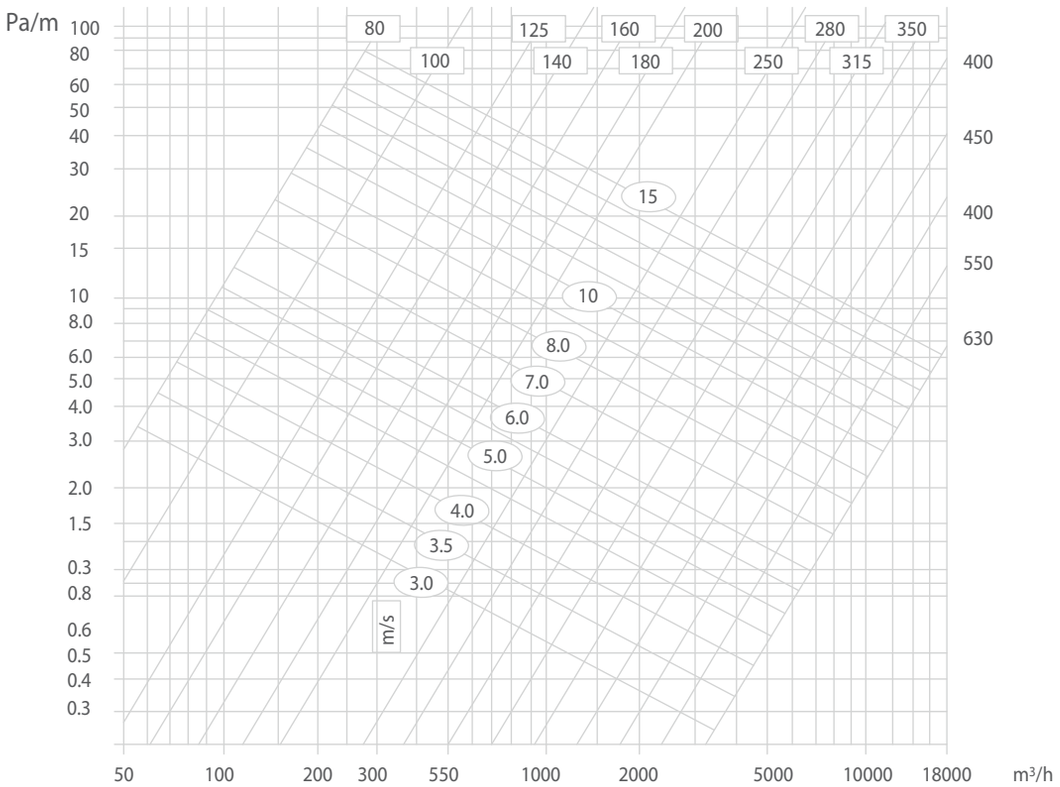
Flexible Ducts

Description
Aluminum flexible duct used in ventilation and air conditioning installations.

Application
Supply/ exhaust of the air.

Characteristics
Flexible tubing is available both non-insulated and insulated.
The standard length is 10 m/pc.
Fiberglass insulation.
The temperature range is - 30°C ÷ 140°C.
Maximum velocity is 20 m/s.
Not suitable for air transport with high concentration of acids and bases or for discharge of combustion gases.

Pressure loss chart



Adapter ADP

Description
Adapter (plenum) made of galvanized sheet, used as an intermediary element between the grille and ventilation ducts.

Characteristics
Adapter can be insulated or non-insulated **ADP IZ/N**
It is made with one or more connections for ventilation ducts.
The connector (socket) is positioned opposite to the grille (standard) or side (optional)
On request adapter is provided with mounting flange

Special Parts

Description
Parts made of galvanized sheet metal for ventilation ducts, type: "T", "Y", sockets, bent pipes, etc.

Characteristics
Can be isolated on request
Dimensions on request

Accessories

Air Filter FIL

Description
Air filter for grille's inlet connection.

Characteristics
The air filter consists of filter material and "U" shaped aluminum frame
Class 2 polyester filter material, with high retention level and thickness of 12 mm.

Mounting Spring

Description
Mounting spring is used for wall or ceiling mounting of the grille.

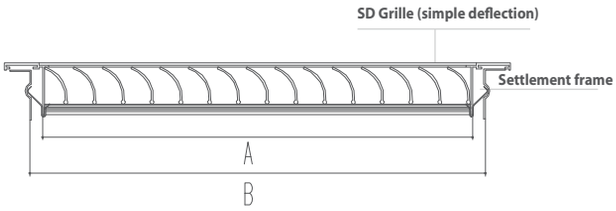
Characteristics
Optional accessory used as a way of mounting.
Not visible after mounting
Spring length is 105 mm.

Settlement Frame For Mounting CC

Description
Settlement frame made of galvanized steel sheet used for wall or ceiling mounting with clip mechanism.

Characteristics
Optional accessory used as a way of mounting.
The mounting of the grille with settlement frame is not visible.
It is mounted in the hole and allows quick detachment of the grille.
The grille is attached to the settlement frame through the clips mounted on the grille's inlet.

Technical design



A - interior size of the grille
B - size of the settlement frame

Mounting Clips

Description
Clip made of steel sheet for mounting the grille in frame.

Ram air flow

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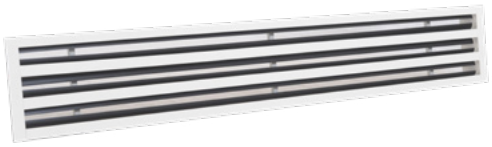
פתרונות מיזוג אוויר מעוצבים

קטלוג מוצרים

סלוטים

סלוט צר - 2 חריצים



סלוט צר - 3 חריצים

סלוט צר

תיאור מוצר	מינימום לחישוב	נתונים טכניים					
		עומק	רוחב	רוחב עם ברגים	שוליים מבחוץ	שוליים מבפנים	אורך מירבי ללא חיבור
סלוט צר חריץ 1	1 מ"א	25 מ"מ	48	52	20 מ"מ	7.5	3.80 מטר
סלוט צר 2 חריצים	1 מ"א	25 מ"מ	95	99	20 מ"מ	0	6 מ'- 9016 . 3.8 מ' -צבעים אחרים
סלוט צר 3 חריצים	1 מ"א	25 מ"מ	142	146	20 מ"מ	0	6 מטר
סלוט צר 4 חריצים	1 מ"א	25 מ"מ	189	193	20 מ"מ	0	6 מטר
סלוט צר צוקל 2 חריצים	1 מ"א	25 מ"מ			0 מ"מ	0	6 מטר
סלוט צר 2 ח' 95*495	יחידה	25 מ"מ			20 מ"מ	9.5	-
סלוט צר 2 ח' 95*795	יחידה	25 מ"מ			20 מ"מ	9.5	-
סלוט צר 2 ח' 95*995	יחידה	25 מ"מ			20 מ"מ	9.5	-
סלוט צר 2 ח' 95*1495	יחידה	25 מ"מ			20 מ"מ	9.5	-
סלוט צר 2 ח' 95*1995	יחידה	25 מ"מ			20 מ"מ	9.5	-
סלוט צר 2 ח' 95*2495	יחידה	25 מ"מ			20 מ"מ	9.5	-
סלוט צר 2 ח' 95*2995	יחידה	25 מ"מ			20 מ"מ	9.5	-

*ישנה אופצייה לסלוט 5 חריצים

סלוט

תיאור מוצר	מינימום לחישוב	נתונים טכניים		אורך מירבי לחתיכה	
		עומק	רוחב אפשרי	שוליים 18 מבחוץ	שוליים 27 מבחוץ
סלוט חריץ 1	1 מ"א	50 מ"מ	40-62	3.8 מטר	3.2 מטר
סלוט 2 חריצים	1 מ"א	50 מ"מ	73-116	9016 -6 מ'. אחר-3.2 מ'	3.2 מטר
סלוט 3 חריצים	1 מ"א	50 מ"מ	117-170	3.8 מטר	3.2 מטר
סלוט 4 חריצים	1 מ"א	50 מ"מ	171-224	3.8 מטר	3.2 מטר
סלוט 5 חריצים	1 מ"א	50 מ"מ	225-278	3.8 מטר	3.2 מטר
סלוט 6 חריצים	1 מ"א	50 מ"מ	201-332	3.8 מטר	3.2 מטר
סלוט 7 חריצים	1 מ"א	50 מ"מ	332-387	3.8 מטר	3.2 מטר
סלוט 8 חריצים	1 מ"א	50 מ"מ	387-441	3.8 מטר	3.2 מטר
סלוט 2 ח' 95*495	יחידה	50 מ"מ			-
סלוט 2 ח' 95*795	יחידה	50 מ"מ			-
סלוט 2 ח' 95*995	יחידה	50 מ"מ			-
סלוט 2 ח' 95*1495	יחידה	50 מ"מ			-
סלוט 2 ח' 95*1995	יחידה	50 מ"מ			-
סלוט 2 ח' 95*2495	יחידה	50 מ"מ			-
סלוט 2 ח' 95*2995	יחידה	50 מ"מ			-

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דקור פרמיום - במידות מיוחדות

שוליים 12 ס"מ

תיאור מוצר	מינימום לחישוב	אורך מירבי ביחידה אחת		
		עומק	9016	אחר
דקור פרמיום גובה 95	1 מ"א	27 מ"מ	5 מטר	5 מטר
דקור פרמיום גובה 145	1 מ"א	27 מ"מ	5 מטר	5 מטר
דקור פרמיום גובה 195	1 מ"א	27 מ"מ	5 מטר	5 מטר
דקור פרמיום לבן על רקע שחור גובה 95	1 מ"א	27 מ"מ	5 מטר	5 מטר
דקור פרמיום לבן על רקע שחור גובה 145	1 מ"א	27 מ"מ	5 מטר	5 מטר
דקור פרמיום לבן על רקע שחור גובה 195	1 מ"א	27 מ"מ	5 מטר	5 מטר
דקור פרמיום עד גובה 200 מ"מ	0.1 מ"ר	27 מ"מ	6 מטר	5 מטר
דקור פרמיום מעל גובה 200 מ"מ	0.1 מ"ר	27 מ"מ	6 מטר	5 מטר
דקור פרמיום עד גובה 200 מ"מ לבן על רקע שחור	0.1 מ"ר	27 מ"מ	6 מטר	5 מטר
דקור פרמיום מעל גובה 200 מ"מ לבן על רקע שחור	0.1 מ"ר	27 מ"מ	6 מטר	5 מטר

דקור

תיאור מוצר	מינימום לחישוב	נתונים טכניים			
		עומק	ללא שוליים	שוליים 18 מ"מ	שוליים 27 / שוליים 10
דקור עד גובה 200 מ"מ	0.1 מ"ר	20		3.8 מטר	3.2 מטר
דקור מעל גובה 200 מ"מ	0.1 מ"ר	20		3.8 מטר	3.2 מטר
דקור שדרות שחורות	0.1 מ"ר	20		3.8 מטר	3.2 מטר
דקור נפתח על ציר + פילטר	0.2 מ"ר	40		3.2 מטר	
דקור עם משקוף עיוור	0.1 מ"ר	49	3.2 מטר		
להבי הטיה אחוריים	0.1 מ"ר	16			
דקור צוקל עד 200 מ"מ	0.1 מ"ר	30	3 מטר		
דקור צוקל מעל 200 מ"מ	0.1 מ"ר	30	3 מטר		
סטריפליין חד/דו כיווני	0.1 מ"ר	24	3.2 מטר		

דקור במידות קבועות במלאי מיידי

תיאור מוצר
דקור 95*500
דקור 95*600
דקור 95*800
דקור 95*995
דקור 95*1495
דקור 95*1995
דקור 95*2495
דקור 95*2995
דקור 95*4000
דקור 200*400
דקור 200*800
דקור 150*400
דקור 150*600
דקור פרמיום 150*1000
דקור פרמיום 150*1500
דקור פרמיום 150*2000
דקור פרמיום 150*2500
דקור פרמיום 150*3000
דקור פרמיום 200*1000
דקור פרמיום 200*1500
דקור פרמיום 200*2000
דקור פרמיום 200*2500
דקור פרמיום 200*3000

דקור פרימיום

דקור

דקור שוליים 0.7

החברה שומרת לעצמה את האפשרות לשנות נתונים טכניים ומחירים ללא הודעה מראש. תיתכן סטייה קלה בין מידות הקטלוג לבין מידות המוצר בפועל.

סלוט צוקל

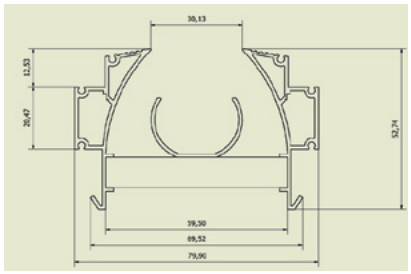
תיאור מוצר	מינימום לחישוב	נתונים טכניים					
		עומק	רוחב	רוחב עם ברגים	שוליים מבחוץ	שוליים מבפנים	אורך מירבי ללא חיבור
סלוט צוקל צר חריץ 1	1 מ"א	25 מ"מ	48				3.80 מטר
סלוט צוקל צר 2 חריצים	1 מ"א	25 מ"מ	95				3.80 מטר
סלוט צוקל צר 3 חריצים	1 מ"א	25 מ"מ	142				3.80 מטר

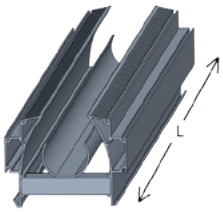
סלוט צוקל

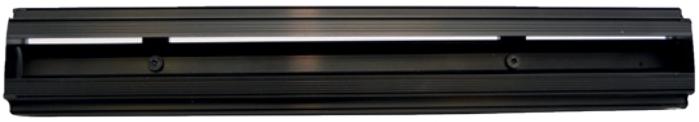
שחר

תיאור מוצר	מינימום לחישוב	נתונים טכניים		אורך מירבי לחתיכה
		עומק	רוחב אפשרי	שוליים 18 מבחוץ
שחר חריץ אחד	1 מ"א	53 מ"מ		6 מטר
שחר שני חריצים	1 מ"א	53 מ"מ		3.2 מטר
שחר MDF	1 מ"א	46 מ"מ		2.4 מטר
תוספת לחיתוך 45 מעלות	יחידה			
קופסאת ניפוח כולל בידוד	1 מ"א			
קופסאת ניפוח ללא בידוד	1 מ"א			

*תוספת חיתוך גרונג - 100₪







תריס אספקה קווים-שחר

החברה שומרת לעצמה את האפשרות לשנות נתונים טכניים ומחירים ללא הודעה מראש. תיתכן סטייה קלה בין מידות הקטלוג לבין מידות המוצר בפועל.

מפזר אוויר תקרתי מרובע
סלוט - SL

דגמים:

- ✔ תקרתי חריץ 1
- ✔ תקרתי 2 חריצים

שניתן לקבל בשני אופנים
חריץ 1 ו - 2 חריצים.

סלוט מרובע

*ניתן לקבל את הליבה עם אלומיניום, תקרה אקוסטית, תאורה.

סלוט צר

תיאור מוצר	מינימום לחישוב	נתונים טכניים			
		עומק	רוחב	שוליים מבחוץ	שוליים מבפנים
סלוט צר חריץ 1 מרובע 600*600 כולל סגירה	יחידה	25 מ"מ	20 מ"מ	20 מ"מ	7.5
סלוט צר חריץ 1 מרובע 600*600 ללא סגירה	יחידה	25 מ"מ	20 מ"מ	20 מ"מ	7.5
סלוט צר 2 ח' מרובע 600*600 כולל סגירה	יחידה	25 מ"מ	20 מ"מ	20 מ"מ	7.5
סלוט צר 2 ח' מרובע 600*600 ללא סגירה	יחידה	25 מ"מ	20 מ"מ	20 מ"מ	7.5

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תריסי אספקה

תיאור מוצר	מינימום לחישוב	שוליים
תריס אספקה 150*300 להב אחורי שחור	1 יחידה	27 מ"מ
תריס אספקה 200*300 להב אחורי שחור	1 יחידה	27 מ"מ
תריס אספקה 300*300 להב אחורי שחור	1 יחידה	27 מ"מ
תריס אספקה 150*400 להב אחורי שחור	1 יחידה	27 מ"מ
תריס אספקה 200*400 להב אחורי שחור	1 יחידה	27 מ"מ
תריס אספקה 150*500 להב אחורי שחור	1 יחידה	27 מ"מ
תריס אספקה 200*500 להב אחורי שחור	1 יחידה	27 מ"מ
תריס אספקה 150*600 להב אחורי שחור	1 יחידה	27 מ"מ
תריס אספקה 200*600 להב אחורי שחור	1 יחידה	27 מ"מ
תריס אספקה 200*800 להב אחורי שחור	1 יחידה	27 מ"מ
אוויר חוזר משולב אספקה 150*600 להב אחורי שחור	1 יחידה	27 מ"מ
אוויר חוזר משולב אספקה 200*600 להב אחורי שחור	1 יחידה	27 מ"מ
אוויר חוזר משולב אספקה 200*800 להב אחורי שחור	1 יחידה	27 מ"מ
תריס אספקה	0.1 מ"ר	27 מ"מ
תריס אספקה להב אחורי שחורי	0.1 מ"ר	27 מ"מ
אספקה לתעלה עגולה	0.1 מ"ר	27 מ"מ
תריס אספקה 220*220	1 יחידה	27 מ"מ
תריס אספקה 300*150	1 יחידה	27 מ"מ
תריס אספקה 300*200	1 יחידה	27 מ"מ
תריס אספקה 300*300	1 יחידה	27 מ"מ
תריס אספקה 400*150	1 יחידה	27 מ"מ
תריס אספקה 400*200	1 יחידה	27 מ"מ
תריס אספקה 500*150	1 יחידה	27 מ"מ
תריס אספקה 500*200	1 יחידה	27 מ"מ
תריס אספקה 600*150	1 יחידה	27 מ"מ
תריס אספקה 600*200	1 יחידה	27 מ"מ
תריס אספקה 800*200	1 יחידה	27 מ"מ
תריס אספקה 1500*100	1 יחידה	27 מ"מ
תריס אספקה 2000*100	1 יחידה	27 מ"מ
אויר חוזר משולב עם אספקה	0.1 מ"ר	27 מ"מ

תריס אספקה שדרות שחורות

אספקה משולב

תריסי אספקה

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עגולים

תקרות נמוכות

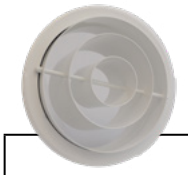
תיאור מוצר	דגם
תקרות נמוכות 6"	RCDL150
תקרות נמוכות 8"	RCDL200
תקרות נמוכות 10"	RCDL250
תקרות נמוכות 12"	RCDL300
תקרות נמוכות 14"	RCDL350
תקרות נמוכות 16"	RCDL400
תקרות נמוכות 18"	RCDL450
תקרות נמוכות 20"	RCDL500

ג'ט - JET

תיאור מוצר	דגם
ג'ט 6"	JET150
ג'ט 8"	JET200
ג'ט 10"	JET250
ג'ט 12"	JET300
ג'ט 14"	JET350
ג'ט 16"	JET400

תקרות גבוהות

תיאור מוצר	דגם
תקרות גבוהות 8"	RCDH200
תקרות גבוהות 10"	RCDH250
תקרות גבוהות 12"	RCDH300
תקרות גבוהות 14"	RCDH350
תקרות גבוהות 16"	RCDH400



ג'ט טבעות- SJ

תיאור מוצר	דגם
ג'ט טבעות 8"	SJ200
ג'ט טבעות 10"	SJ250
ג'ט טבעות 12"	SJ300
ג'ט טבעות 14"	SJ350
ג'ט טבעות 16"	SJ400
ג'ט טבעות 20"	SJ450

תקרות גבוהות מתכוונן

תיאור מוצר	דגם
תקרתי עגול מתכוונן 12"	VSR-12
תקרתי עגול מתכוונן 16"	VSR-16
תקרתי עגול מתכוונן 20"	VSR-20

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ממ"ד

תיאור מוצר
ממ"ד 8"
ממ"ד 8" מסגרת 350*350
ממ"ד 8" מסגרת 380*380

יניקה / אספקה

תיאור מוצר	דגם
אספקה / יניקה 4"	SV/EV-100
אספקה / יניקה 5"	SV/EV-125
אספקה / יניקה 6"	SV/EV-150
אספקה / יניקה 8"	SV/EV-200

תקרתי 4 כיוונים

תיאור מוצר
תקרתי 6"
תקרתי 9"
תקרתי 12"
תקרתי 15"
תקרתי 18"
תקרתי 21"

תקרתי 4 כיוונים

תריס ממ"ד

מחליפי אריח

תיאור מוצר
מחליף אריח 6"
מחליף אריח 9"
מחליף אריח 12"
מחליף אריח 15"
מחליף אריח 18"

תריס ממ"ד מסגרת

תריס אספקה/יניקה

לב אספקה

תיאור מוצר
תקרתי + לב אספקה 9"
תקרתי + לב אספקה 12"
תקרתי + לב אספקה 15"
תקרתי + לב אספקה 18"
מחליף אריח + לב אספקה 9"
מחליף אריח + לב אספקה 12"
מחליף אריח + לב אספקה 15"

לב אספקה

מחליפי אריח

תיאור מוצר	מינימום לחישוב
תקרתי	0.085 מ"ר
תקרתי לב אספקה	0.085 מ"ר

מחליף אריח

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לוח מערבולת

דגם	מינימום לחישוב
SWCI-24	יחידה
SWCI-28	יחידה
SWCI-36	יחידה
SWCI-48	יחידה
SWCI-48/2	יחידה

לוח מערבולת

סלוט צר 2 חריצים

דקור

דקור פרימיום

דקור פרימיום שחור לבן

שחר 1 חריץ

ג'ט

תריס נגד גשם

תריס אספקה עגול

תריס יניקה עגול

תריס אספקה
שדרות שחורות

תריס אוויר חוזר
נפתח על ציר

תקרתי 4 כיוונים

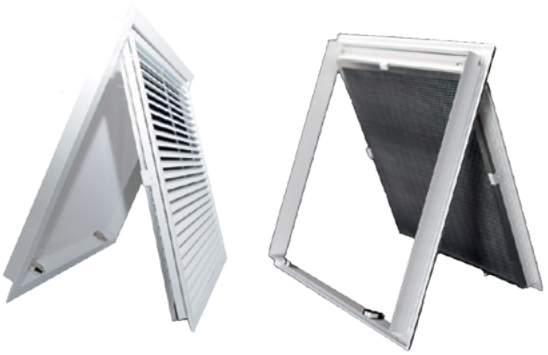
החברה שומרת לעצמה את האפשרות לשנות נתונים טכניים ומחירים ללא הודעה מראש. תיתכן סטייה קלה בין מידות הקטלוג לבין מידות המוצר בפועל.

אוויר חוזר

תיאור מוצר	מינימום לחישוב
תריס אוויר חוזר	0.1 מ"ר
אוויר חוזר + ווסת	0.1 מ"ר
אוויר חוזר קבוע + פילטר	0.1 מ"ר

אוויר חוזר נפתח על ציר

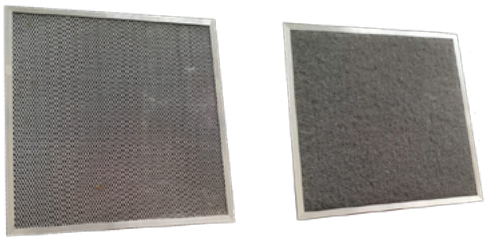
תיאור מוצר	מינימום לחישוב
אוויר חוזר נפתח על ציר + פילטר	0.2 מ"ר
אויר חוזר ליבה נשלפת ללא ציר עם פרפרים	0.2 מ"ר
אוייר חוזר נ.ע.צ + פילטר 540 * 540	יחידה
אוייר חוזר נ.ע.צ + פילטר 540 * 540 פינליין	יחידה
אויר חוזר פלסטיק עם פרפרים 540 * 540	יחידה
אוייר חוזר נ.ע.צ + פילטר 545 * 545 - חוץ 600	יחידה
אוייר חוזר נ.ע.צ + פילטר 550 * 550 - חוץ 605	יחידה
אוייר חוזר נ.ע.צ + פילטר 590 * 590	יחידה
אוייר חוזר נ.ע.צ + פילטר 595 * 595	יחידה
אוייר חוזר 540 * 540 - חוץ 595	יחידה
אוייר חוזר 540 פינליין	יחידה
אויר חוזר 545 * 545 - חוץ 600	יחידה
אויר חוזר 595 * 595	יחידה
אויר חוזר 140 * 140	יחידה
אויר חוזר 190 * 190	יחידה



אוויר חוזר נפתח על ציר

פילטר

תיאור מוצר	מינימום לחישוב
פילטר חסין אש עם מסגרת	0.2 מ"ר



אוויר חוזר נגד גשם

תיאור מוצר	מינימום לחישוב
אויר חוזר נגד גשם להב רחב	0.1 מ"ר
אויר חוזר נגד גשם להב צר	0.1 מ"ר
מחיר לרשת	0.25 מ"ר
אויר חוזר נגד גשם עגול 4"	יחידה
אויר חוזר נגד גשם עגול 6"	יחידה
אויר חוזר נגד גשם עגול 8"	יחידה
אויר חוזר נגד גשם עגול 10"	יחידה
אויר חוזר נגד גשם עגול 12"	יחידה
אויר חוזר נגד גשם עגול 14"	יחידה
אויר חוזר נגד גשם כיפתי 4"	יחידה
אויר חוזר נגד גשם כיפתי 6"	יחידה
אויר חוזר נגד גשם כיפתי 8"	יחידה

תריס נגד גשם

תריס נגד גשם

החברה שומרת לעצמה את האפשרות לשנות נתונים טכניים ומחירים ללא הודעה מראש. תיתכן סטייה קלה בין מידות הקטלוג לבין מידות המוצר בפועל.

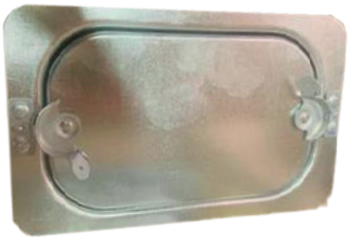
דלת שירות - AD

תיאור מוצר	מינימום לחישוב
דלת שירות	0.2 מ"ר
דלת שירות ללא ציר	0.1 מ"ר
דלת שירות 400*400	יחידה
דלת שירות 500*500	יחידה
דלת שירות 600*600	יחידה
דלת שירות 540*540	יחידה

דלת שרות AD

דלת שירות למדף אש - AC

תיאור מוצר	מינימום לחישוב
דלת שירות מדף אש 295*205	יחידה
דלת שירות מדף אש 295*430	יחידה



דף שרות למדף אש - AC

דלת גבס - DG

תיאור מוצר	מינימום לחישוב
דלת שרות גבס 400*400	יחידה
דלת שרות גבס 500*500	יחידה
דלת שרות גבס 600*600	יחידה

דלת גבס - DG

ווסתים - OBD-VCD

תיאור מוצר	מינימום לחישוב
ווסת כמות	0.1 מ"ר
דמפר וויסות ידני גלגלי שיניים נסתרים	0.25 מ"ר
דמפר וויסות גלגלי שניים נסתרים הכנה למנוע/מנוע נסתר	0.25 מ"ר
מדף ויסות עגול 6"	יחידה
מדף ויסות עגול 8"	יחידה
מדף ויסות עגול 10"	יחידה

ווסתים OBD-VCD

אל חזור

תיאור מוצר	מינימום לחישוב
תריס/דמפר אל חזור	0.25 מ"ר

*להב קצר בתוספת 30%

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RAM
Airflow

פתרונות מיזוג אוויר מעוצבים

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